Supplementary Figures for Chapter 4.

Comorbidities in Juvenile-onset Rheumatic Diseases: a systematic review and meta-analysis

Shall be considered in conjunction with the

Thesis submitted in accordance with the requirements of the

University of Liverpool

for the degree of Doctor of Philosophy by

Syed Abu Bakar Siddiq

Department of Health Data Science

2026

=For reproducibility only in academic research and development=

Supplementary Figures S4.1-S4.75. Meta-analysis: Pooled estimates, Forest plots and Funnel plots 3
Childhood/JIA 3
Adulthood/JIA 27
Childhood/jSLE 37
Adulthood/jSLE 54
Childhood/JDM 66
Adulthood/JDM 76

=For reproducibility only in academic research and development=

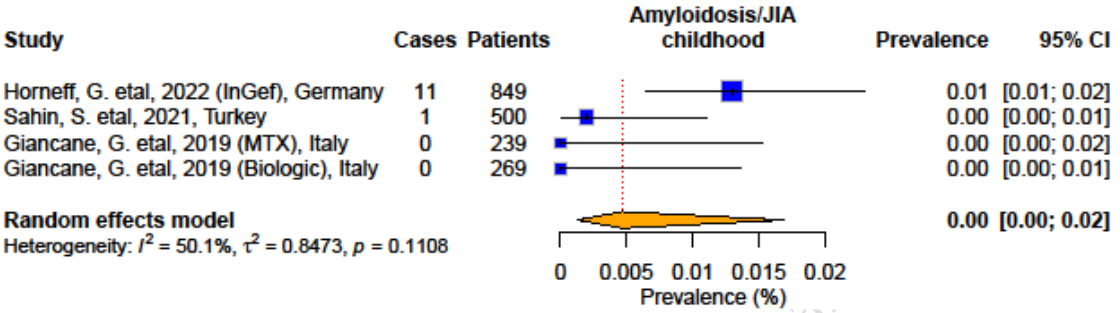
Supplementary Figures S4.1-S4.75. Meta-analysis: Pooled estimates, Forest plots and Funnel plots

Childhood/JIA

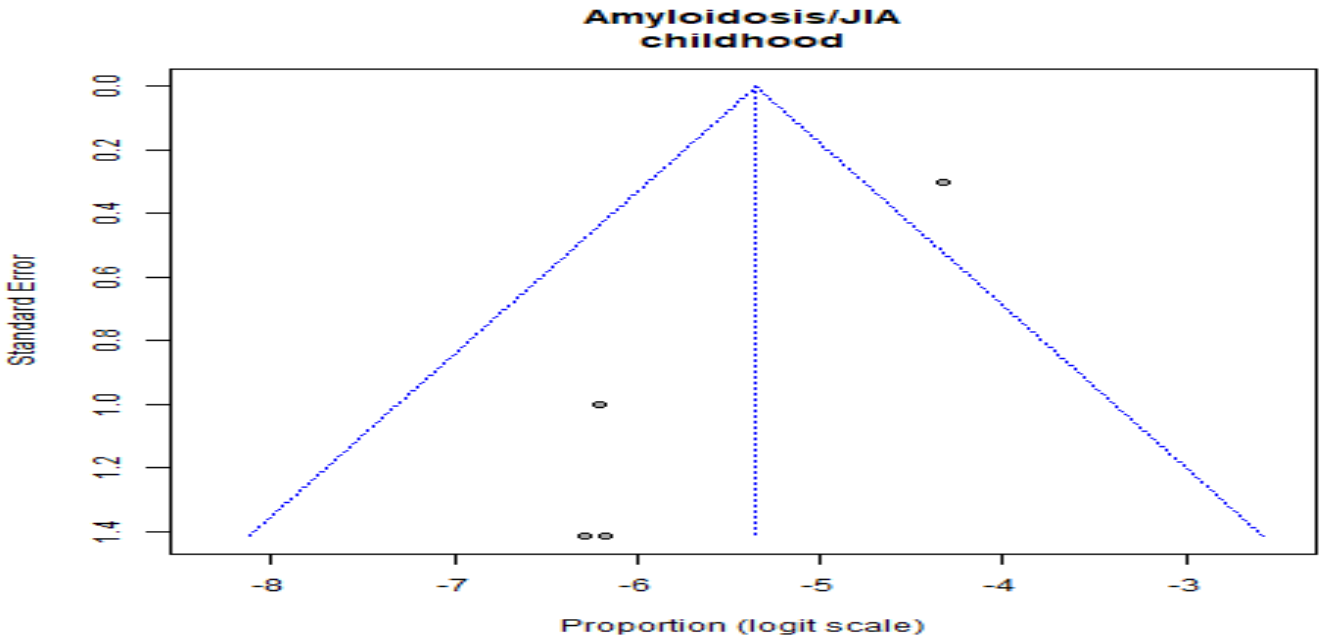
Supplementary Figure S4.1: Forest and Funnel Plots of studies reporting the prevalence of amyloidosis in children with Juvenile Idiopathic Arthritis

Amyloidosis

Forest



Funnel



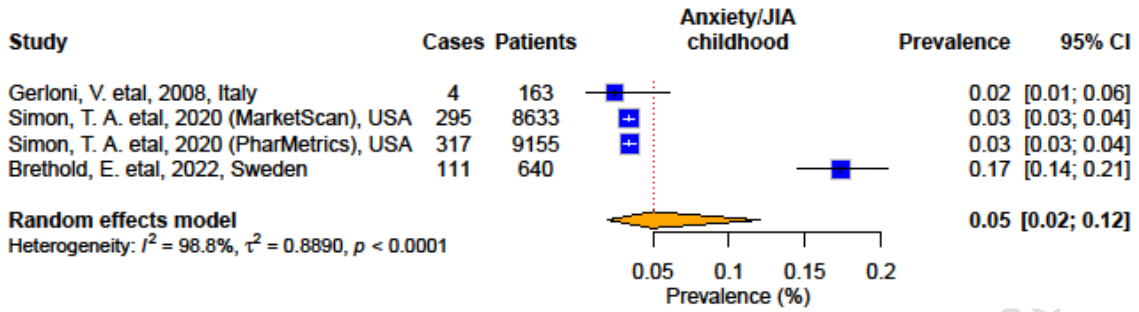
Summary statistics

```
                                        proportion           95%-CI     %W(random)
Horneff G. etal, 2022 (InGef), Germany      0.0130 [0.0065; 0.0231]     46.1
Sahin, S. etal, 2021, Turkey                0.0020 [0.0001; 0.0111]     23.4
Giancane G etal, 2019 (MTX), Italy          0.0000 [0.0000; 0.0153]     15.2
Giancane G etal, 2019 (Biologic), Italy     0.0000 [0.0000; 0.0136]     15.2
Number of studies: k = 4; Number of observations: o = 1857; Number of events: e = 12
                    proportion           95%-CI
Random effects model     0.0047 [0.0013; 0.0170]
Quantifying heterogeneity (with 95%-CIs): tau^2 = 0.8473 [0.0000; 11.9786]; tau = 0.9205 [0.0000; 3.4610]
 I^2 = 50.1% [0.0%; 83.5%]; H = 1.42 [1.00; 2.46]
Test of heterogeneity:
    Q d.f. p-value
 6.02    3  0.1108
```

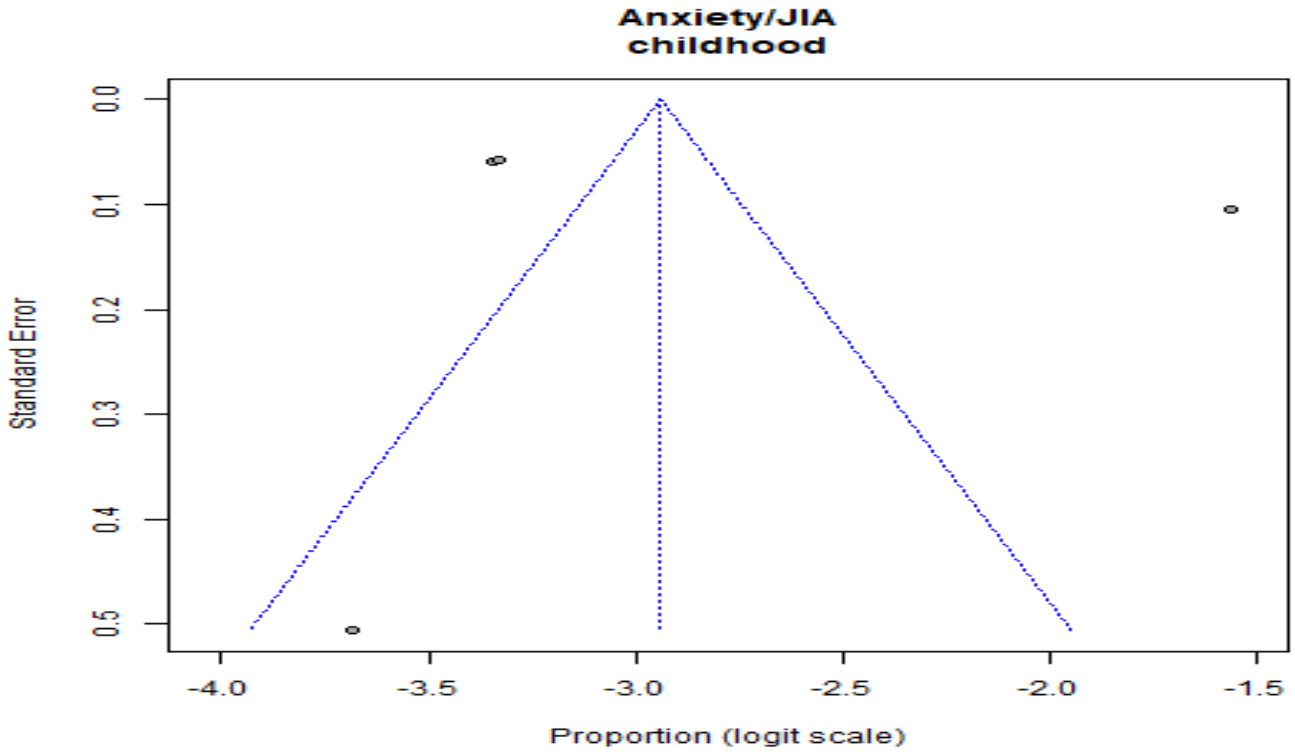
Supplementary Figure S4.2: Forest and Funnel Plots of studies reporting the prevalence of anxiety in children with Juvenile Idiopathic Arthritis

Anxiety

Forest



Funnel



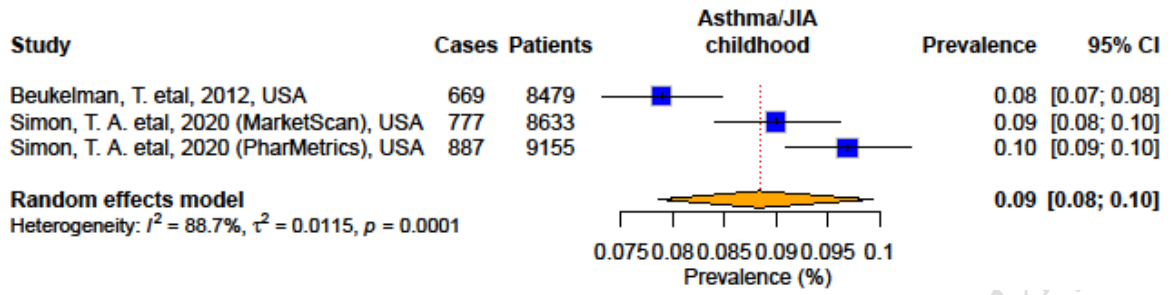
Summary statistics

```
                                      proportion           95%-CI  %W(random)
Gerloni, V. etal, 2008, Italy             0.0245 [0.0067; 0.0616]  20.7
Simon, T. A. etal, 2020 (MarketScan), USA 0.0342 [0.0304; 0.0382]  26.5
Simon, T. A. etal, 2020 (PharMetrics), USA 0.0346 [0.0310; 0.0386]  26.5
Brethold E. etal, 2022, Sweden            0.1734 [0.1449; 0.2050]  26.3
Number of studies: k = 4; Number of observations: o = 18591; Number of events: e = 727
                    proportion           95%-CI
Random effects model     0.0502 [0.0200; 0.1206]
Quantifying heterogeneity (with 95%-CIs): tau^2 = 0.8890 [0.2565; 12.7255]; tau = 0.9429 [0.5065; 3.5673]
 I^2 = 98.8% [98.2%; 99.2%]; H = 9.15 [7.38; 11.34]
Test of heterogeneity:
      Q d.f.  p-value
 251.03    3 < 0.0001
```

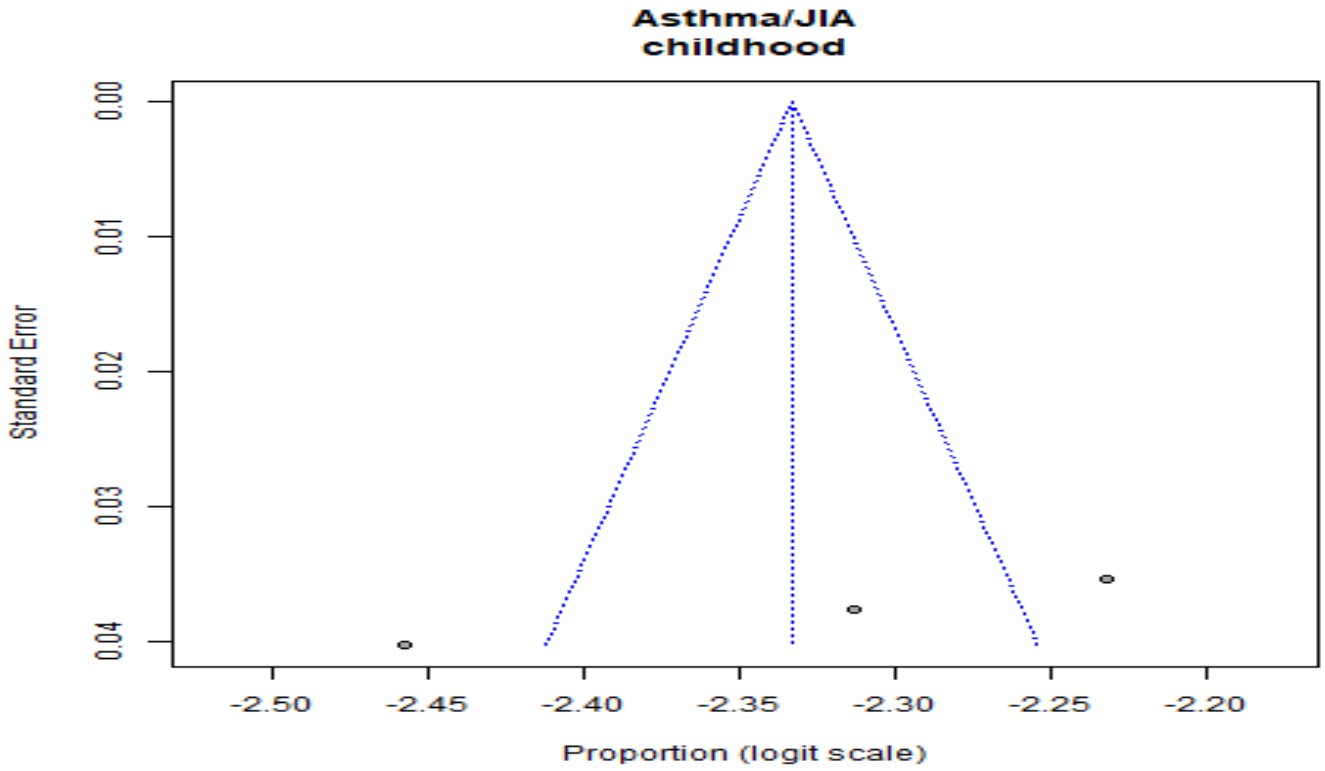
Supplementary Figure S4.3: Forest and Funnel Plots of studies reporting the prevalence of asthma in children with Juvenile Idiopathic Arthritis

Asthma

Forest



Funnel



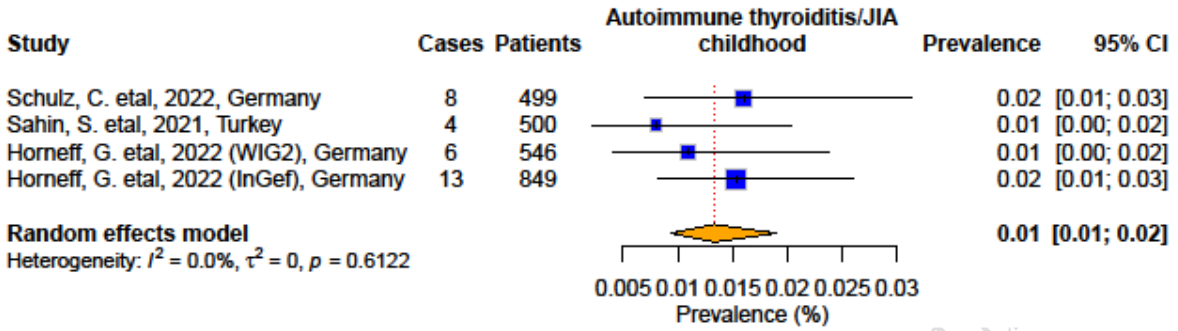
Summary statistics

```
                                        proportion           95%-CI  %W(random)
Beukelman, T. etal, 2012, USA               0.0789 [0.0732; 0.0848]  32.8
Simon, T. A. etal, 2020 (MarketScan), USA   0.0900 [0.0840; 0.0962]  33.4
Simon, T. A. etal, 2020 (PharMetrics), USA  0.0969 [0.0909; 0.1031]  33.8
Number of studies: k = 3; Number of observations: o = 26267; Number of events: e = 2333
                     proportion           95%-CI
Random effects model     0.0884 [0.0786; 0.0993]
Quantifying heterogeneity (with 95%-CIs):
 tau^2 = 0.0115 [0.0021; 0.5116]; tau = 0.1072 [0.0454; 0.7153]
 I^2 = 88.7% [69.0%; 95.9%]; H = 2.98 [1.80; 4.94]
Test of heterogeneity:      Q d.f. p-value
                        17.77    2  0.0001
```

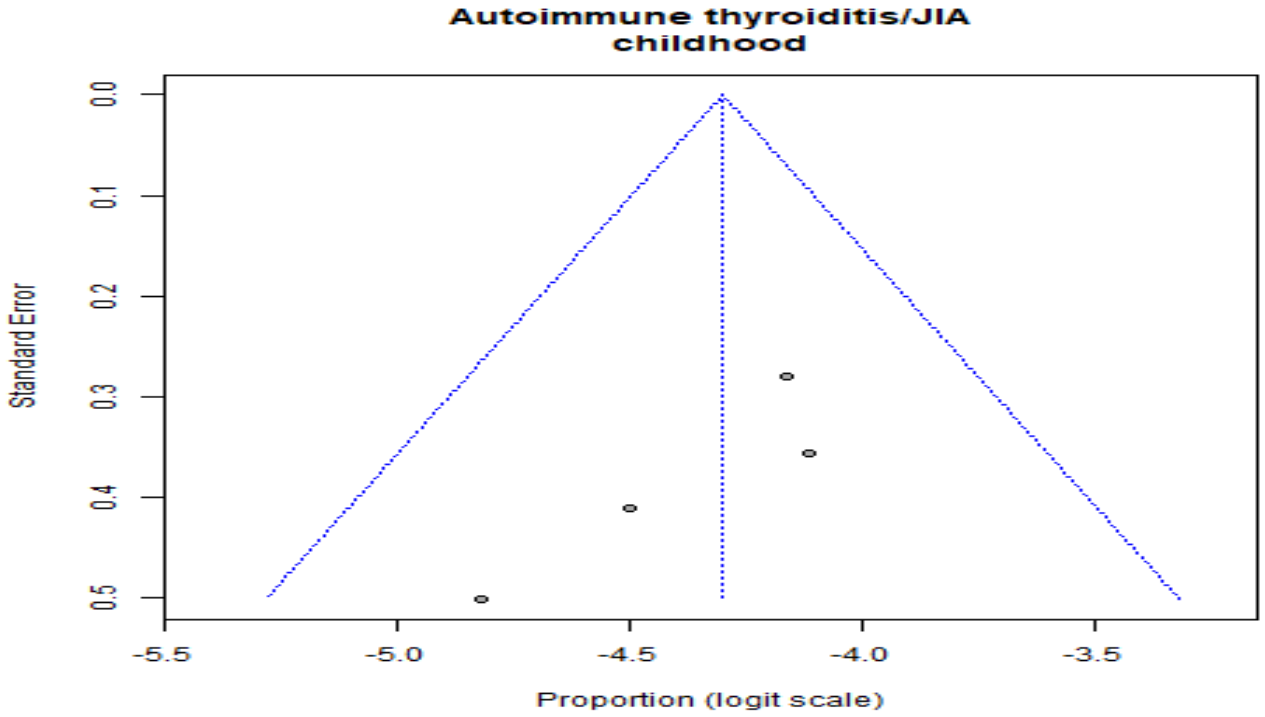
Supplementary Figure S4.4: Forest and Funnel Plots of studies reporting the prevalence of autoimmune thyroiditis in children with Juvenile Idiopathic Arthritis

Autoimmune thyroiditis

Forest



Funnel



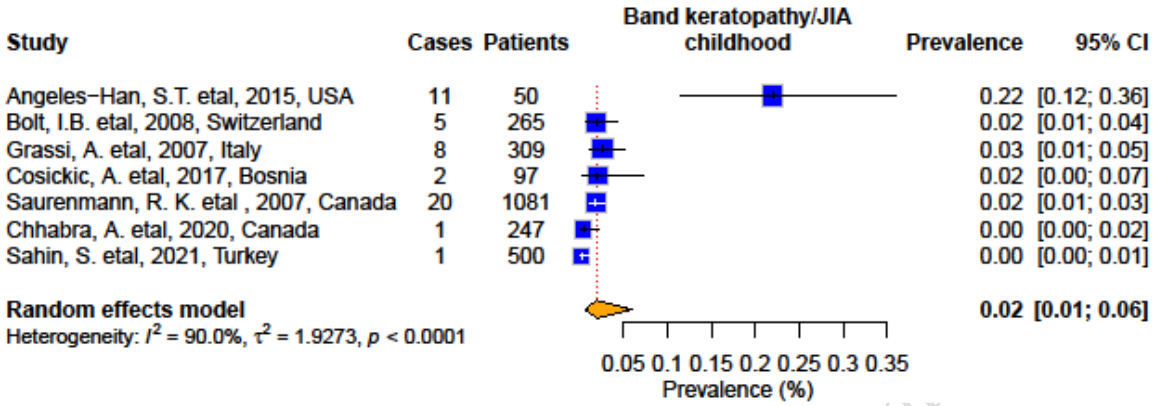
Summary statistics

```
                                  proportion           95%-CI      %W(random)
Schulz, C. etal, 2022, Germany        0.0160 [0.0069; 0.0313]      25.7
Sahin, S. etal, 2021, Turkey          0.0080 [0.0022; 0.0204]      13.0
Horneff G. etal, 2022 (WIG2), Germany 0.0110 [0.0040; 0.0238]      19.4
Horneff G. etal, 2022 (InGef), Germany 0.0153 [0.0082; 0.0260]     41.9
Number of studies: k = 4; Number of observations: o = 2394; Number of events: e = 31
                     proportion           95%-CI
Random effects model     0.0134 [0.0094; 0.0189]
Quantifying heterogeneity (with 95%-CIs):
 tau^2 = 0 [0.0000; 1.3075]; tau = 0 [0.0000; 1.1435]
 I^2 = 0.0% [0.0%; 84.7%]; H = 1.00 [1.00; 2.56]
Test of heterogeneity:     Q d.f. p-value
                        1.81    3  0.6122
```

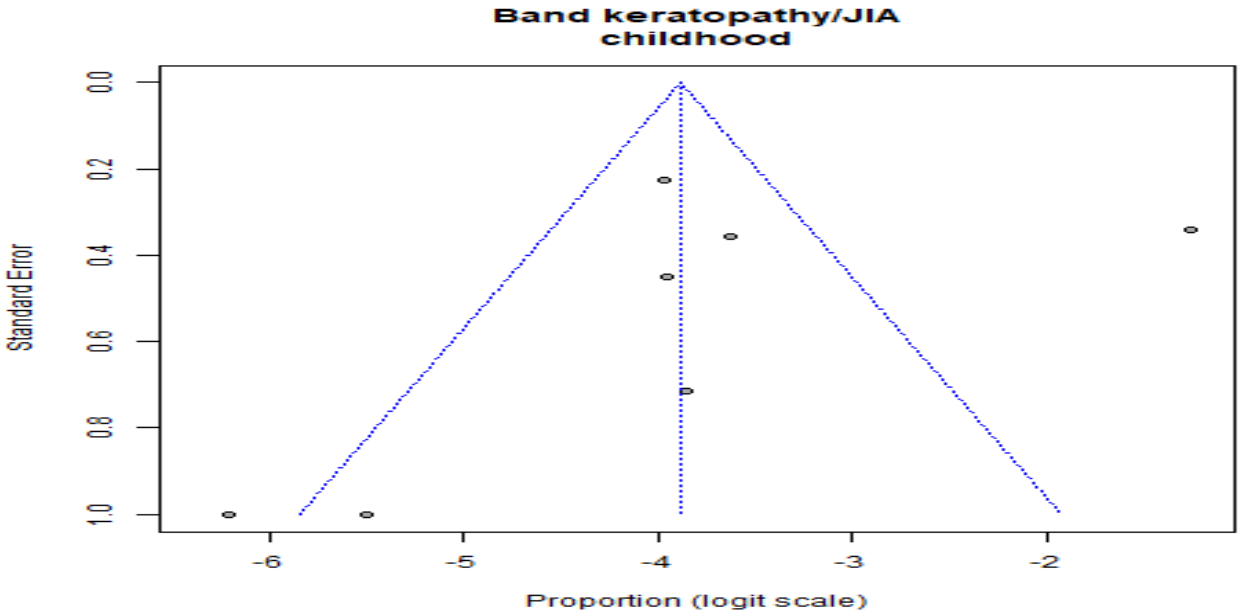
Supplementary Figure S4.5: Forest and Funnel Plots of studies reporting the prevalence of band keratopathy in children with Juvenile Idiopathic Arthritis

Band keratopathy

Forest



Funnel



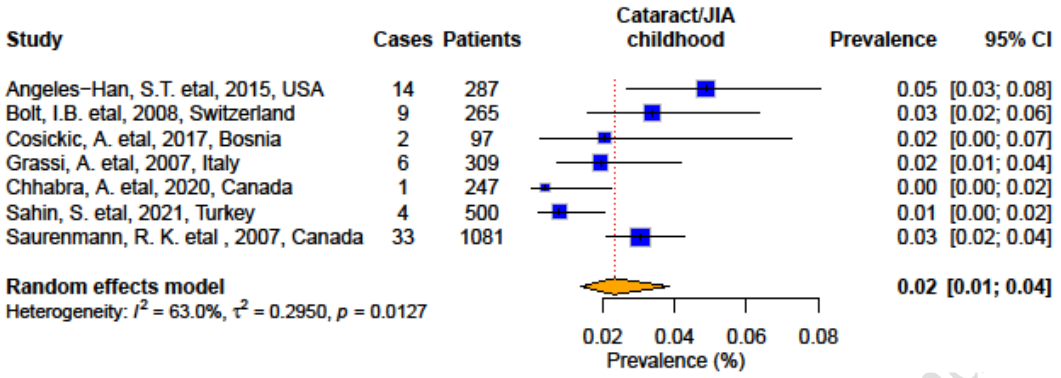
Summary statistics

```
                                    proportion           95%-CI       %W(random)
Angeles-Han S.T. etal, 2015, USA        0.2200 [0.1153; 0.3596]       16.1
Bolt I.B. etal, 2008, Switzerland       0.0189 [0.0062; 0.0435]       15.4
Grassi, A. etal, 2007, Italy            0.0259 [0.0112; 0.0504]       16.0
Cosickic, A. etal, 2017, Bosnia         0.0206 [0.0025; 0.0725]       13.5
Saurenmann, R. K. etal , 2007, Canada   0.0185 [0.0113; 0.0284]       16.6
Chhabra, A. etal, 2020, Canada          0.0040 [0.0001; 0.0223]       11.2
Sahin, S. etal, 2021, Turkey            0.0020 [0.0001; 0.0111]       11.2
Number of studies: k = 7; Number of observations: o = 2549; Number of events: e = 48
                    proportion           95%-CI
Random effects model     0.0201 [0.0066; 0.0593]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.9273 [0.6009; 11.4288]; tau = 1.3883 [0.7752; 3.3806]
 I^2 = 90.0% [81.9%; 94.5%]; H = 3.16 [2.35; 4.25]
Test of heterogeneity:      Q d.f.  p-value
                        | 59.88    6 < 0.0001
```

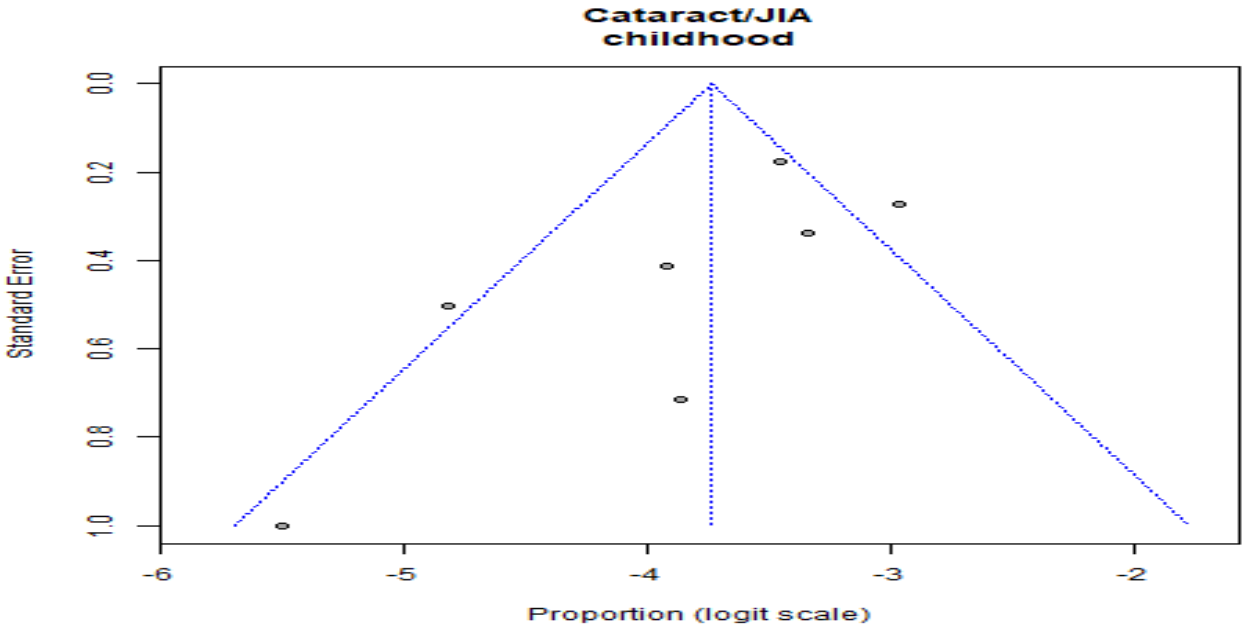
Supplementary Figure S4.6: Forest and Funnel Plots of studies reporting the prevalence of cataracts in children with Juvenile Idiopathic Arthritis

Cataract

Forest



Funnel



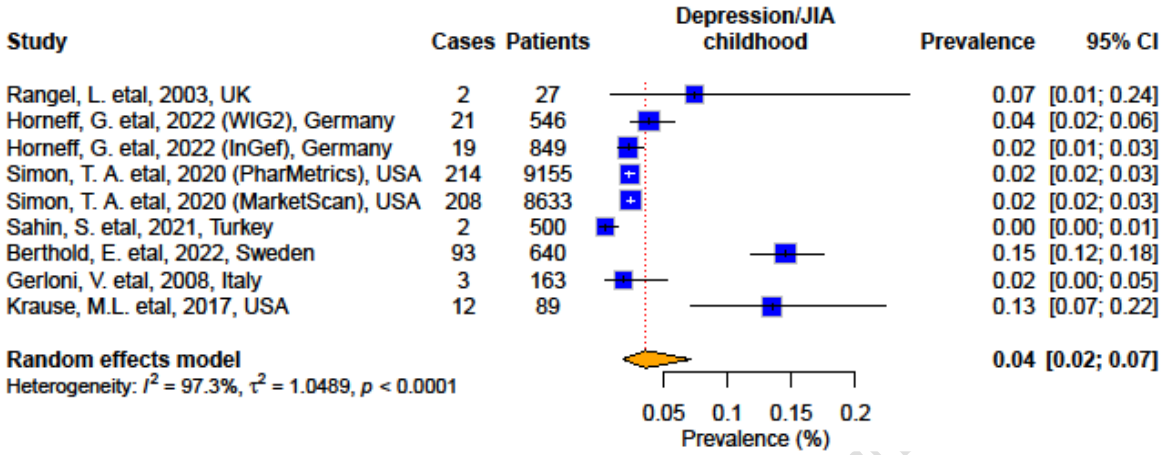
Summary statistics

```
                                  proportion           95%-CI       %W(random)
Angeles-Han S.T. etal, 2015, USA      0.0488 [0.0269; 0.0805]       19.0
Bolt I.B. etal, 2008, Switzerland     0.0340 [0.0156; 0.0635]       17.2
Cosickic, A. etal, 2017, Bosnia       0.0206 [0.0025; 0.0725]       8.7
Grassi, A. etal, 2007, Italy          0.0194 [0.0072; 0.0418]       15.2
Chhabra, A. etal, 2020, Canada        0.0040 [0.0001; 0.0223]       5.4
Sahin, S. etal, 2021, Turkey          0.0080 [0.0022; 0.0204]       12.9
Saurenmann, R. K. etal , 2007, Canada 0.0305 [0.0211; 0.0426]       21.6
Number of studies: k = 7; Number of observations: o = 2786; Number of events: e = 69
                    proportion           95%-CI
Random effects model     0.0232 [0.0139; 0.0385]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.2950 [0.0217; 3.3077]; tau = 0.5432 [0.1474; 1.8187]
 I^2 = 63.0% [16.0%; 83.7%]; H = 1.64 [1.09; 2.48]
Test of heterogeneity:      Q d.f. p-value
                        16.21    6  0.0127
```

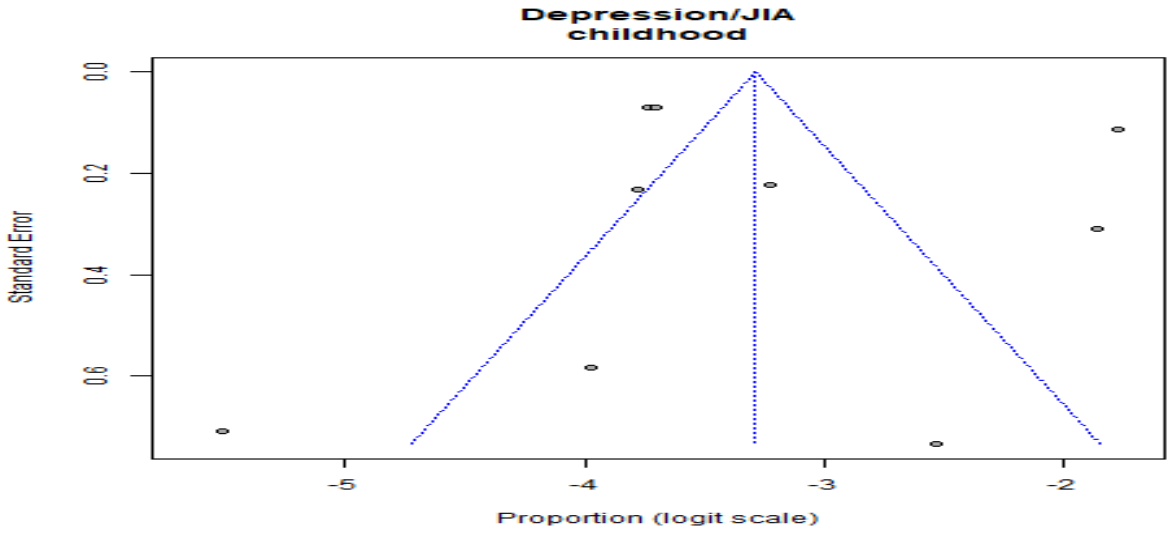
Supplementary Figure S4.7: Forest and Funnel Plots of studies reporting the prevalence of depression in children with Juvenile Idiopathic Arthritis

Depression

Forest



Funnel



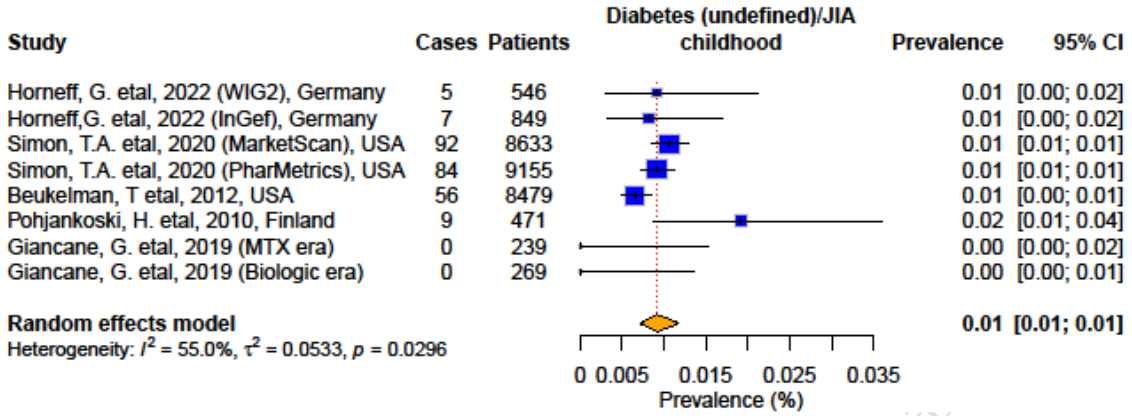
Summary statistics

```
                                          proportion           95%-CI   %W(random)
Rangel, L. etal, 2003, UK                     0.0741 [0.0091; 0.2429]  8.4
Horneff G. etal, 2022 (WIG2), Germany         0.0385 [0.0240; 0.0582]  12.1
Horneff G. etal, 2022 (InGef), Germany        0.0224 [0.0135; 0.0347]  12.1
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0234 [0.0204; 0.0267]  12.6
Simon, T. A. etal, 2020 (MarketScan), USA     0.0241 [0.0210; 0.0276]  12.6
Sahin, S. etal, 2021                          0.0040 [0.0005; 0.0144]  8.6
Berthold, E. etal, 2022                       0.1453 [0.1189; 0.1750]  12.5
Gerloni, V. etal, 2008, Italy                 0.0184 [0.0038; 0.0528]  9.6
Krause ML et al, 2017                         0.1348 [0.0717; 0.2237]  11.6
Number of studies: k = 9; Number of observations: o = 20602; Number of events: e = 574
                    proportion           95%-CI
Random effects model     0.0361 [0.0180; 0.0711]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.0489 [0.4100; 4.7484]; tau = 1.0242 [0.6403; 2.1791]
 I^2 = 97.3% [96.1%; 98.1%]; H = 6.04 [5.07; 7.18]
Test of heterogeneity:       Q d.f.  p-value
                      291.44    8 < 0.0001
```

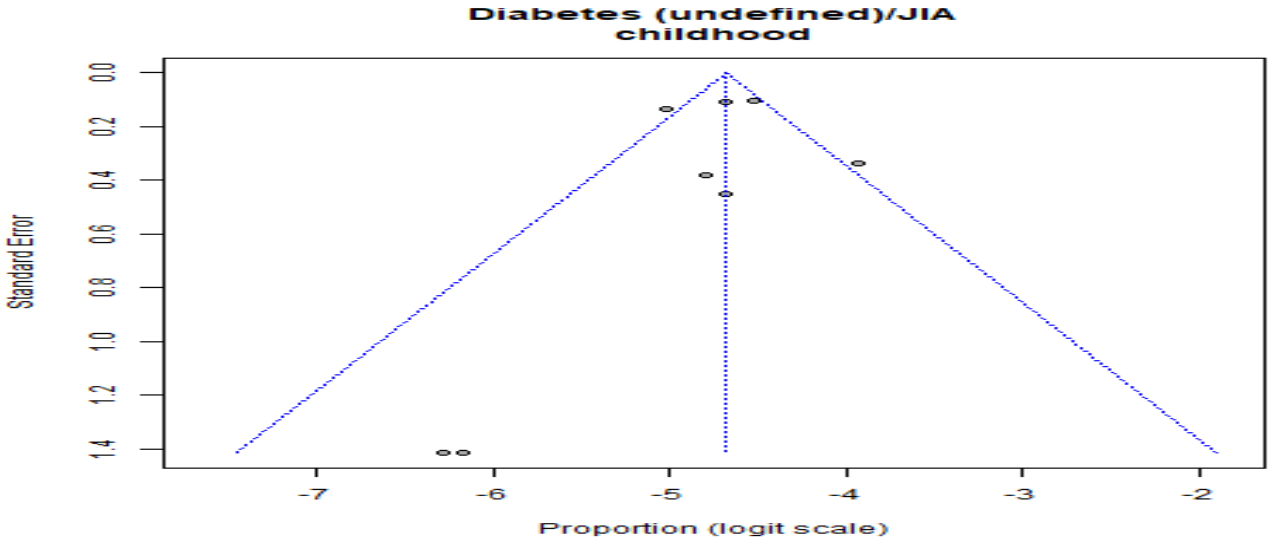
Supplementary Figure S4.8: Forest and Funnel Plots of studies reporting the prevalence of diabetes (undefined) in children with Juvenile Idiopathic Arthritis

Diabetes (undefined)

Forest



Funnel



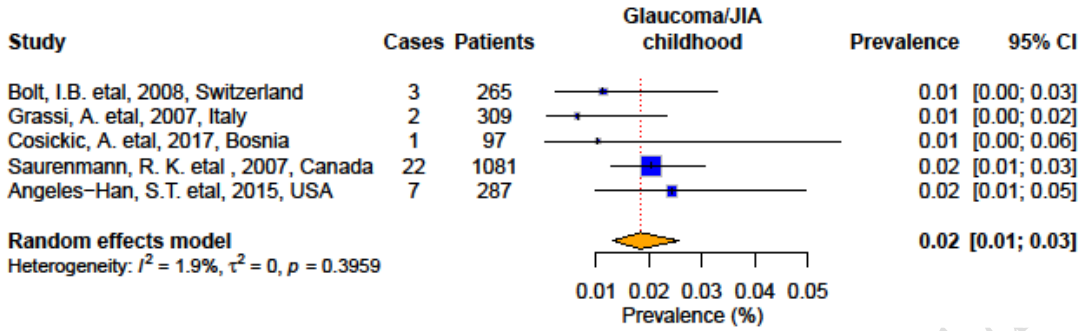
Summary statistics

```
                                        proportion           95%-CI    %W(random)
Horneff, G. etal, 2022 (WIG2), Germany      0.0092 [0.0030; 0.0212]    6.4
Horneff,G. etal, 2022 (InGef), Germany      0.0082 [0.0033; 0.0169]    8.3
Simon, TA. etal, 2020 (MarketScan), USA     0.0107 [0.0086; 0.0131]   25.6
Simon, TA. etal, 2020 (PharMetrics), USA    0.0092 [0.0073; 0.0113]   25.2
Beukelman, T etal, 2012, USA                0.0066 [0.0050; 0.0086]   23.1
Pohjankoski, H. etal, 2010, Finland         0.0191 [0.0088; 0.0360]    9.9
Giancane G etal, 2019 (MTX era)             0.0000 [0.0000; 0.0153]    0.8
Giancane G etal, 2019 (Biologic era)        0.0000 [0.0000; 0.0136]    0.8
Number of studies: k = 8; Number of observations: o = 28641; Number of events: e = 253
                    proportion           95%-CI
Random effects model     0.0092 [0.0072; 0.0118]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.0533 [0.0000; 1.5500]; tau = 0.2309 [0.0000; 1.2450]
 I^2 = 55.0% [0.4%; 79.6%]; H = 1.49 [1.00; 2.22]
Test of heterogeneity:      Q d.f. p-value
                        15.55    7  0.0296
```

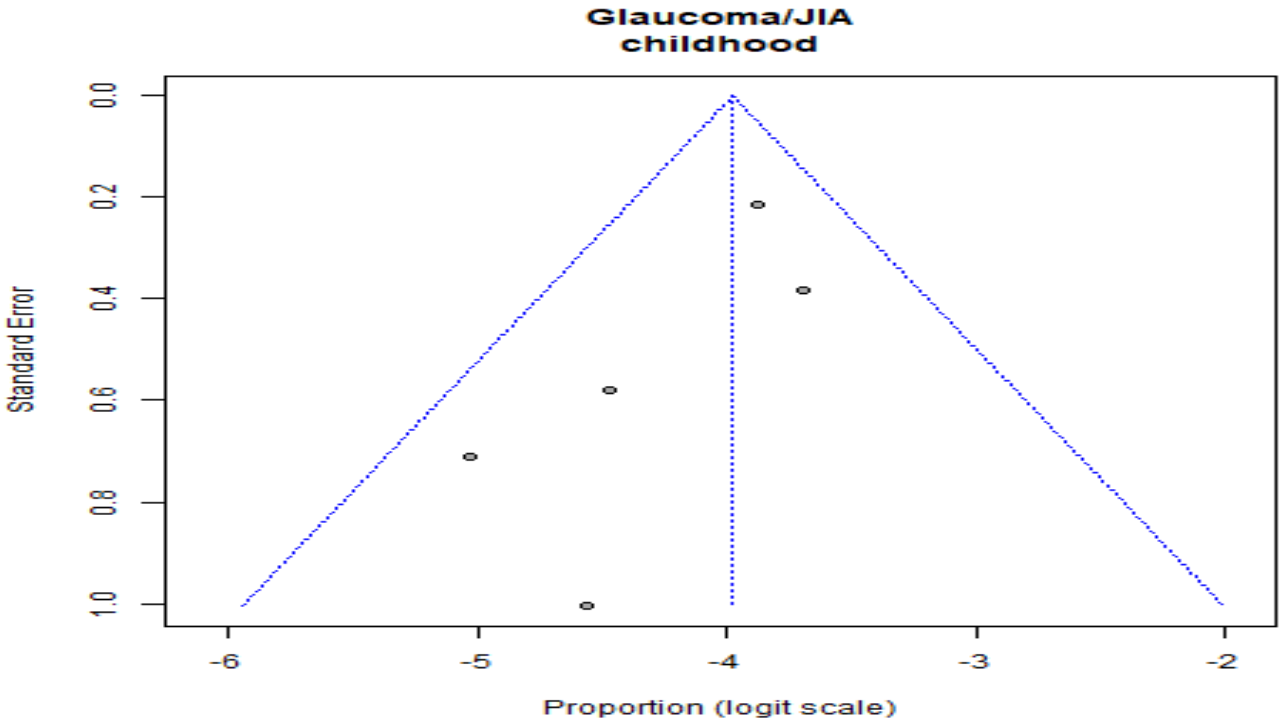
Supplementary Figure S4.9: Forest and Funnel Plots of studies reporting the prevalence of glaucoma in children with Juvenile Idiopathic Arthritis

Glaucoma

Forest



Funnel



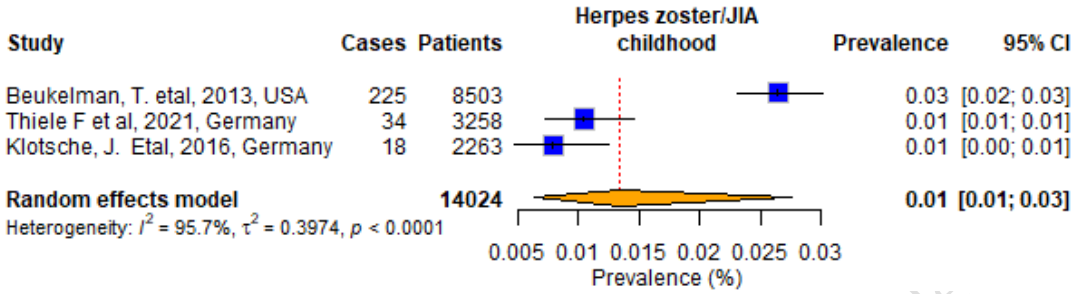
Summary statistics

```
                                    proportion           95%-CI       %W(random)
Bolt I.B. etal, 2008, Switzerland       0.0113 [0.0023; 0.0327]       8.6
Grassi, A. etal, 2007, Italy            0.0065 [0.0008; 0.0232]       5.8
Cosickic, A. etal, 2017, Bosnia         0.0103 [0.0003; 0.0561]       2.9
Saurenmann, R. K. etal , 2007, Canada   0.0204 [0.0128; 0.0307]       62.8
Angeles-Han ST et al, 2015, USA         0.0244 [0.0099; 0.0496]       19.9
Number of studies: k = 5; Number of observations: o = 2039; Number of events: e = 35
                     proportion           95%-CI
Random effects model     0.0184 [0.0133; 0.0256]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 2.1342]; tau = 0 [0.0000; 1.4609]
 I^2 = 1.9% [0.0%; 79.6%]; H = 1.01 [1.00; 2.21]
Test of heterogeneity:     Q d.f. p-value
                        4.08    4  0.3959
```

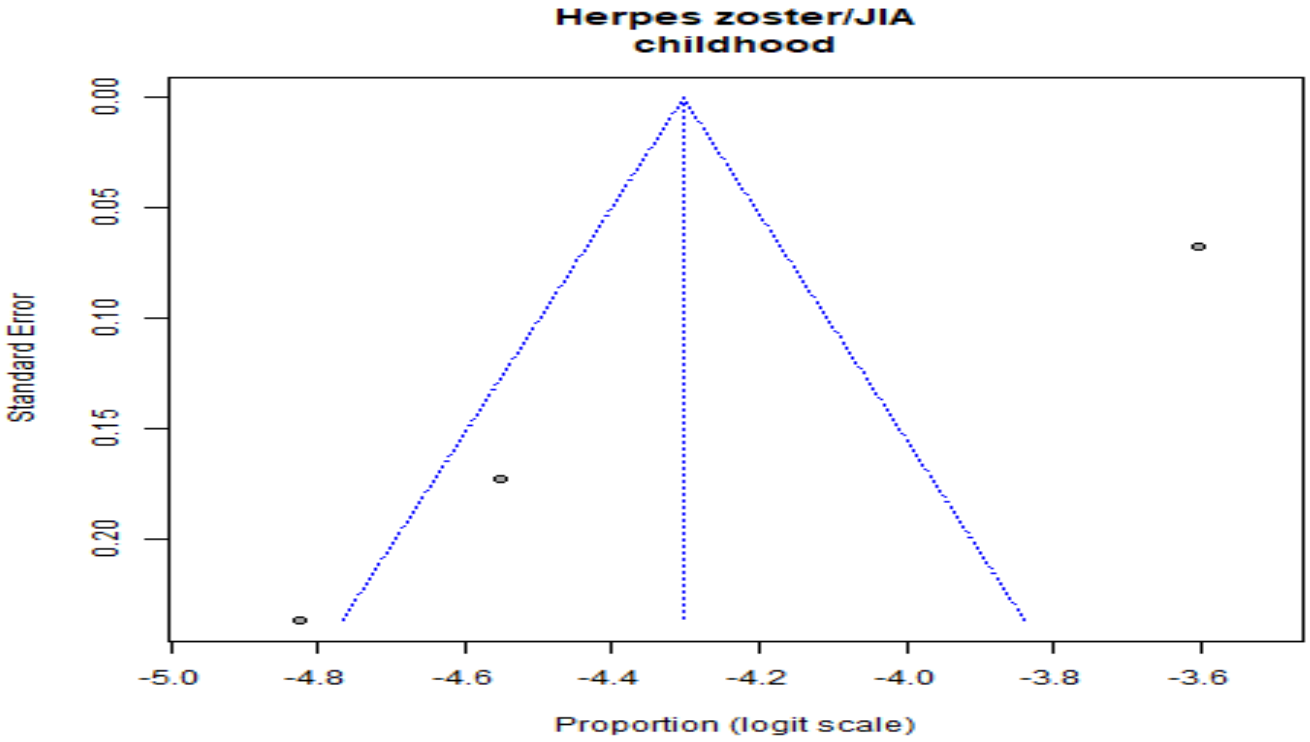
Supplementary Figure S4.10: Forest and Funnel Plots of studies reporting the prevalence of herpes zoster in children with Juvenile Idiopathic Arthritis

Herpes zoster

Forest



Funnel



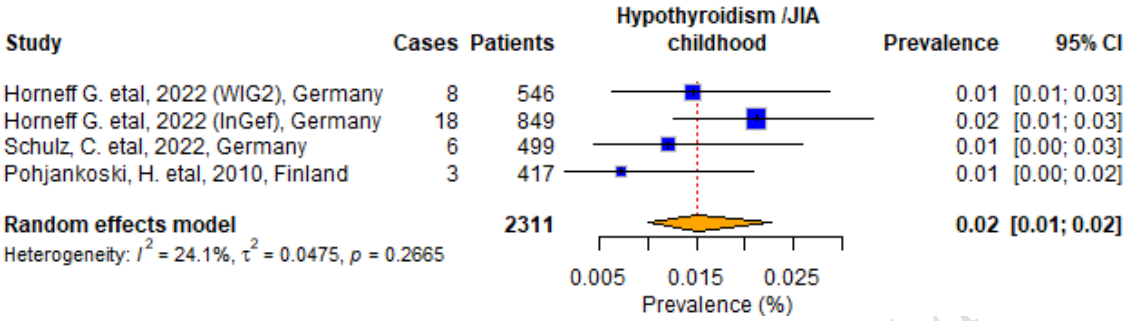
Summary statistics

```
                                proportion           95%-CI   %W(random)
Beukelman, T. etal, 2013, USA       0.0265 [0.0232; 0.0301]       35.4
Thiele F et al, 2021, Germany       0.0104 [0.0072; 0.0146]       33.3
Klotsche, J.  Etal, 2016, Germany   0.0080 [0.0047; 0.0125]       31.4
Number of studies: k = 3; Number of observations: o = 14024; Number of events: e = 277
                     proportion           95%-CI
Random effects model     0.0133 [0.0064; 0.0275]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.3974 [0.0907; 16.1850]; tau = 0.6304 [0.3012; 4.0231]
 I^2 = 95.7% [90.5%; 98.0%]; H = 4.80 [3.25; 7.10]
Test of heterogeneity:      Q d.f.  p-value
                        46.13    2 < 0.0001
```

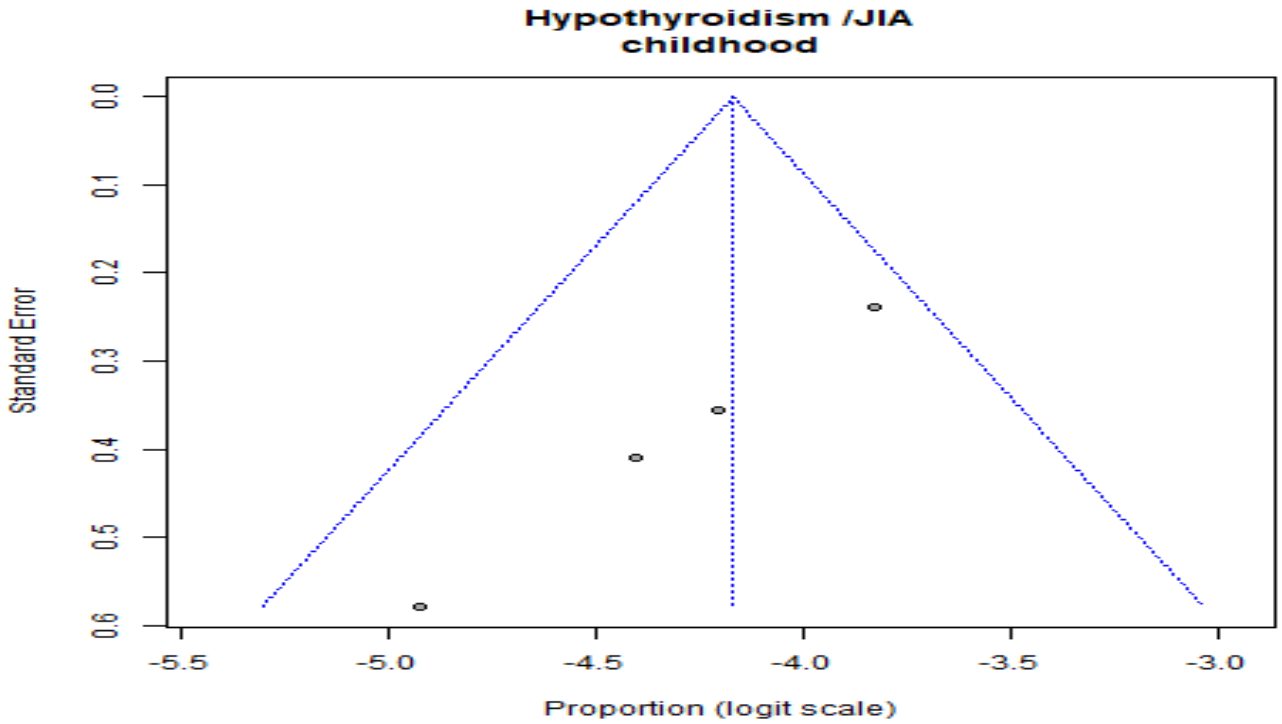
Supplementary Figure S4.11: Forest and Funnel Plots of studies reporting the prevalence of hypothyroidism in children with Juvenile Idiopathic Arthritis

Hypothyroidism

Forest



Funnel



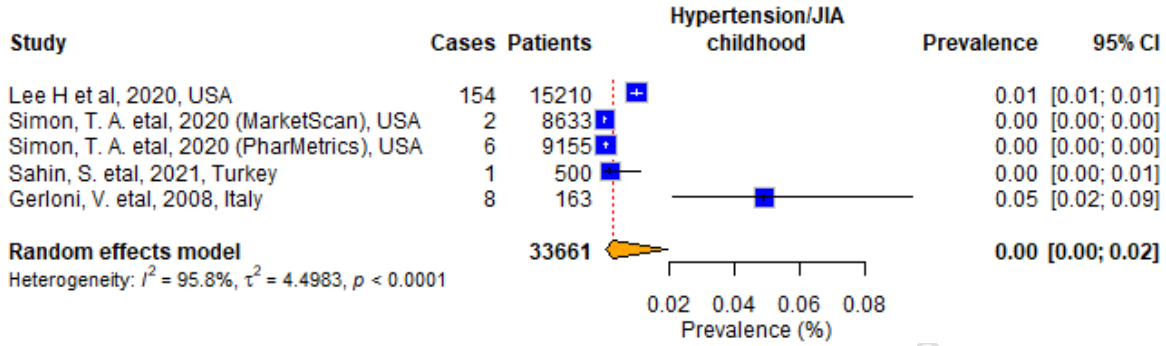
Summary statistics

```
                                        proportion           95%-CI      %W(random)
Horneff G. etal, 2022 (WIG2), Germany       0.0147 [0.0063; 0.0287]      25.4
Horneff G. etal, 2022 (InGef), Germany      0.0212 [0.0126; 0.0333]      42.5
Schulz, C. etal, 2022, Germany              0.0120 [0.0044; 0.0260]      20.5
Pohjankoski, H. etal, 2010, Finland         0.0072 [0.0015; 0.0209]      11.6
Number of studies: k = 4; Number of observations: o = 2311; Number of events: e = 35
                     proportion           95%-CI
Random effects model     0.0152 [0.0101; 0.0228]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.0475 [0.0000; 2.6861]; tau = 0.2179 [0.0000; 1.6389]
 I^2 = 24.1% [0.0%; 88.4%]; H = 1.15 [1.00; 2.93]
Test of heterogeneity:      Q d.f. p-value
                         | 3.95    3  0.2665
```

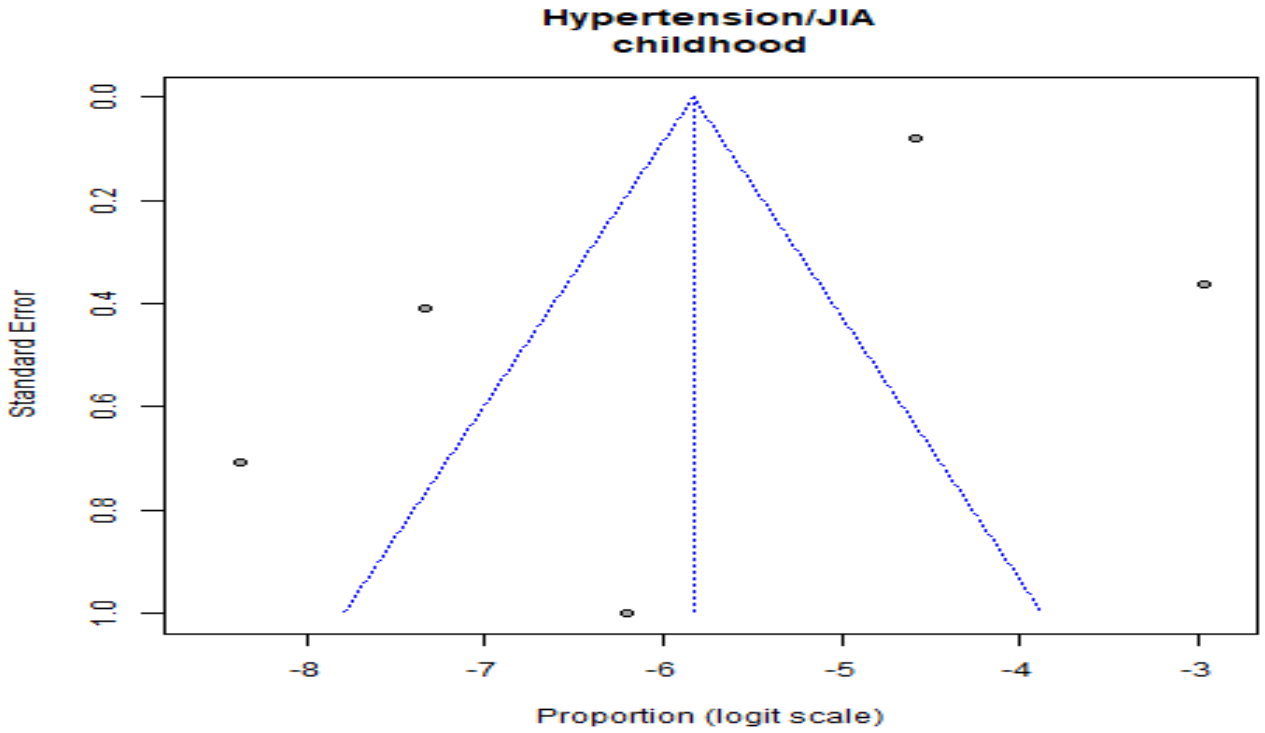
Supplementary Figure S4.12: Forest and Funnel Plots of studies reporting the prevalence of hypertension in children with Juvenile Idiopathic Arthritis

Hypertension

Forest



Funnel



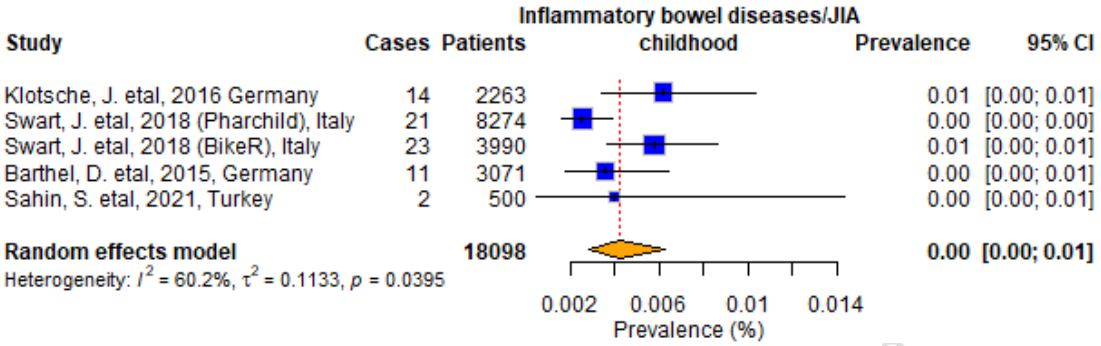
Summary statistics

```
                                          proportion           95%-CI  %W(random)
Lee H et al, 2020, USA                        0.0101 [0.0086; 0.0118]  21.5
Simon, T. A. etal, 2020 (MarketScan), USA     0.0002 [0.0000; 0.0008]  19.3
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0007 [0.0002; 0.0014]  20.7
Sahin, S. etal, 2021, Turkey                  0.0020 [0.0001; 0.0111]  17.6
Gerloni, V. etal, 2008, Italy                 0.0491 [0.0214; 0.0944]  20.9
Number of studies: k = 5; Number of observations: o = 33661; Number of events: e = 171
                    proportion          95%-CI
Random effects model     0.0029 [0.0004; 0.0197]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 4.4983 [1.4273; 38.1419]; tau = 2.1209 [1.1947; 6.1759]
 I^2 = 95.8% [92.7%; 97.6%]; H = 4.89 [3.70; 6.45]
Test of heterogeneity:      Q d.f.  p-value
                        95.46    4 < 0.0001
```

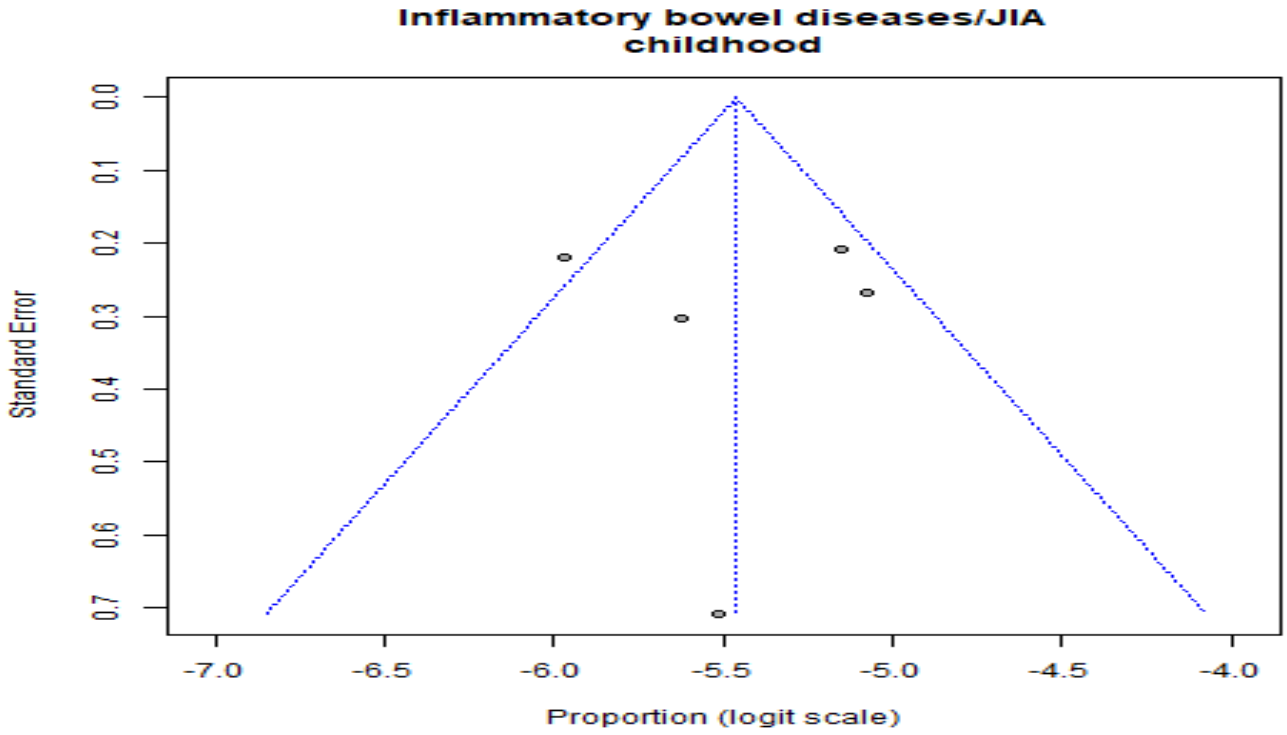
Supplementary Figure S4.13: Forest and Funnel Plots of studies reporting the prevalence of Inflammatory Bowel Diseases in children with Juvenile Idiopathic Arthritis

Inflammatory bowel diseases

Forest



Funnel



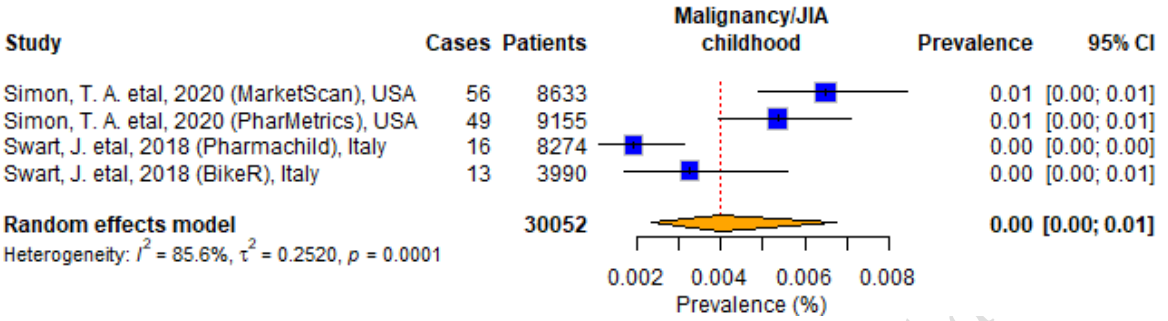
Summary statistics

```
                                        proportion           95%-CI     %W(random)
Klotsche, J. etal, 2016 Germany             0.0062 [0.0034; 0.0104]     22.0
Swart, J. etal, 2018 (Pharchild), Italy     0.0025 [0.0016; 0.0039]     25.4
Swart, J. etal, 2018 (BikeR), Italy         0.0058 [0.0037; 0.0086]     26.0
Barthel, D. etal, 2015, Germany             0.0036 [0.0018; 0.0064]     20.0
Sahin, S. etal, 2021, Turkey                0.0040 [0.0005; 0.0144]     6.6
Number of studies: k = 5; Number of observations: o = 18098; Number of events: e = 71
                     proportion           95%-CI
Random effects model     0.0042 [0.0028; 0.0063]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.1133 [0.0000; 1.0495]; tau = 0.3366 [0.0000; 1.0245]
 I^2 = 60.2% [0.0%; 85.1%]; H = 1.59 [1.00; 2.59]
Test of heterogeneity:       Q d.f. p-value
                         10.06    4  0.0395
```

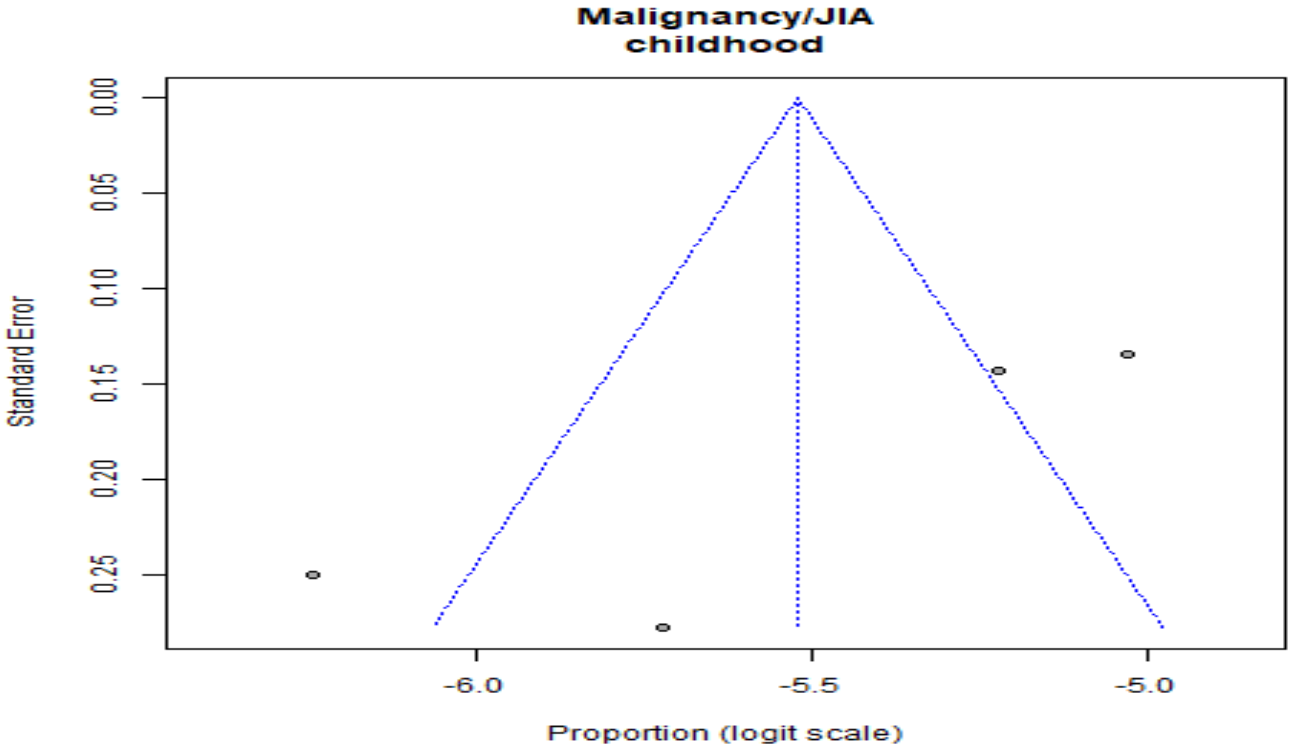
Supplementary Figure S4.14: Forest and Funnel Plots of studies reporting the prevalence of malignancy in children with Juvenile Idiopathic Arthritis

Malignancy

Forest



Funnel



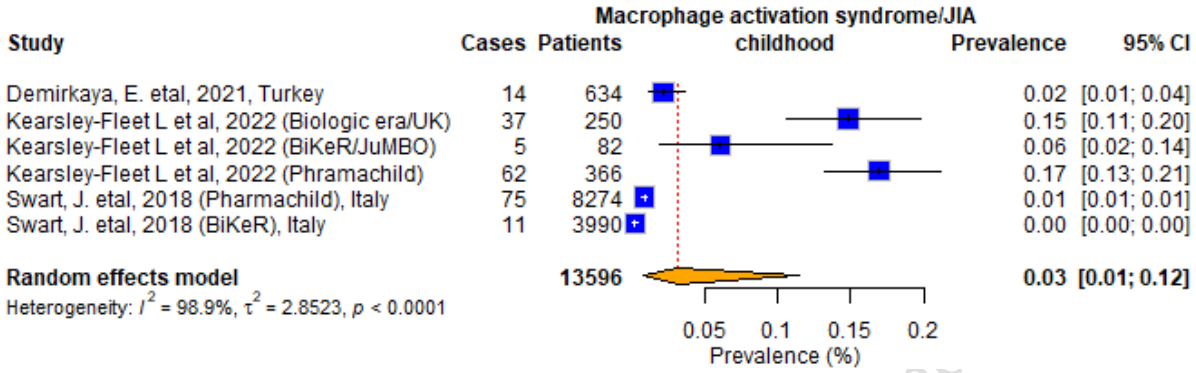
Summary statistics

```
                                        proportion           95%-CI  %W(random)
Simon, T. A. etal, 2020 (MarketScan), USA     0.0065 [0.0049; 0.0084]  27.3
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0054 [0.0040; 0.0071]  27.0
Swart, J. etal, 2018 (Pharmachild), Italy     0.0019 [0.0011; 0.0031]  23.4
Swart, J. etal, 2018 (BikeR), Italy           0.0033 [0.0017; 0.0056]  22.4
Number of studies: k = 4; Number of observations: o = 30052; Number of events: e = 134
                     proportion           95%-CI
Random effects model     0.0040 [0.0023; 0.0068]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.2520 [0.0517; 4.0772]; tau = 0.5020 [0.2273; 2.0192]
 I^2 = 85.6% [64.7%; 94.2%]; H = 2.64 [1.68; 4.14]
Test of heterogeneity:      Q d.f. p-value
                        20.88    3  0.0001
```

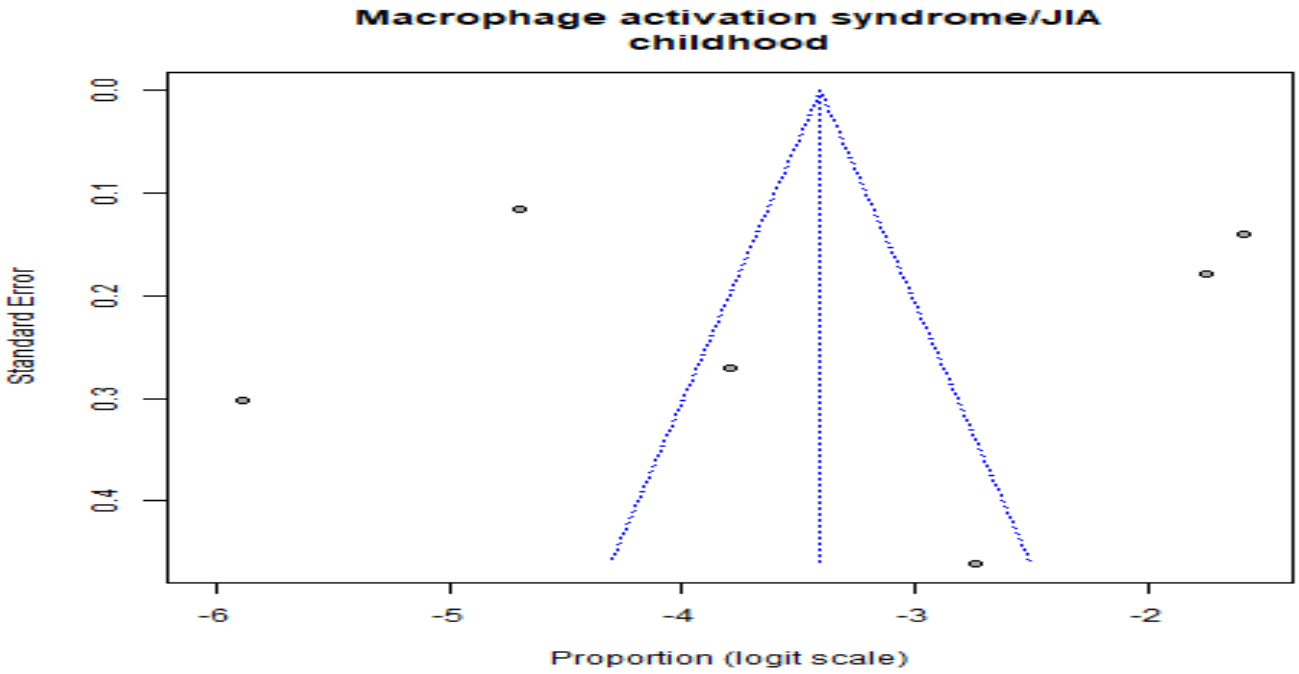
Supplementary Figure S4.15: Forest and Funnel Plots of studies reporting the prevalence of macrophage activation syndrome in children with Juvenile Idiopathic Arthritis

Macrophage activation syndrome

Forest



Funnel



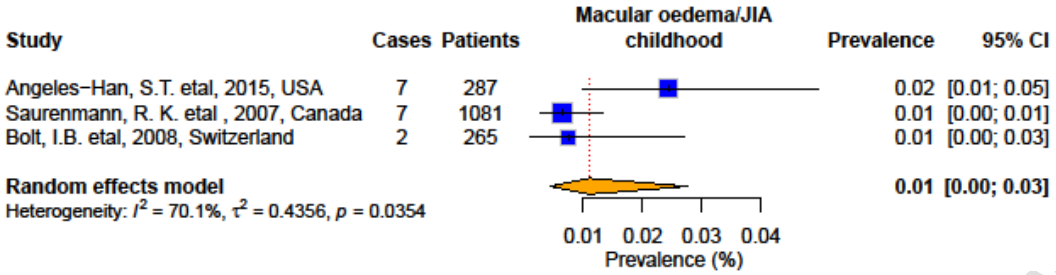
Summary statistics

```
                                          proportion           95%-CI %W(random)
Demirkaya, E. etal, 2021, Turkey              0.0221 [0.0121; 0.0368]       16.7
Kearsley-Fleet L et al, 2022 (Biologic era/UK)  0.1480 [0.1064; 0.1982]       16.9
Kearsley-Fleet L et al, 2022 (BiKeR/JuMBO)    0.0610 [0.0201; 0.1366]       15.9
Kearsley-Fleet L et al, 2022 (Phramachild)    0.1694 [0.1324; 0.2118]       17.0
Swart, J. etal, 2018 (Pharmachild), Italy     0.0091 [0.0071; 0.0113]       17.0
Swart, J. etal, 2018 (BiKeR), Italy           0.0028 [0.0014; 0.0049]       16.6
Number of studies: k = 6; Number of observations: o = 13596; Number of events: e = 204
                    proportion           95%-CI
Random effects model     0.0321 [0.0084; 0.1153]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 2.8523 [1.0712; 17.3534]; tau = 1.6889 [1.0350; 4.1657]
 I^2 = 98.9% [98.5%; 99.2%]; H = 9.48 [8.05; 11.16]
Test of heterogeneity:       Q d.f.  p-value
                      449.21    5 < 0.0001
```

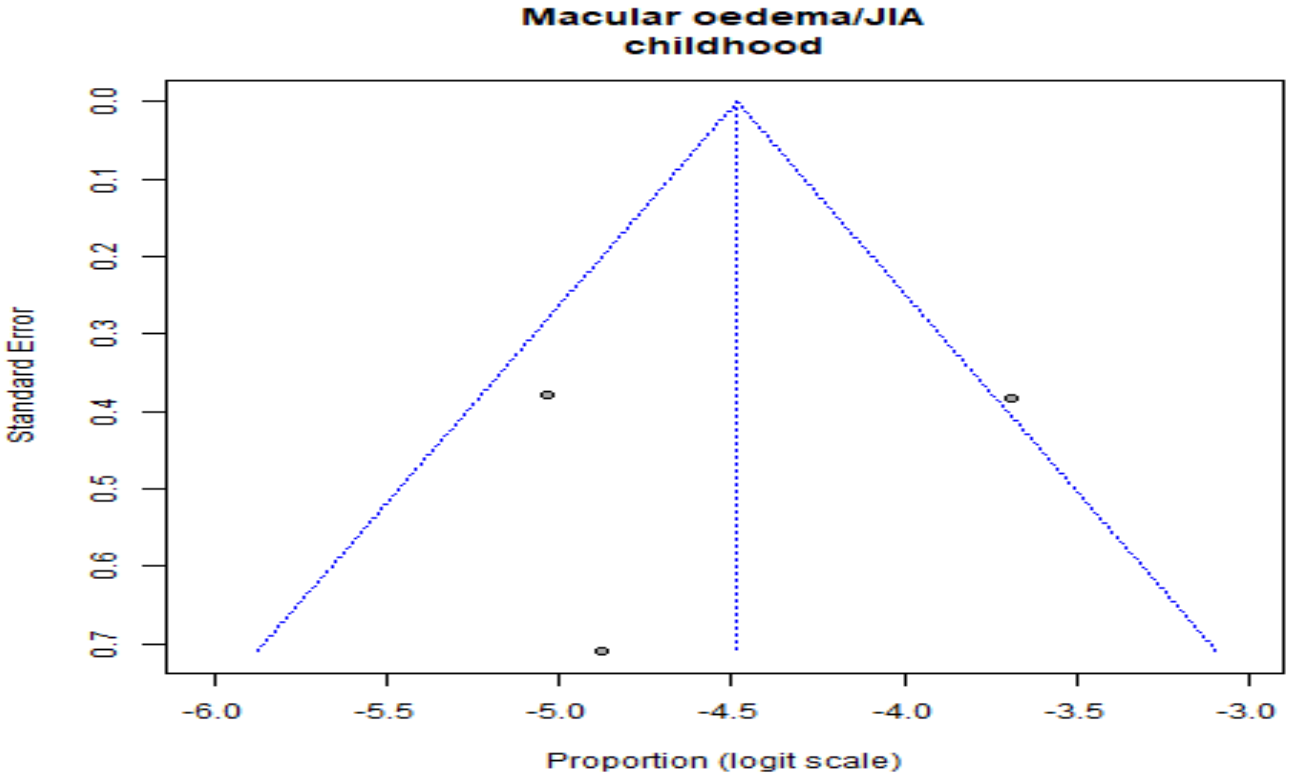
Supplementary Figure S4.16: Forest and Funnel Plots of studies reporting the prevalence of macular oedema in children with Juvenile Idiopathic Arthritis

Macular Oedema

Forest



Funnel



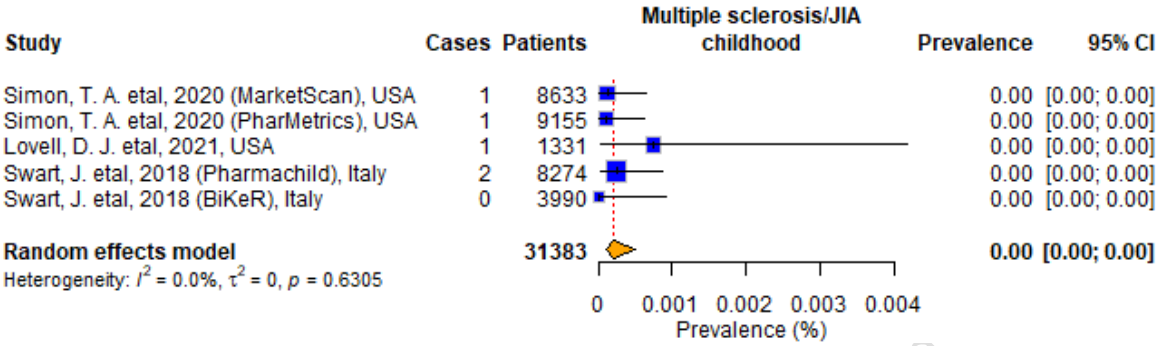
Summary statistics

```
                                    proportion           95%-CI %W(random)
Angeles-Han S.T. etal, 2015, USA        0.0244 [0.0099; 0.0496]       38.1
Saurenmann, R. K. etal , 2007, Canada   0.0065 [0.0026; 0.0133]       38.3
Bolt I.B. etal, 2008, Switzerland       0.0075 [0.0009; 0.0270]       23.6
Number of studies: k = 3; Number of observations: o = 1633; Number of events: e = 16
                     proportion           95%-CI
Random effects model     0.0112 [0.0045; 0.0276]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4356 [0.0000; 21.1933]; tau = 0.6600 [0.0000; 4.6036]
 I^2 = 70.1% [0.0%; 91.2%]; H = 1.83 [1.00; 3.38]
Test of heterogeneity:     Q d.f. p-value
                        6.68    2  0.0354
```

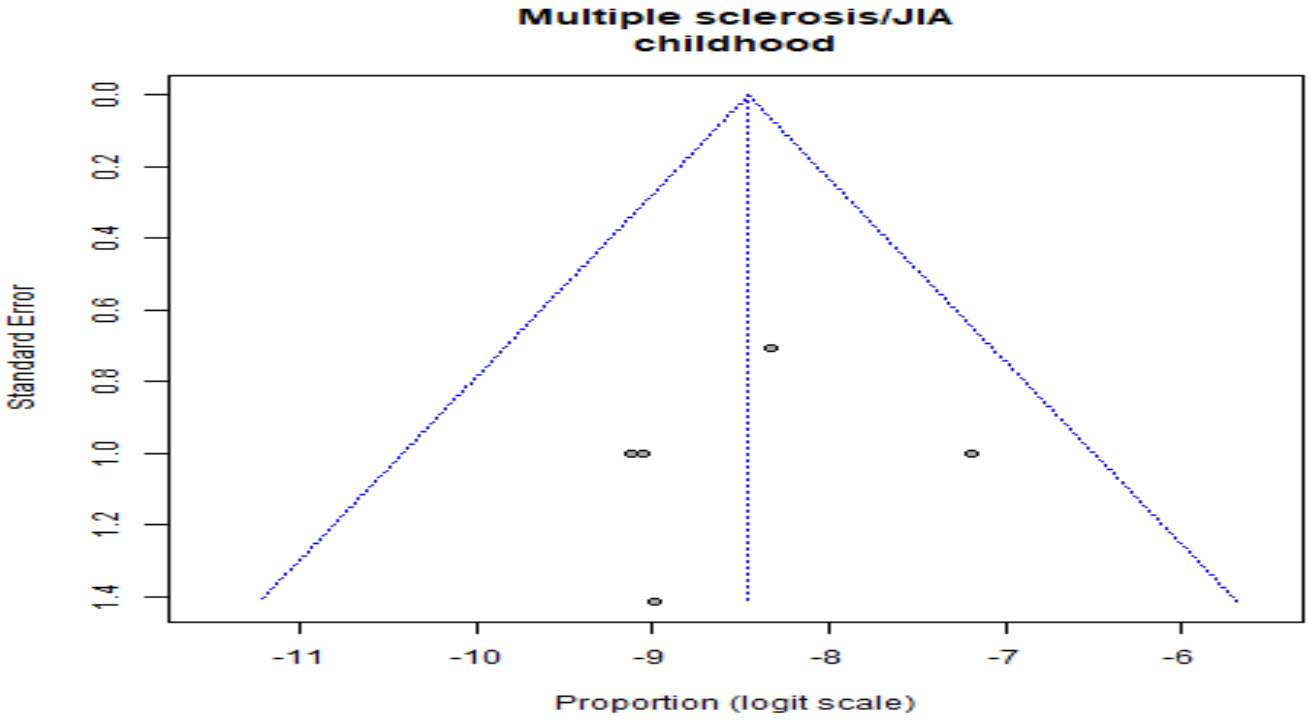
Supplementary Figure S4.17: Forest and Funnel Plots of studies reporting the prevalence of Multiple Sclerosis in children with Juvenile Idiopathic Arthritis

Multiple sclerosis

Forest



Funnel



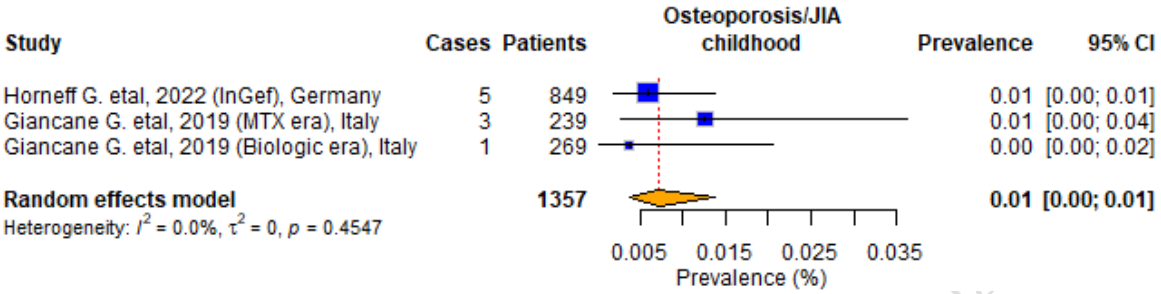
Summary statistics

```
                                          proportion           95%-CI %W(random)
Simon, T. A. etal, 2020 (MarketScan), USA     0.0001 [0.0000; 0.0006]       18.2
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0001 [0.0000; 0.0006]       18.2
Lovell, D. J. etal, 2021, USA                 0.0008 [0.0000; 0.0042]       18.2
Swart, J. etal, 2018 (Pharmachild), Italy     0.0002 [0.0000; 0.0009]       36.4
Swart, J. etal, 2018 (BiKeR), Italy           0.0000 [0.0000; 0.0009]        9.1
Number of studies: k = 5; Number of observations: o = 31383; Number of events: e = 5
                     proportion           95%-CI
Random effects model     0.0002 [0.0001; 0.0005]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 4.4533]; tau = 0 [0.0000; 2.1103]
 I^2 = 0.0% [0.0%; 79.2%]; H = 1.00 [1.00; 2.19]
Test of heterogeneity:     Q d.f. p-value
                        2.58    4  0.6305
```

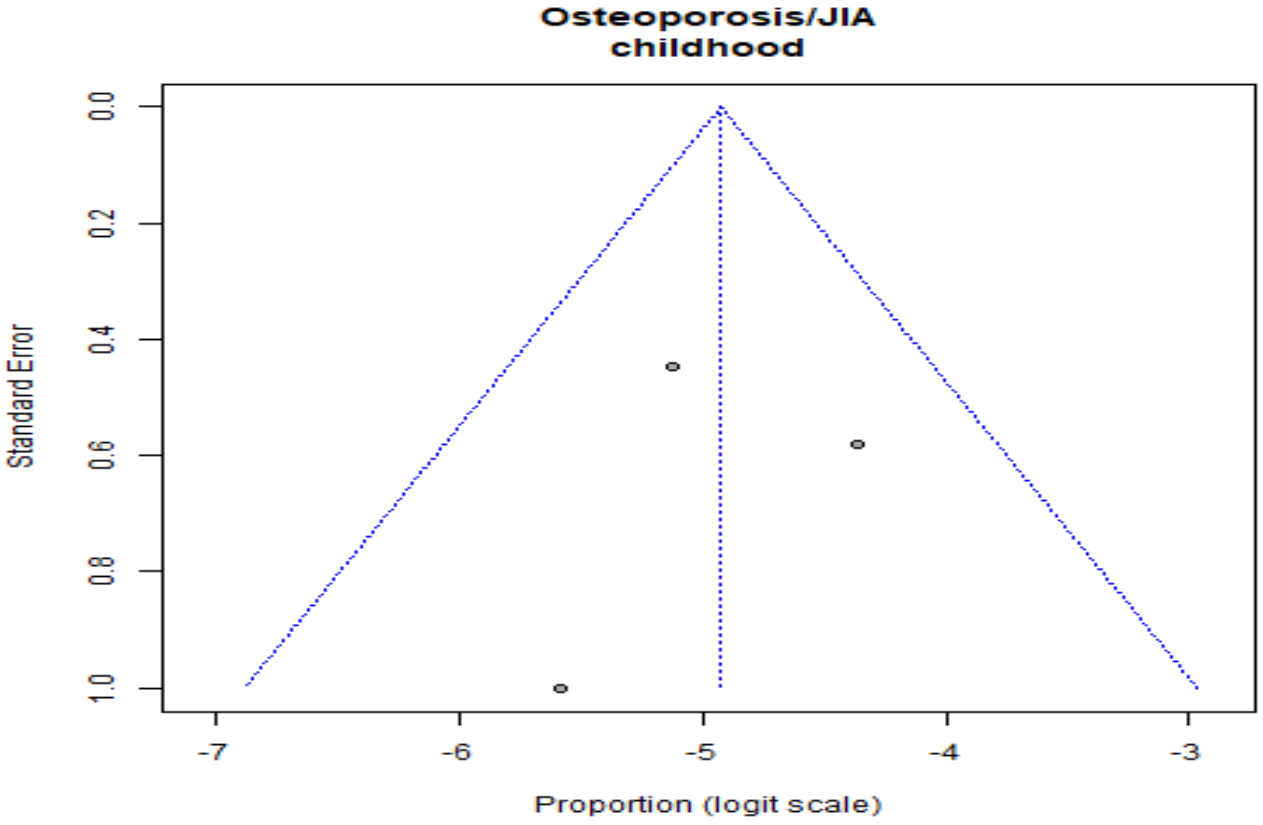
Supplementary Figure S4.18: Forest and Funnel Plots of studies reporting the prevalence of osteoporosis in children with Juvenile Idiopathic Arthritis

Osteoporosis

Forest



Funnel



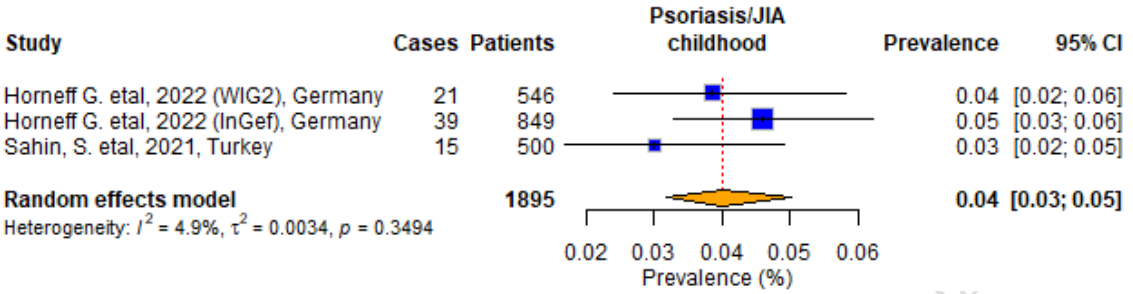
Summary statistics

```
                                          proportion           95%-CI %W(random)
Horneff G. etal, 2022 (InGef), Germany        0.0059 [0.0019; 0.0137]       55.7
Giancane G. etal, 2019 (MTX era), Italy       0.0126 [0.0026; 0.0362]       33.2
Giancane G. etal, 2019 (Biologic era), Italy  0.0037 [0.0001; 0.0205]       11.2
Number of studies: k = 3; Number of observations: o = 1357; Number of events: e = 9
                    proportion           95%-CI
Random effects model     0.0072 [0.0037; 0.0138]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 14.5276]; tau = 0 [0.0000; 3.8115]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:      Q d.f. p-value
                        | 1.58    2  0.4547
```

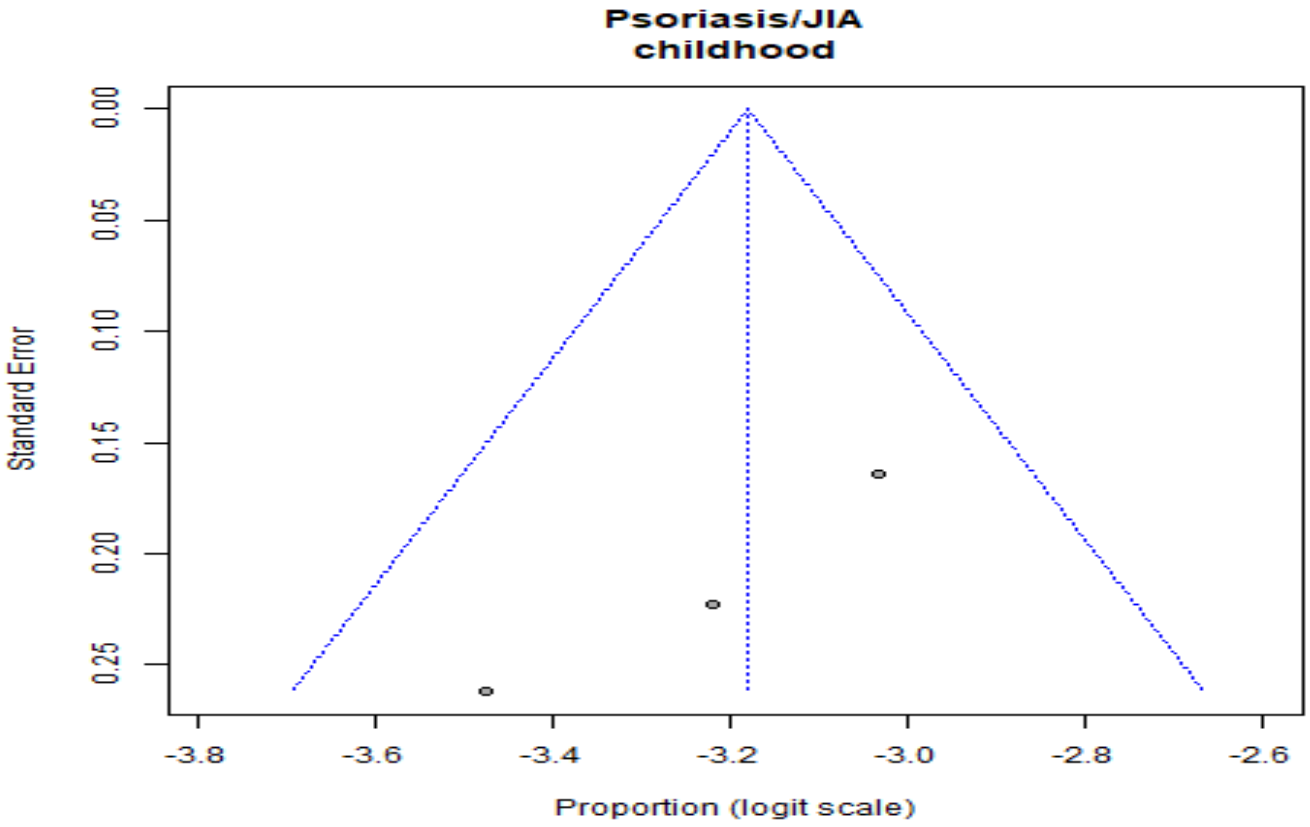
Supplementary Figure S4.19: Forest and Funnel Plots of studies reporting the prevalence of psoriasis in children with Juvenile Idiopathic Arthritis

Psoriasis

Forest



Funnel



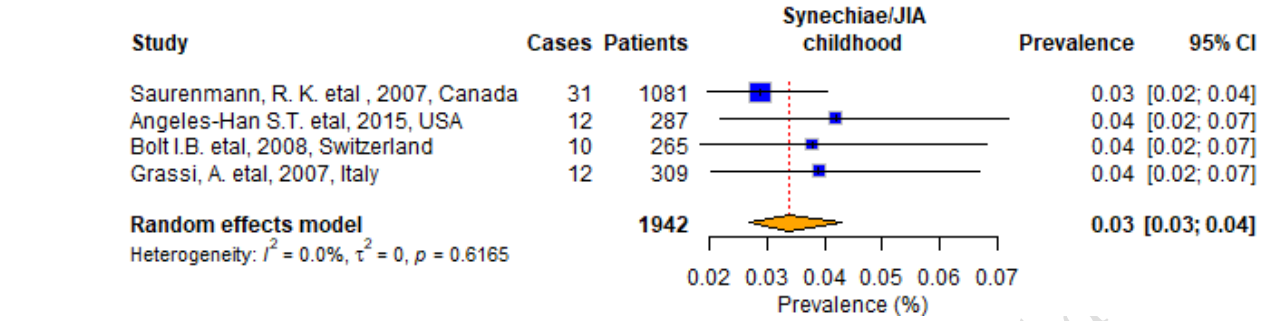
Summary statistics

```
                                    proportion           95%-CI %W(random)
Horneff G. etal, 2022 (WIG2), Germany     0.0385 [0.0240; 0.0582]       28.7
Horneff G. etal, 2022 (InGef), Germany    0.0459 [0.0329; 0.0623]       50.2
Sahin, S. etal, 2021, Turkey              0.0300 [0.0169; 0.0490]       21.1
Number of studies: k = 3; Number of observations: o = 1895; Number of events: e = 75
                      proportion           95%-CI
Random effects model     0.0399 [0.0316; 0.0503]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.0034 [0.0000; 1.9016]; tau = 0.0579 [0.0000; 1.3790]
 I^2 = 4.9% [0.0%; 90.1%]; H = 1.03 [1.00; 3.18]
Test of heterogeneity:     Q d.f. p-value
                        2.10    2  0.3494
```

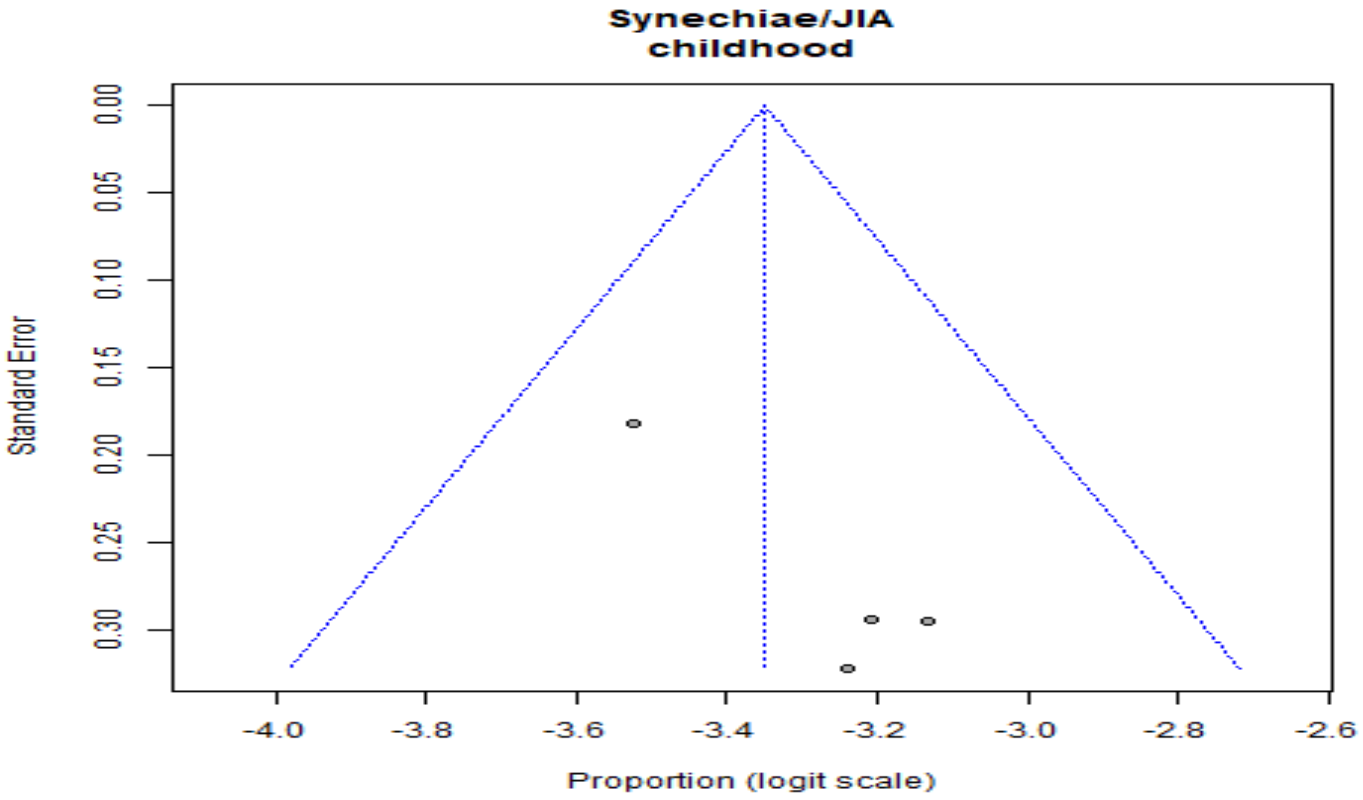
Supplementary Figure S4.20: Forest and Funnel Plots of studies reporting the prevalence of synechiae in children with Juvenile Idiopathic Arthritis

Synechiae

Forest



Funnel



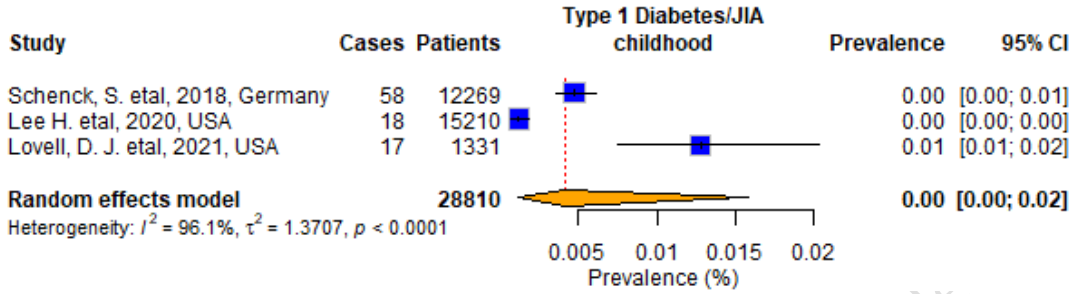
Summary statistics

```
                                    proportion           95%-CI %W(random)
Saurenmann, R. K. etal , 2007, Canada     0.0287 [0.0196; 0.0405]       48.0
Angeles-Han S.T. etal, 2015, USA          0.0418 [0.0218; 0.0719]       18.3
Bolt I.B. etal, 2008, Switzerland         0.0377 [0.0182; 0.0683]       15.3
Grassi, A. etal, 2007, Italy              0.0388 [0.0202; 0.0669]       18.4
Number of studies: k = 4; Number of observations: o = 1942; Number of events: e = 65
                     proportion           95%-CI
Random effects model     0.0339 [0.0267; 0.0430]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 0.3558]; tau = 0 [0.0000; 0.5965]
 I^2 = 0.0% [0.0%; 84.7%]; H = 1.00 [1.00; 2.56]
Test of heterogeneity:     Q d.f. p-value
                        1.79    3  0.6165
```

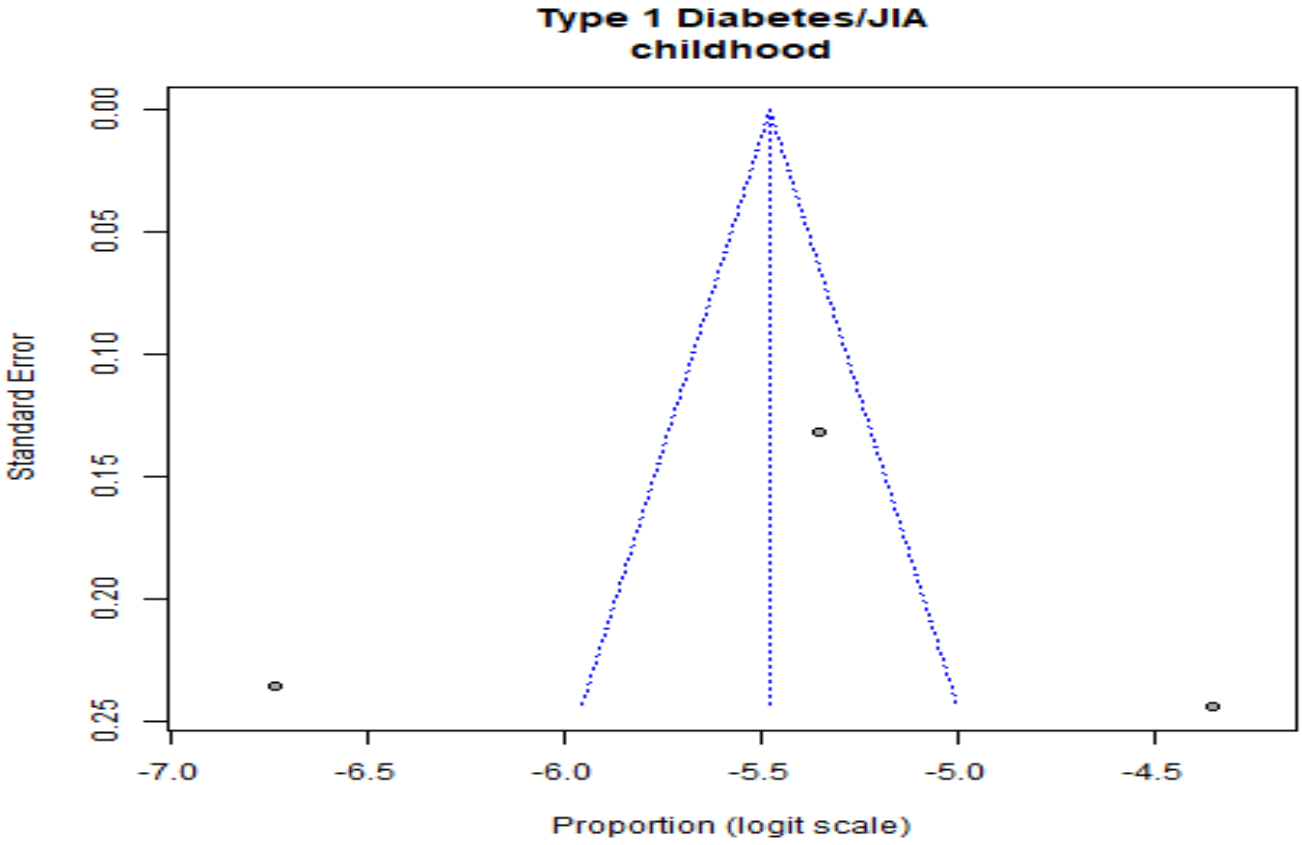
Supplementary Figure S4.21: Forest and Funnel Plots of studies reporting the prevalence of Type 1 diabetes in children with Juvenile Idiopathic Arthritis

Type 1 Diabetes

Forest



Funnel



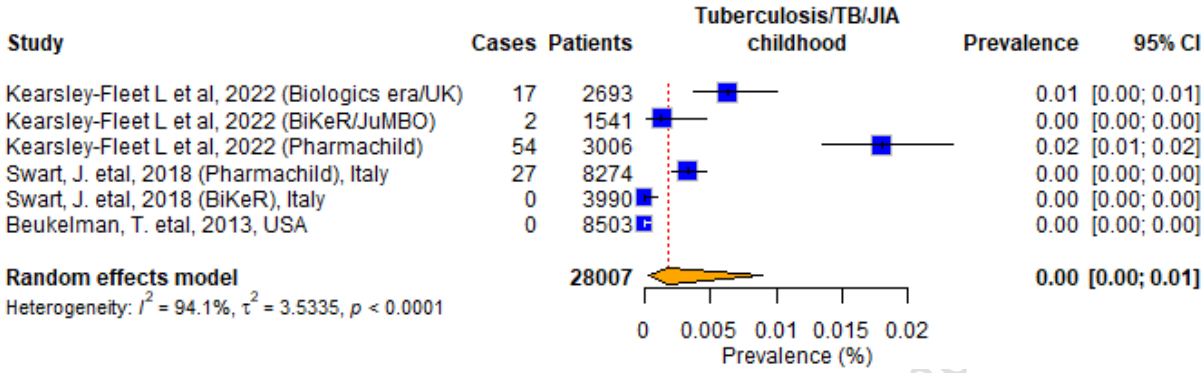
Summary statistics

```
                                proportion           95%-CI %W(random)
Schenck, S. etal, 2018, Germany     0.0047 [0.0036; 0.0061]       34.0
Lee H. etal, 2020, USA              0.0012 [0.0007; 0.0019]       33.1
Lovell, D. J. etal, 2021, USA       0.0128 [0.0075; 0.0204]       33.0
Number of studies: k = 3; Number of observations: o = 28810; Number of events: e = 93
                     proportion           95%-CI
Random effects model     0.0042 [0.0011; 0.0158]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.3707 [0.3335; 56.8637]; tau = 1.1708 [0.5775; 7.5408]
 I^2 = 96.1% [91.6%; 98.2%]; H = 5.04 [3.45; 7.37]
Test of heterogeneity:      Q d.f.  p-value
                        50.80    2 < 0.0001
```

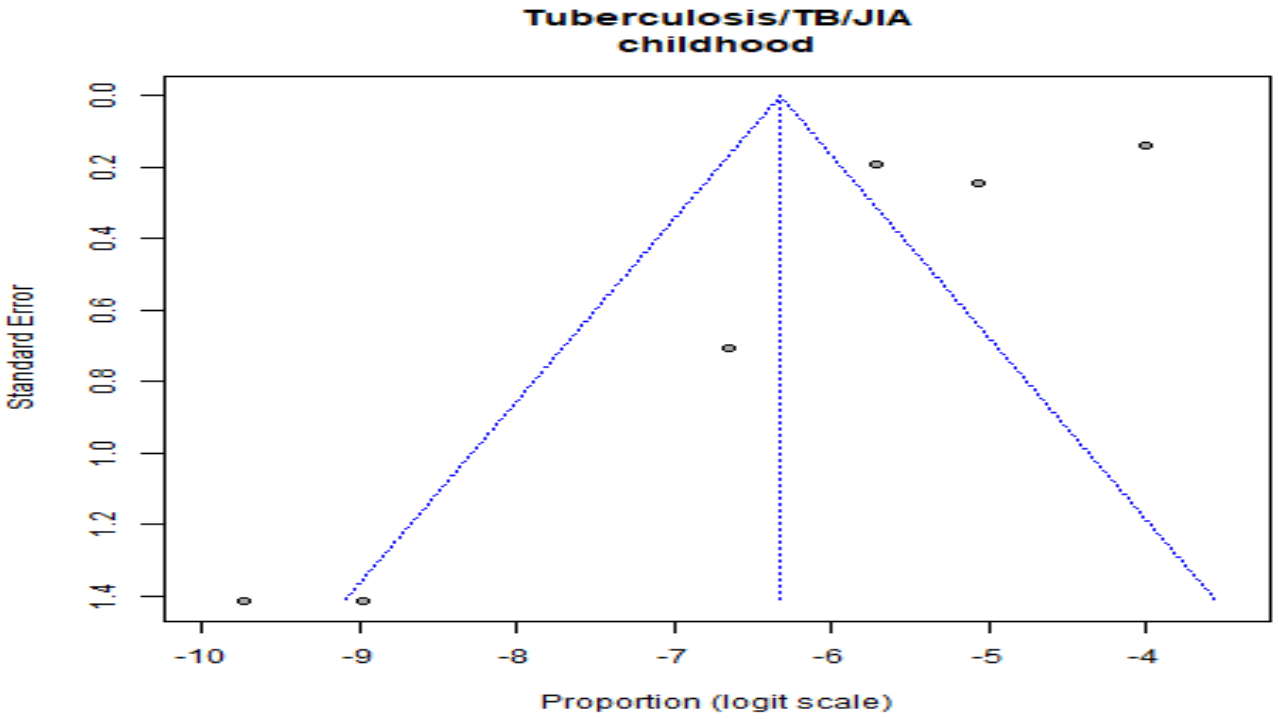
Supplementary Figure S4.22: Forest and Funnel Plots of studies reporting the prevalence of tuberculosis in children with Juvenile Idiopathic Arthritis

Tuberculosis/TB

Forest



Funnel



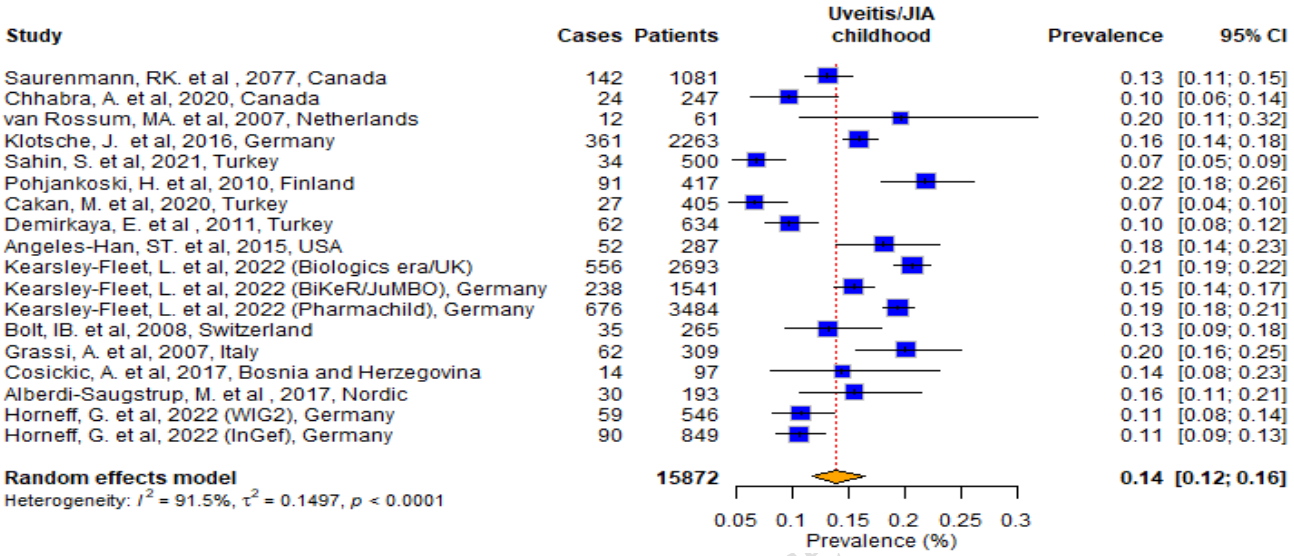
Summary statistics

```
                                                proportion           95%-CI %W(random)
Kearsley-Fleet L et al, 2022 (Biologics era/UK)     0.0063 [0.0037; 0.0101]       19.2
Kearsley-Fleet L et al, 2022 (BiKeR/JuMBO)          0.0013 [0.0002; 0.0047]       17.1
Kearsley-Fleet L et al, 2022 (Pharmachild)          0.0180 [0.0135; 0.0234]       19.4
Swart, J. etal, 2018 (Pharmachild), Italy           0.0033 [0.0022; 0.0047]       19.3
Swart, J. etal, 2018 (BiKeR), Italy                 0.0000 [0.0000; 0.0009]       12.5
Beukelman, T. etal, 2013, USA                       0.0000 [0.0000; 0.0004]       12.5
Number of studies: k = 6; Number of observations: o = 28007; Number of events: e = 100
                     proportion           95%-CI
Random effects model     0.0018 [0.0004; 0.0090]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 3.5335 [0.9812; 29.4146]; tau = 1.8797 [0.9905; 5.4235]
 I^2 = 94.1% [89.8%; 96.6%]; H = 4.13 [3.13; 5.45]
Test of heterogeneity:      Q d.f.  p-value
                        85.36    5 < 0.0001
```

Supplementary Figure S4.23: Forest and Funnel Plots of studies reporting the prevalence of uveitis in children with Juvenile Idiopathic Arthritis

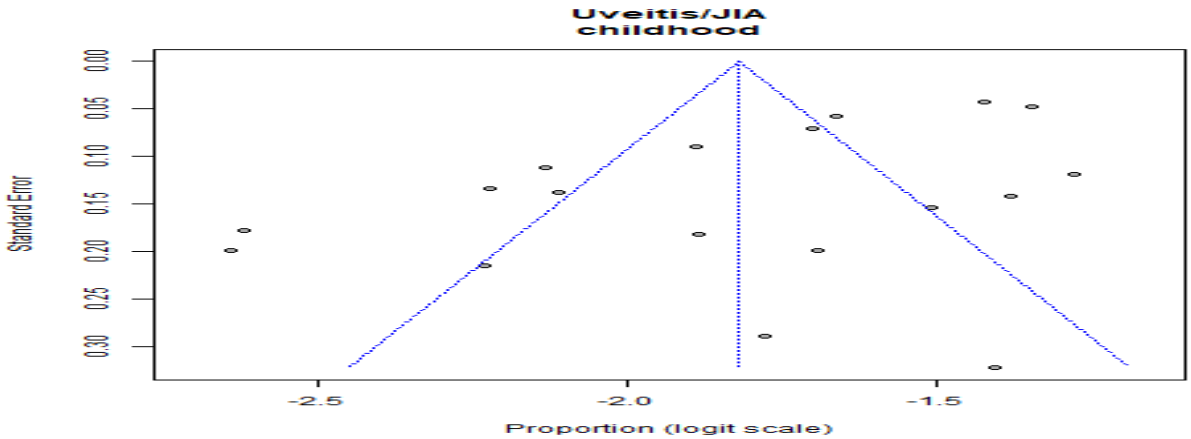
Uveitis

Forest



Funnel

Funnel



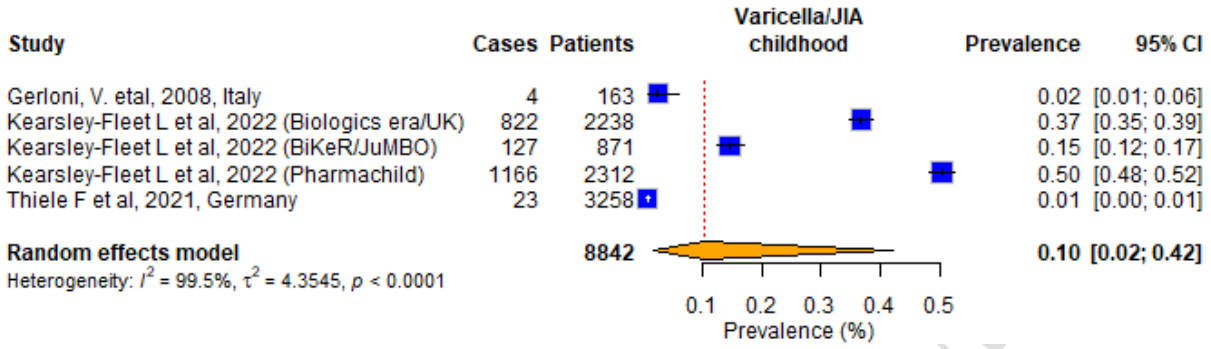
Summary statistics

```
                                                    proportion           95%-CI %W(random)
Saurenmann, RK. et al , 2077, Canada                    0.1314 [0.1118; 0.1530]        6.1
Chhabra, A. et al, 2020, Canada                         0.0972 [0.0633; 0.1411]        4.9
van Rossum, MA. et al, 2007, Netherlands                0.1967 [0.1060; 0.3184]        3.8
Klotsche, J.  et al, 2016, Germany                      0.1595 [0.1447; 0.1753]        6.3
Sahin, S. et al, 2021, Turkey                           0.0680 [0.0475; 0.0937]        5.3
Pohjankoski, H. et al, 2010, Finland                    0.2182 [0.1795; 0.2610]        5.9
Cakan, M. et al, 2020, Turkey                           0.0667 [0.0444; 0.0955]        5.1
Demirkaya, E. et al , 2011, Turkey                      0.0978 [0.0758; 0.1236]        5.8
Angeles-Han, ST. et al, 2015, USA                       0.1812 [0.1384; 0.2307]        5.6
Kearsley-Fleet, L. et al, 2022 (Biologics era/UK)       0.2065 [0.1913; 0.2222]        6.4
Kearsley-Fleet, L. et al, 2022 (BiKeR/JuMBO), Germany   0.1544 [0.1367; 0.1735]        6.3
Kearsley-Fleet, L. et al, 2022 (Pharmachild), Germany   0.1940 [0.1810; 0.2076]        6.4
Bolt, IB. et al, 2008, Switzerland                      0.1321 [0.0937; 0.1789]        5.3
Grassi, A. et al, 2007, Italy                           0.2006 [0.1574; 0.2497]        5.7
Cosickic, A. et al, 2017, Bosnia and Herzegovina        0.1443 [0.0812; 0.2303]        4.2
Alberdi-Saugstrup, M. et al , 2017, Nordic              0.1554 [0.1074; 0.2144]        5.1
Horneff, G. et al, 2022 (WIG2), Germany                 0.1081 [0.0833; 0.1372]        5.7
Horneff, G. et al, 2022 (InGef), Germany                0.1060 [0.0861; 0.1287]        6.0
Number of studies: k = 18; Number of observations: o = 15872; Number of events: e = 2565
                         proportion           95%-CI
Random effects model         0.1395 [0.1179; 0.1643]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.1497 [0.0741; 0.3706]; tau = 0.3870 [0.2721; 0.6088]
 I^2 = 91.5% [88.0%; 93.9%]; H = 3.42 [2.89; 4.05]
Test of heterogeneity:       Q d.f.  p-value
                      | 198.93    17 < 0.0001
```

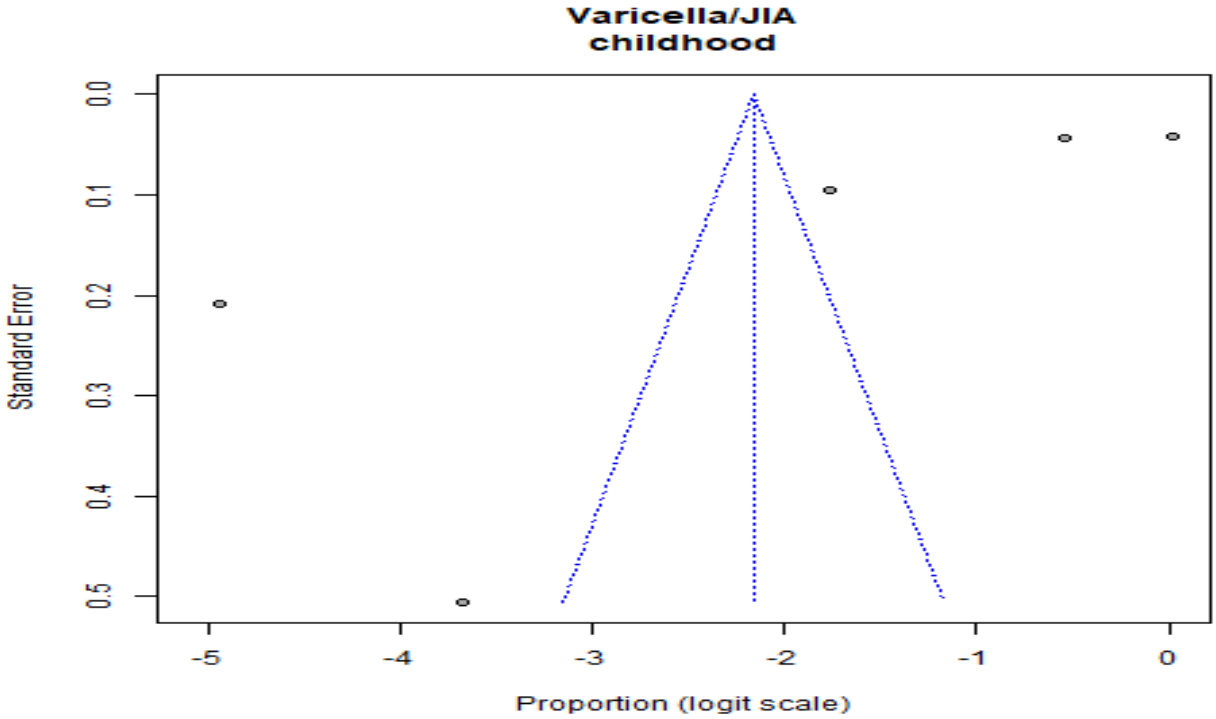
Supplementary Figure S4.24: Forest and Funnel Plots of studies reporting the prevalence of varicella in children with Juvenile Idiopathic Arthritis

Varicella

Forest



Funnel



Summary statistics

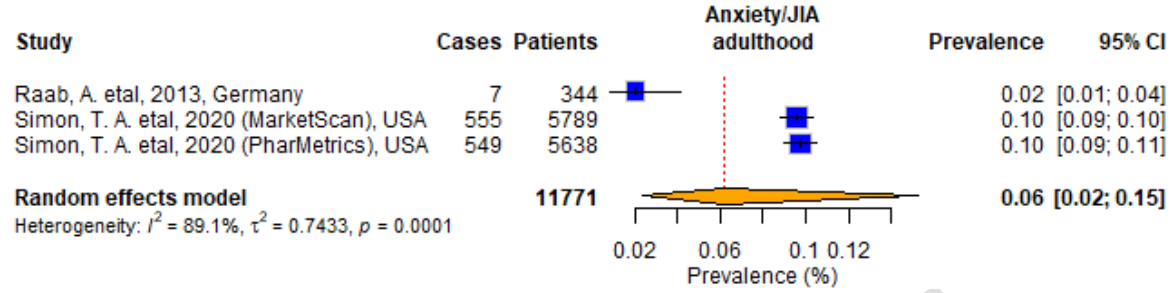
```
                                          proportion           95%-CI %W(random)
Gerloni, V. etal, 2008, Italy                 0.0245 [0.0067; 0.0616]       19.2
Kearsley-Fleet L et al, 2022 (Biologics era/UK)  0.3673 [0.3473; 0.3877]       20.3
Kearsley-Fleet L et al, 2022 (BiKeR/JuMBO)    0.1458 [0.1230; 0.1710]       20.2
Kearsley-Fleet L et al, 2022 (Pharmachild)    0.5043 [0.4837; 0.5249]       20.3
Thiele F et al, 2021, Germany                 0.0071 [0.0045; 0.0106]       20.1
Number of studies: k = 5; Number of observations: o = 8842; Number of events: e = 2142
                     proportion           95%-CI
Random effects model     0.1031 [0.0179; 0.4204]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 4.3545 [1.5277; 36.2489]; tau = 2.0868 [1.2360; 6.0207]
 I^2 = 99.5% [99.4%; 99.6%]; H = 14.32 [12.48; 16.44]
Test of heterogeneity:       Q d.f.  p-value
                      820.56    4 < 0.0001
```

Adulthood/JIA

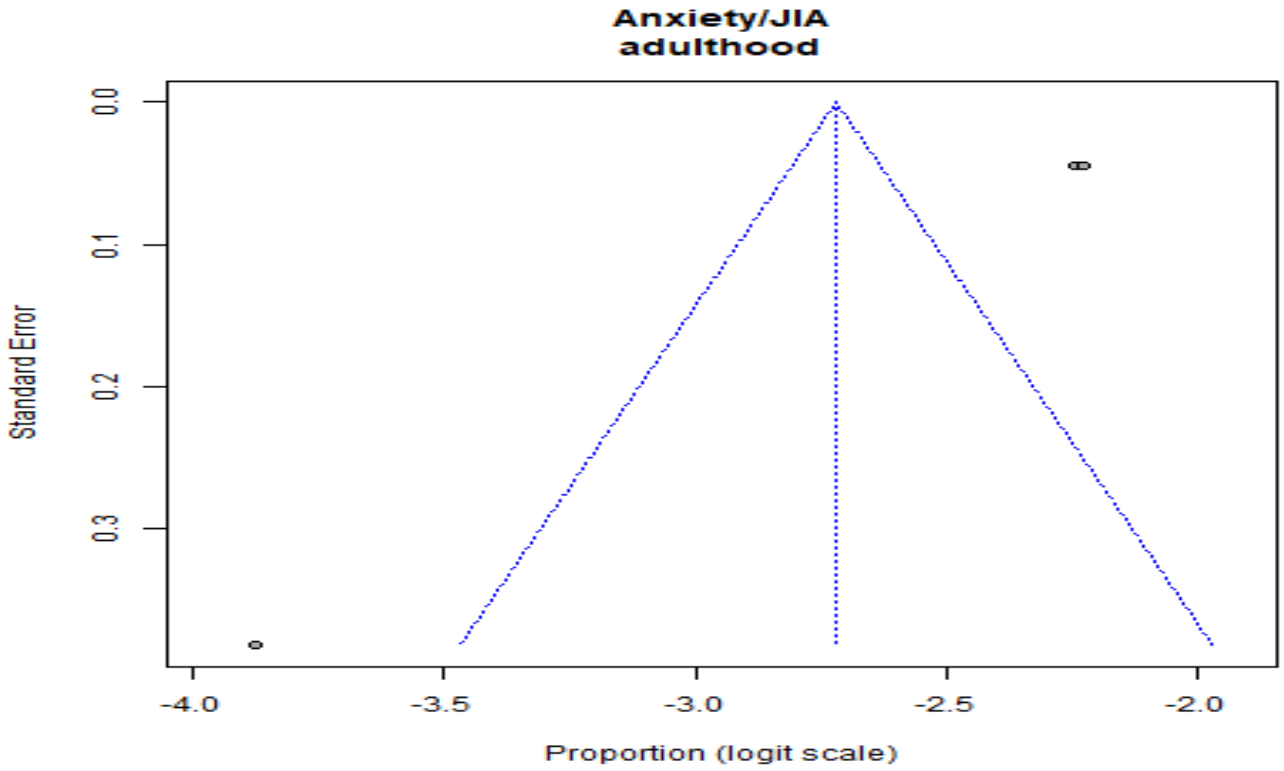
Supplementary Figure S4.25: Forest and Funnel Plots of studies reporting the prevalence of anxiety in adults with Juvenile Idiopathic Arthritis

Anxiety

Forest



Funnel



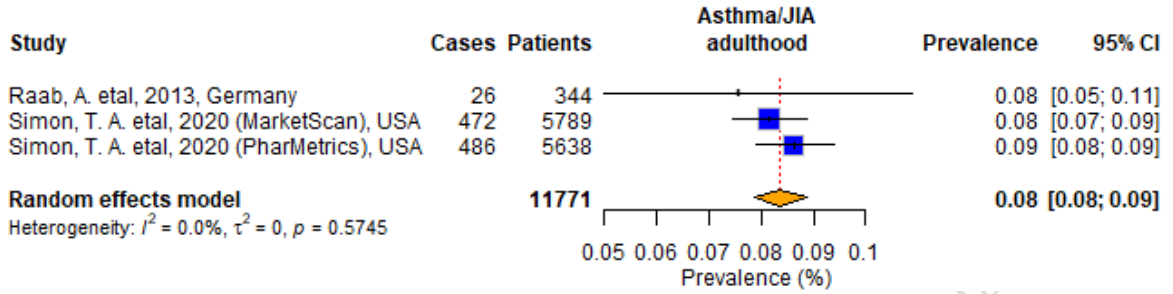
Summary statistics

```
                                        proportion           95%-CI %W(random)
Raab, A. etal, 2013, Germany                0.0203 [0.0082; 0.0415]       29.5
Simon, T. A. etal, 2020 (MarketScan), USA   0.0959 [0.0884; 0.1037]       35.2
Simon, T. A. etal, 2020 (PharMetrics), USA  0.0974 [0.0898; 0.1054]       35.2
Number of studies: k = 3; Number of observations: o = 11771; Number of events: e = 1111
                    proportion           95%-CI
Random effects model     0.0618 [0.0236; 0.1525]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.7433 [0.1448; 35.2654]; tau = 0.8622 [0.3806; 5.9385]
 I^2 = 89.1% [70.3%; 96.0%]; H = 3.03 [1.83; 5.01]
Test of heterogeneity:      Q d.f. p-value
                        18.36    2  0.0001
```

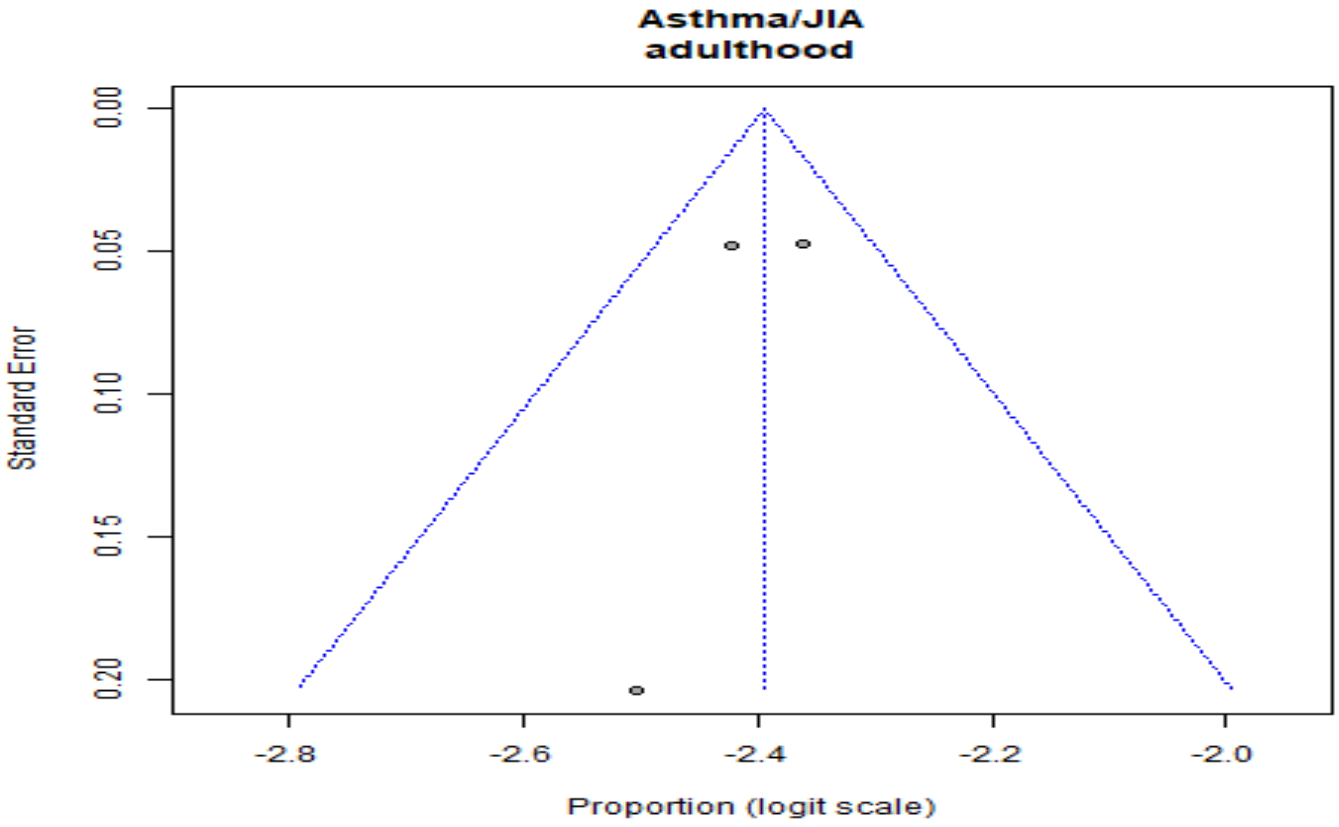
Supplementary Figure S4.26: Forest and Funnel Plots of studies reporting the prevalence of asthma in adults with Juvenile Idiopathic Arthritis

Asthma

Forest



Funnel



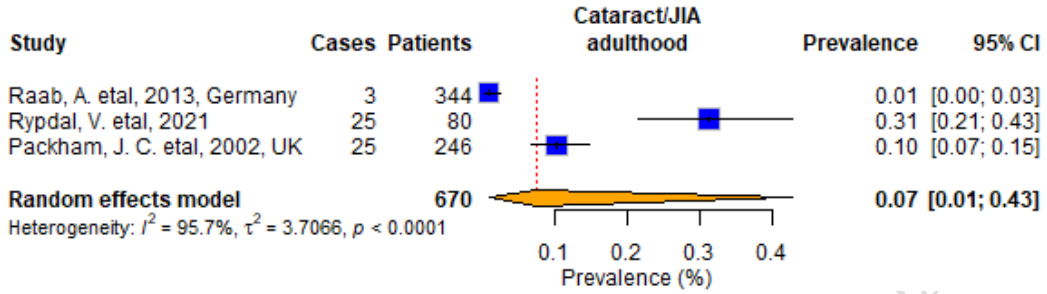
Summary statistics

```
                                        proportion           95%-CI %W(random)
Raab, A. etal, 2013, Germany                0.0756 [0.0500; 0.1088]        2.7
Simon, T. A. etal, 2020 (MarketScan), USA   0.0815 [0.0746; 0.0889]       48.1
Simon, T. A. etal, 2020 (PharMetrics), USA  0.0862 [0.0790; 0.0938]       49.3
Number of studies: k = 3; Number of observations: o = 11771; Number of events: e = 984
                     proportion           95%-CI
Random effects model     0.0836 [0.0788; 0.0888]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 0.1803]; tau = 0 [0.0000; 0.4246]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                         | 1.11    2  0.5745
```

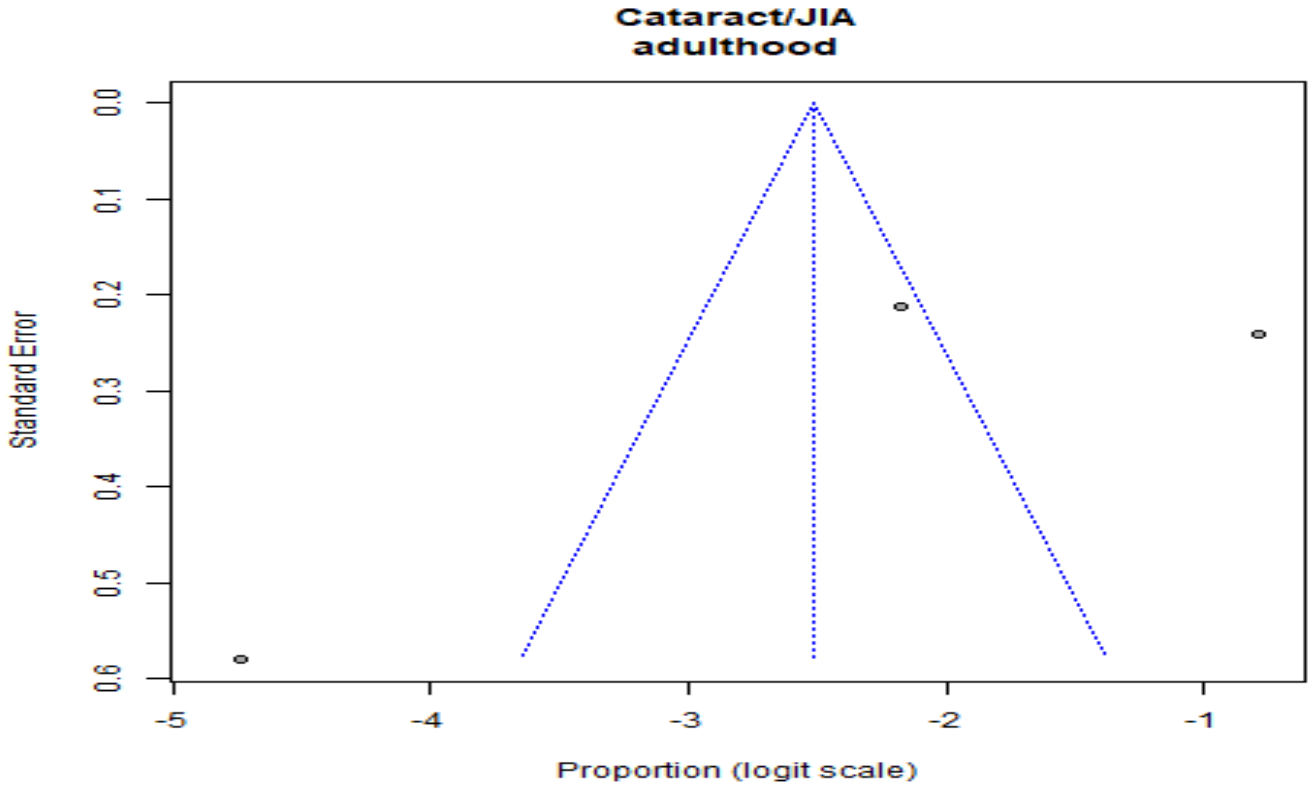
Supplementary Figure S4.27: Forest and Funnel Plots of studies reporting the prevalence of cataract in adults with Juvenile Idiopathic Arthritis

Cataract

Forest



Funnel



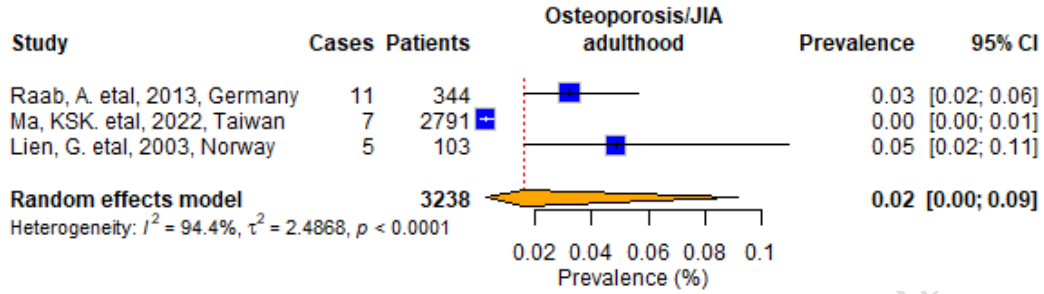
Summary statistics

```
                              proportion           95%-CI %W(random)
Raab, A. etal, 2013, Germany      0.0087 [0.0018; 0.0253]       31.7
Rypdal, V. etal, 2021             0.3125 [0.2135; 0.4259]       34.1
Packham, J. C. etal, 2002, UK     0.1016 [0.0669; 0.1464]       34.2
Number of studies: k = 3
Number of observations: o = 670
Number of events: e = 53
                     proportion           95%-CI
Random effects model     0.0748 [0.0087; 0.4266]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 3.7066 [0.8680; >100.0000]; tau = 1.9252 [0.9316; >10.0000]
 I^2 = 95.7% [90.6%; 98.0%]; H = 4.82 [3.26; 7.11]
Test of heterogeneity:       Q d.f.  p-value
                         46.40    2 < 0.0001
```

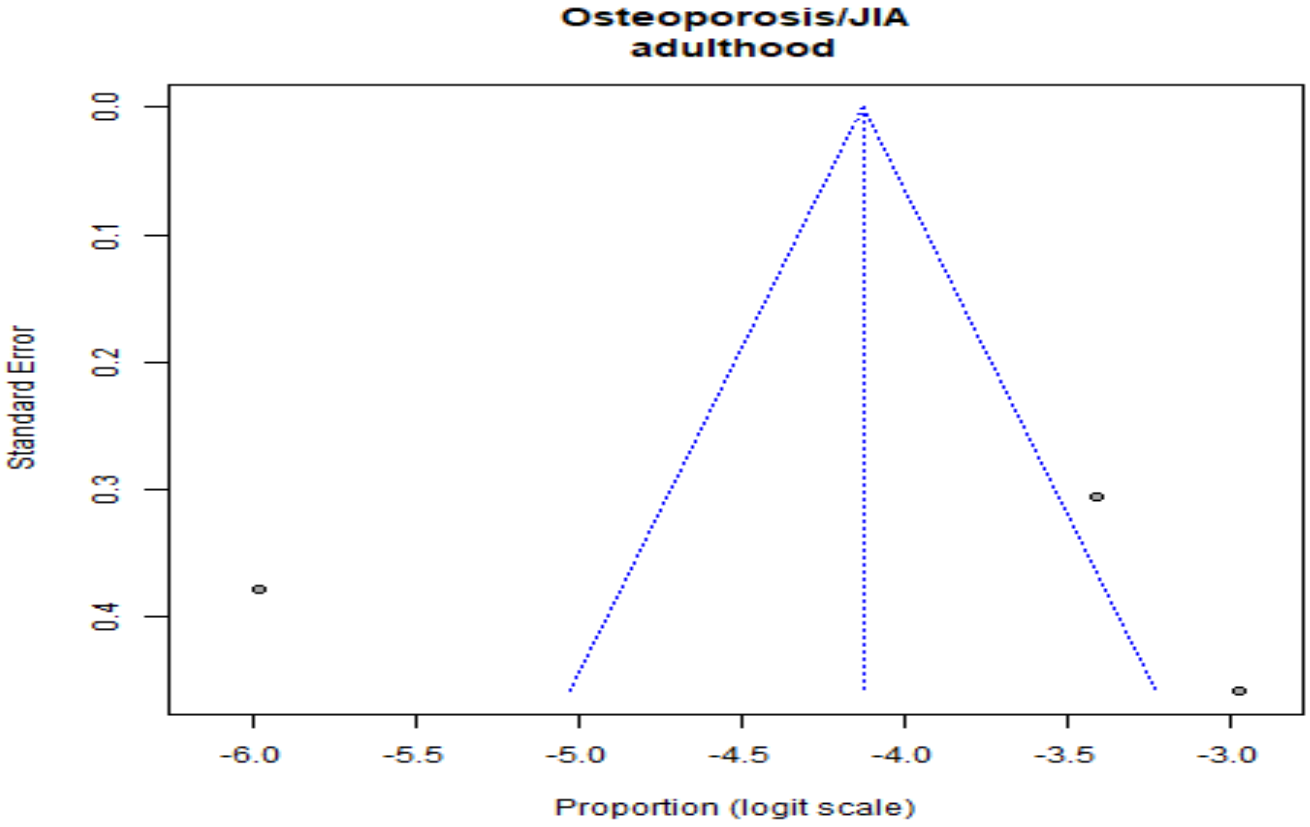
Supplementary Figure S4.28: Forest and Funnel Plots of studies reporting the prevalence of osteoporosis in adults with Juvenile Idiopathic Arthritis

Osteoporosis

Forest



Funnel



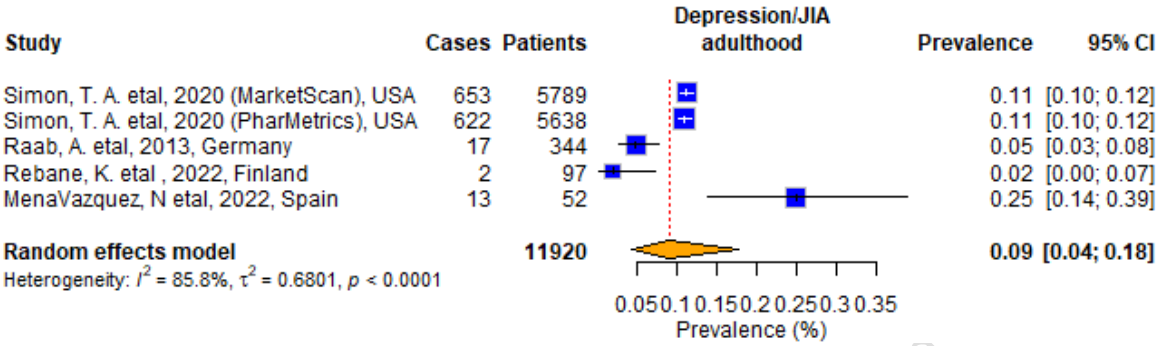
Summary statistics

```
                           proportion           95%-CI %W(random)
Raab, A. etal, 2013, Germany     0.0320 [0.0161; 0.0565]       34.0
Ma, KSK. etal, 2022, Taiwan      0.0025 [0.0010; 0.0052]       33.4
Lien, G. etal, 2003, Norway      0.0485 [0.0159; 0.1097]       32.6
Number of studies: k = 3; Number of observations: o = 3238; Number of events: e = 23
                     proportion           95%-CI
Random effects model     0.0158 [0.0026; 0.0918]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 2.4868 [0.5640; >100.0000]; tau = 1.5770 [0.7510; >10.0000]
 I^2 = 94.4% [87.1%; 97.6%]; H = 4.24 [2.78; 6.45]
Test of heterogeneity:      Q d.f.  p-value
                        35.90    2 < 0.0001
```

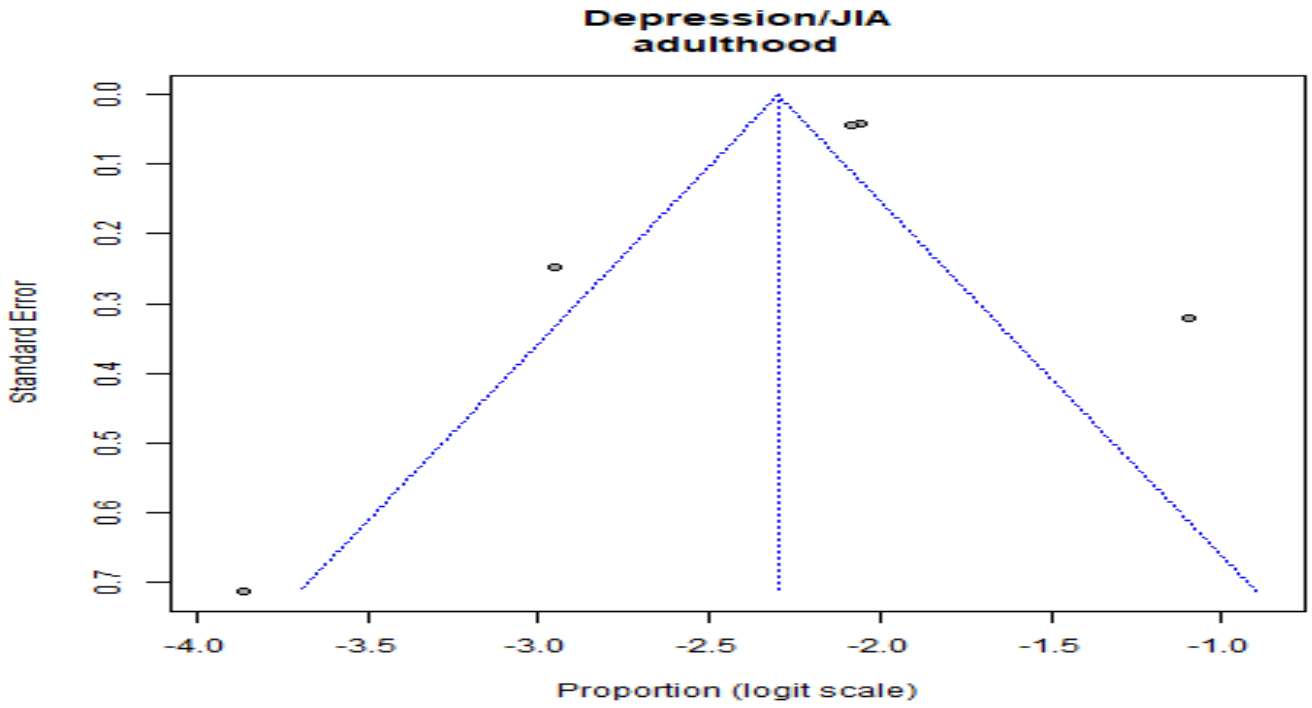
Supplementary Figure S4.29: Forest and Funnel Plots of studies reporting the prevalence of depression in adults with Juvenile Idiopathic Arthritis

Depression

Forest



Funnel



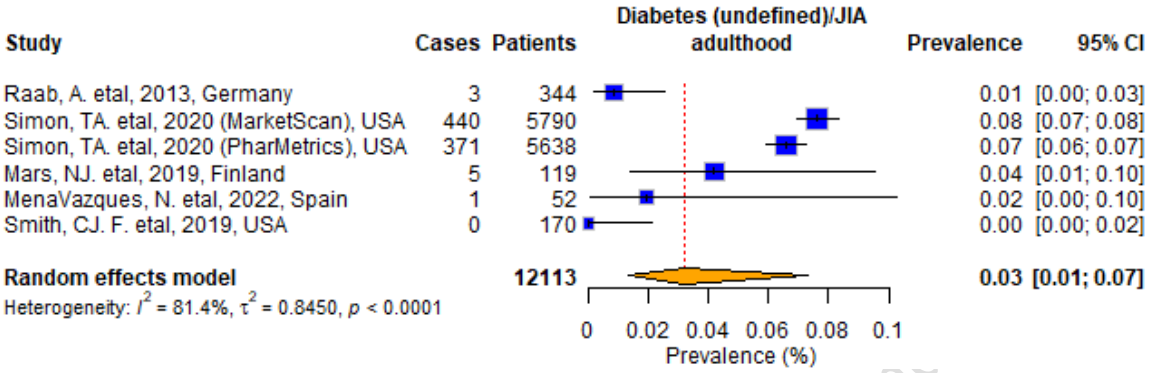
Summary statistics

```
                                        proportion           95%-CI %W(random)
Simon, T. A. etal, 2020 (MarketScan), USA     0.1128 [0.1048; 0.1212]       22.9
Simon, T. A. etal, 2020 (PharMetrics), USA    0.1103 [0.1023; 0.1188]       22.9
Raab, A. etal, 2013, Germany                  0.0494 [0.0290; 0.0779]       21.1
Rebane, K. etal , 2022, Finland               0.0206 [0.0025; 0.0725]       13.1
MenaVazquez, N etal, 2022, Spain              0.2500 [0.1403; 0.3895]       20.0
Number of studies: k = 5; Number of observations: o = 11920; Number of events: e = 1307
                    proportion           95%-CI
Random effects model     0.0911 [0.0442; 0.1787]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.6801 [0.1703; 8.6878]; tau = 0.8247 [0.4127; 2.9475]
 I^2 = 85.8% [68.9%; 93.5%]; H = 2.66 [1.79; 3.94]
Test of heterogeneity:      Q d.f.  p-value
                        28.23    4 < 0.0001
```

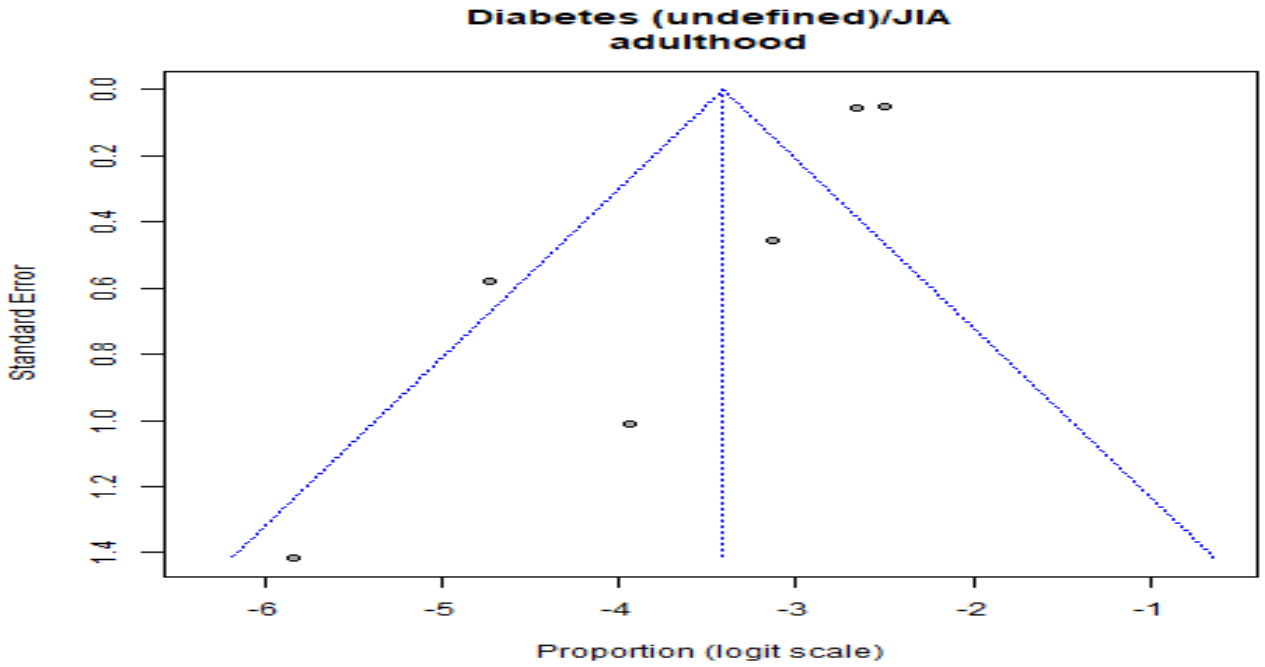
Supplementary Figure S4.30: Forest and Funnel Plots of studies reporting the prevalence of diabetes (undefined) in adults with Juvenile Idiopathic Arthritis

Diabetes (undefined)

Forest



Funnel



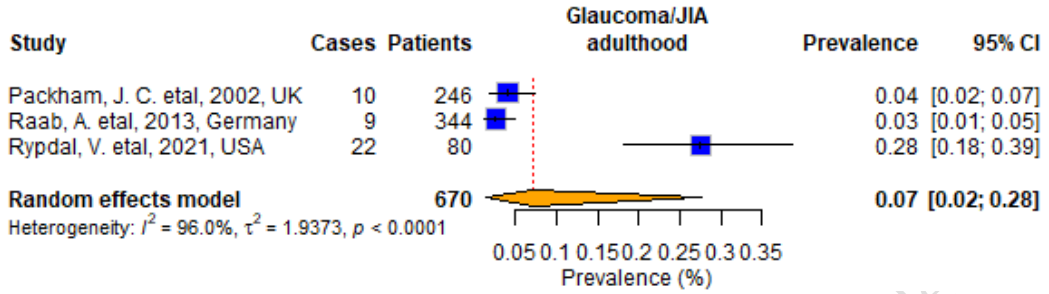
Summary statistics

```
                                        proportion           95%-CI %W(random)
Raab, A. etal, 2013, Germany                0.0087 [0.0018; 0.0253]       16.8
Simon, TA. etal, 2020 (MarketScan), USA     0.0760 [0.0693; 0.0831]       23.4
Simon, TA. etal, 2020 (PharMetrics), USA    0.0658 [0.0595; 0.0726]       23.4
Mars, NJ. etal, 2019, Finland               0.0420 [0.0138; 0.0953]       18.8
MenaVazques, N. etal, 2022, Spain           0.0192 [0.0005; 0.1026]       10.6
Smith, CJ. F. etal, 2019, USA               0.0000 [0.0000; 0.0215]        7.0
Number of studies: k = 6; Number of observations: o = 12113; Number of events: e = 820
                     proportion           95%-CI
Random effects model     0.0319 [0.0136; 0.0731]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.8450 [0.1589; 9.2267]; tau = 0.9192 [0.3986; 3.0375]
 I^2 = 81.4% [60.1%; 91.3%]; H = 2.32 [1.58; 3.39]
Test of heterogeneity:      Q d.f.  p-value
                        26.84    5 < 0.0001
```

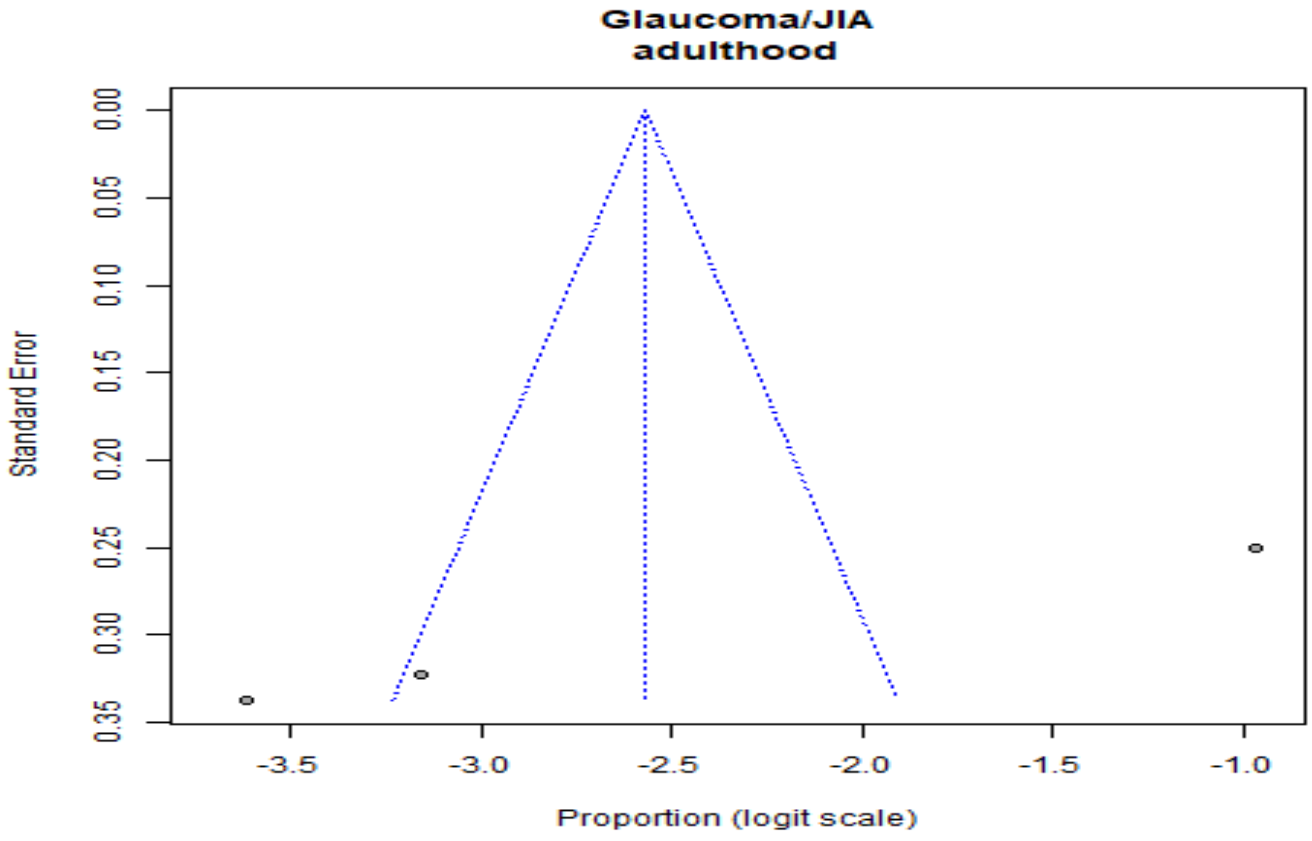
Supplementary Figure S4.31: Forest and Funnel Plots of studies reporting the prevalence of glaucoma in adults with Juvenile Idiopathic Arthritis

Glaucoma

Forest



Funnel



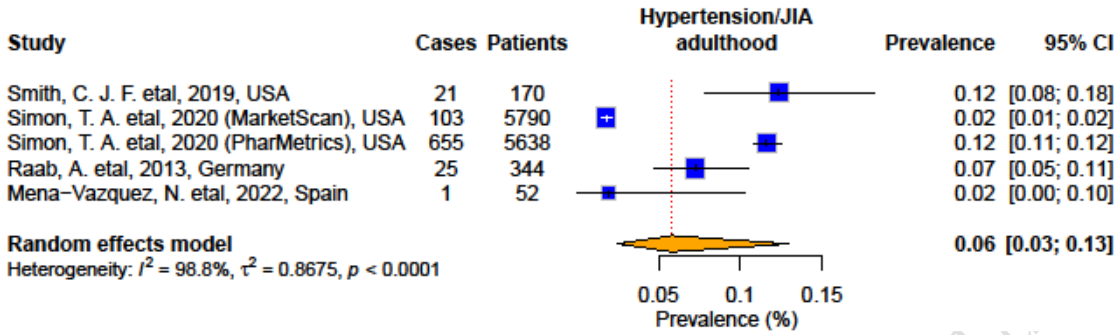
Summary statistics

```
                           proportion           95%-CI %W(random)
Packham, J. C. etal, 2002, UK     0.0407 [0.0197; 0.0735]       33.2
Raab, A. etal, 2013, Germany      0.0262 [0.0120; 0.0491]       33.0
Rypdal, V. etal, 2021, USA        0.2750 [0.1810; 0.3862]       33.8
Number of studies: k = 3; Number of observations: o = 670; Number of events: e = 41
                     proportion           95%-CI
Random effects model     0.0711 [0.0150; 0.2774]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.9373 [0.4633; 79.0548]; tau = 1.3919 [0.6807; 8.8913]
 I^2 = 96.0% [91.5%; 98.1%]; H = 5.02 [3.43; 7.35]
Test of heterogeneity:      Q d.f.  p-value
                        50.50    2 < 0.0001
```

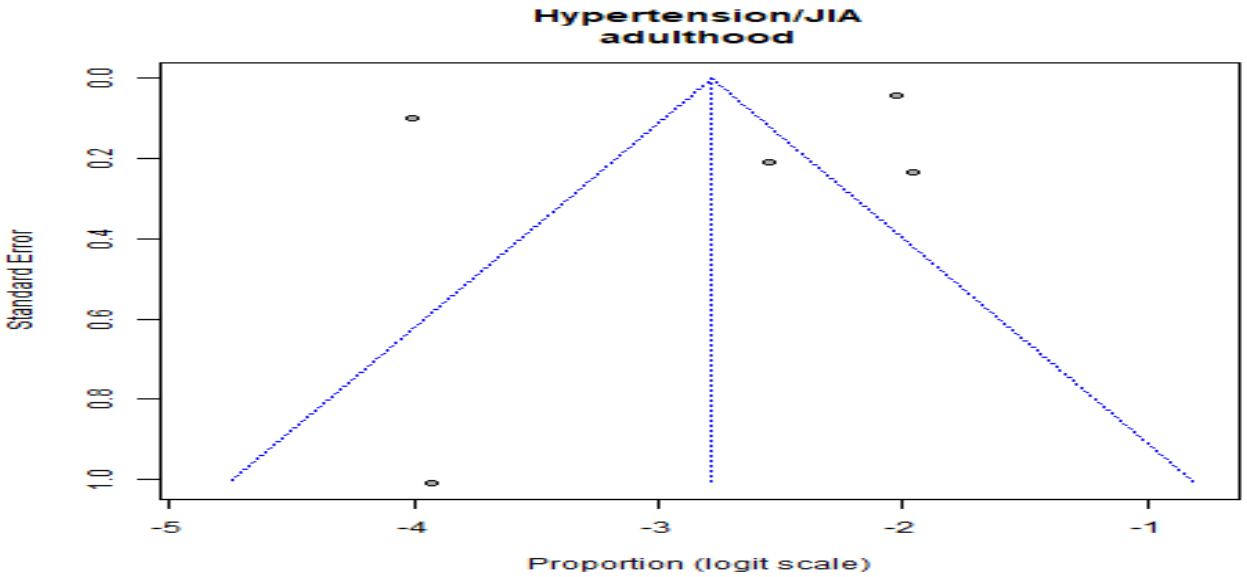
Supplementary Figure S4.32: Forest and Funnel Plots of studies reporting the prevalence of hypertension in adults with Juvenile Idiopathic Arthritis

Hypertension

Forest



Funnel



Summary statistics

```
                                          proportion   95% CI                 %W(random)
Smith, C. J. F. etal, 2019, USA               0.1235   [0.0781; 0.1826]         21.7
Simon, T. A. etal, 2020 (MarketScan), USA     0.0178   [0.0145; 0.0215]         22.8
Simon, T. A. etal, 2020 (PharMetrics), USA    0.1162   [0.1079; 0.1248]         23.0
Raab, A. etal, 2013, Germany                  0.0727   [0.0476; 0.1054]         21.9
Mena-Vazquez, N. etal, 2022, Spain            0.0192   [0.0005; 0.1026]         10.6

Number of studies: k = 5
Number of observations: o = 11994
Number of events: e = 805

                     proportion         95% CI
Random effects model     0.0584         [0.0252; 0.1296]

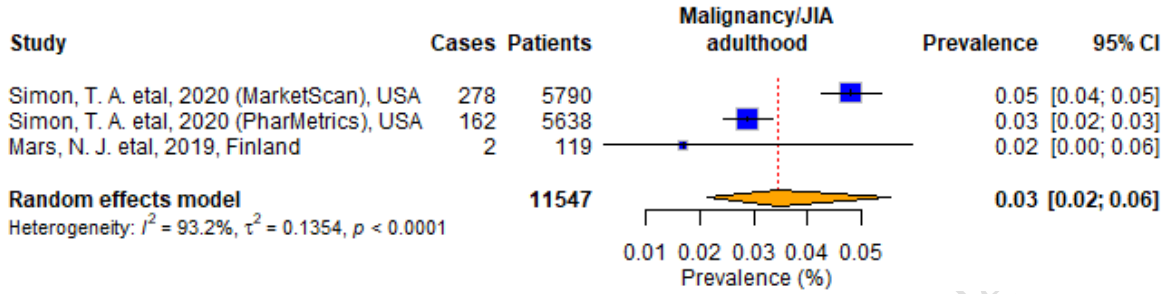
Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.8675 [0.2585; 8.1278]; tau = 0.9314 [0.5084; 2.8509]
 I^2 = 98.8% [98.3%; 99.2%]; H = 9.28 [7.71; 11.17]

Test of heterogeneity:
      Q d.f.  p-value
 344.45    4 < 0.0001
```

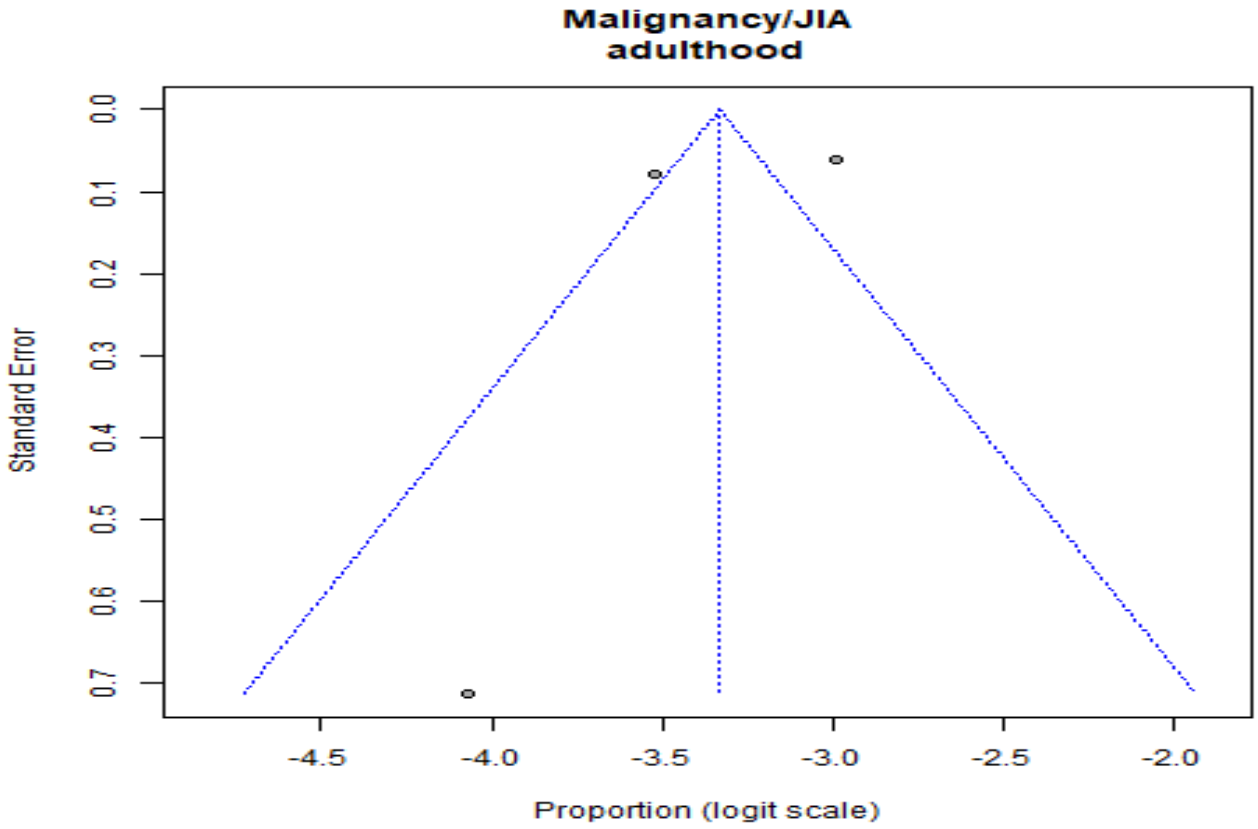
Supplementary Figure S4.33: Forest and Funnel Plots of studies reporting the prevalence of malignancy in adults with Juvenile Idiopathic Arthritis

Malignancy

Forest



Funnel



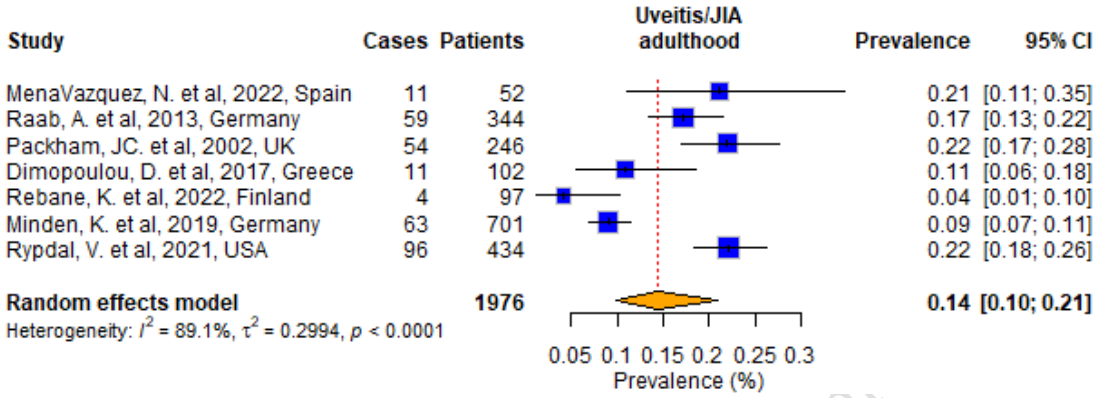
Summary statistics

```
                                        proportion           95%-CI %W(random)
Simon, T. A. etal, 2020 (MarketScan), USA     0.0480 [0.0426; 0.0538]       45.5
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0287 [0.0245; 0.0334]       44.7
Mars, N. J. etal, 2019, Finland               0.0168 [0.0020; 0.0594]        9.8
Number of studies: k = 3; Number of observations: o = 11547; Number of events: e = 442
                    proportion           95%-CI
Random effects model     0.0345 [0.0214; 0.0553]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.1354 [0.0182; 11.3036]; tau = 0.3679 [0.1350; 3.3621]
 I^2 = 93.2% [83.6%; 97.2%]; H = 3.85 [2.47; 5.99]
Test of heterogeneity:      Q d.f.  p-value
                        29.61    2 < 0.0001
```

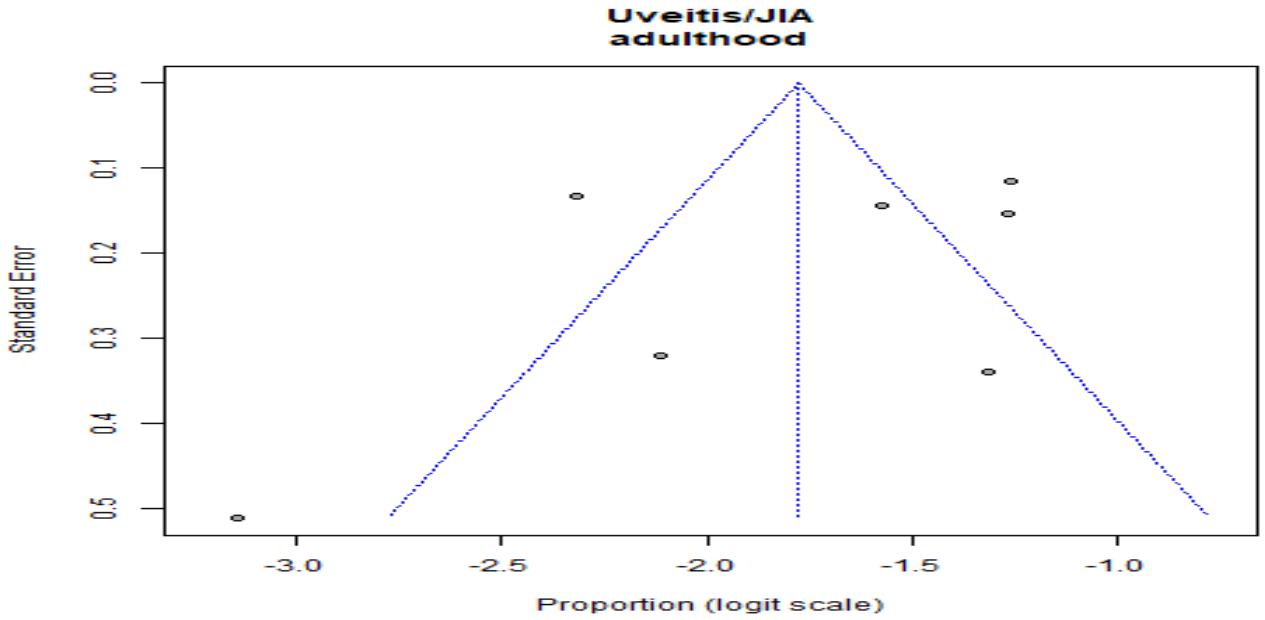
Supplementary Figure S4.34: Forest and Funnel Plots of studies reporting the prevalence of uveitis in adults with Juvenile Idiopathic Arthritis

Uveitis

Forest



Funnel



Summary statistics

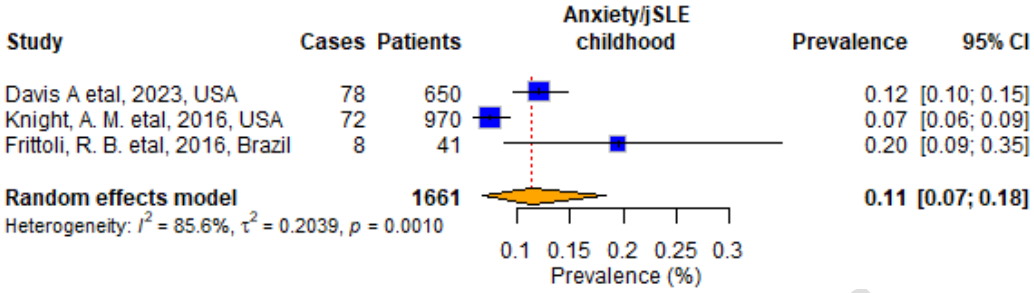
```
                                proportion           95%-CI %W(random)
MenaVazquez, N. et al, 2022, Spain     0.2115 [0.1106; 0.3470]       12.5
Raab, A. et al, 2013, Germany          0.1715 [0.1332; 0.2156]       16.2
Packham, JC. et al, 2002, UK           0.2195 [0.1694; 0.2765]       16.1
Dimopoulou, D. et al, 2017, Greece     0.1078 [0.0551; 0.1848]       12.9
Rebane, K. et al, 2022, Finland        0.0412 [0.0113; 0.1022]        9.3
Minden, K. et al, 2019, Germany        0.0899 [0.0698; 0.1135]       16.4
Rypdal, V. et al, 2021, USA            0.2212 [0.1830; 0.2632]       16.6
Number of studies: k = 7; Number of observations: o = 1976; Number of events: e = 298
                    proportion           95%-CI
Random effects model     0.1446 [0.0976; 0.2090]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.2994 [0.0932; 2.2766]; tau = 0.5472 [0.3053; 1.5089]
 I^2 = 89.1% [79.9%; 94.0%]; H = 3.02 [2.23; 4.09]
Test of heterogeneity:      Q d.f.  p-value
                        54.84    6 < 0.0001
```

Childhood/jSLE

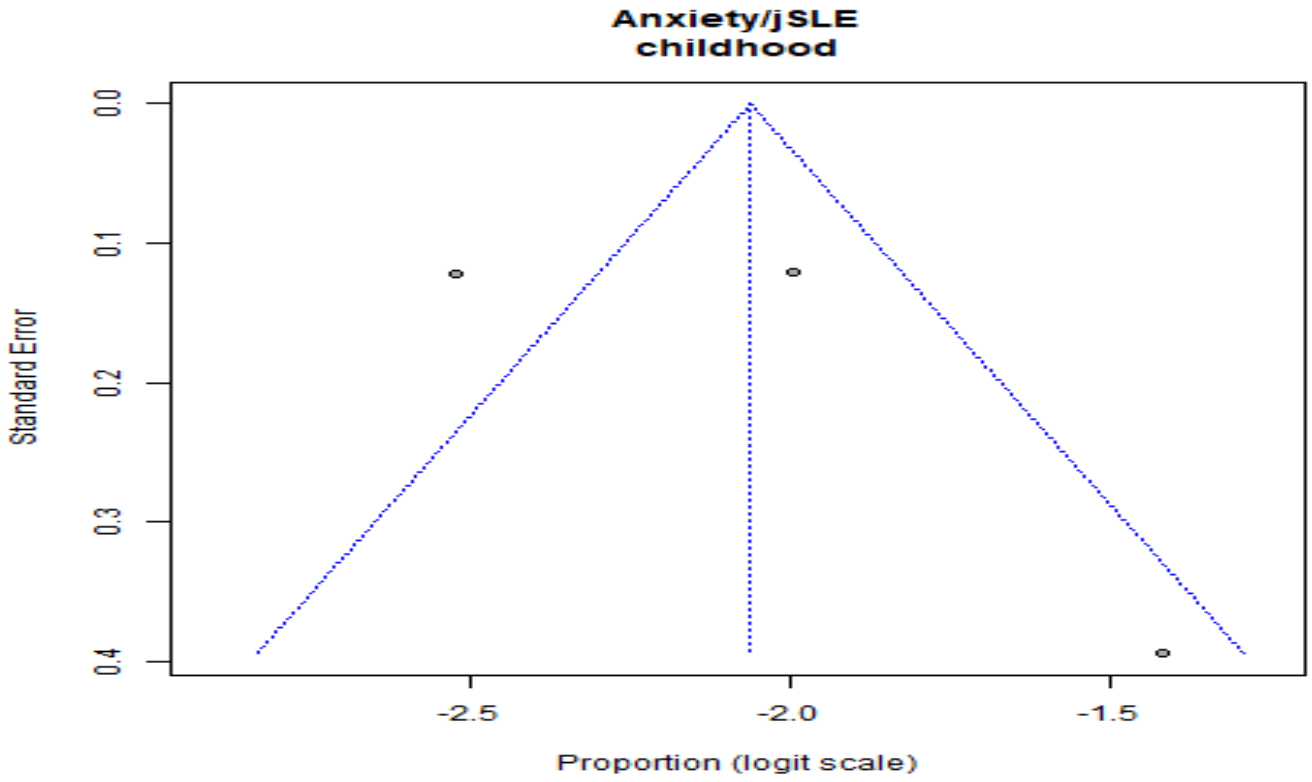
Supplementary Figure S4.35: Forest and Funnel Plots of studies reporting the prevalence of anxiety in children with juvenile-onset Systemic Lupus Erythematosus

Anxiety

Forest



Funnel



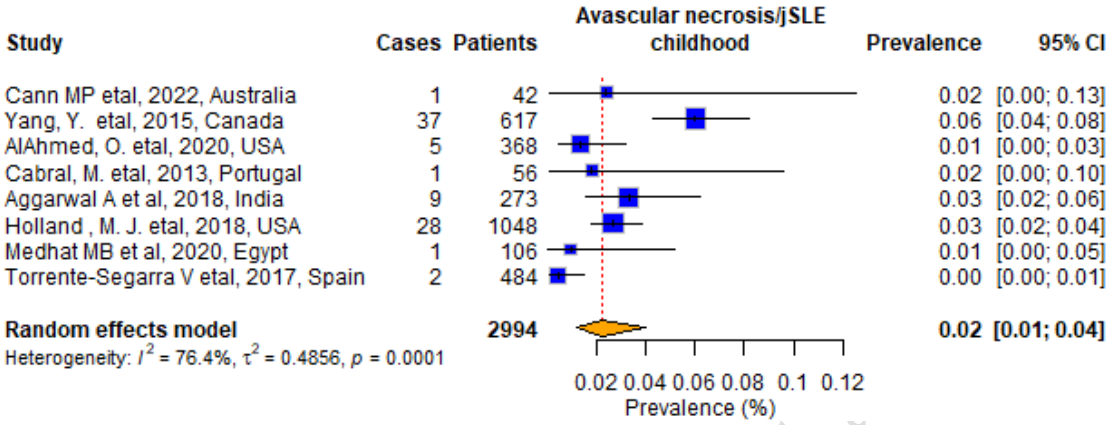
Summary statistics

```
                                proportion           95%-CI %W(random)
Davis A etal, 2023, USA             0.1200 [0.0960; 0.1475]       38.4
Knight, A. M. etal, 2016, USA       0.0742 [0.0585; 0.0926]       38.3
Frittoli, R. B. etal, 2016, Brazil  0.1951 [0.0882; 0.3487]       23.3
Number of studies: k = 3; Number of observations: o = 1661; Number of events: e = 158
                     proportion           95%-CI
Random effects model     0.1129 [0.0673; 0.1833]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.2039 [0.0224; 12.0079]; tau = 0.4515 [0.1498; 3.4652]
 I^2 = 85.6% [57.7%; 95.1%]; H = 2.63 [1.54; 4.50]
Test of heterogeneity:      Q d.f. p-value
                        13.84    2  0.0010
```

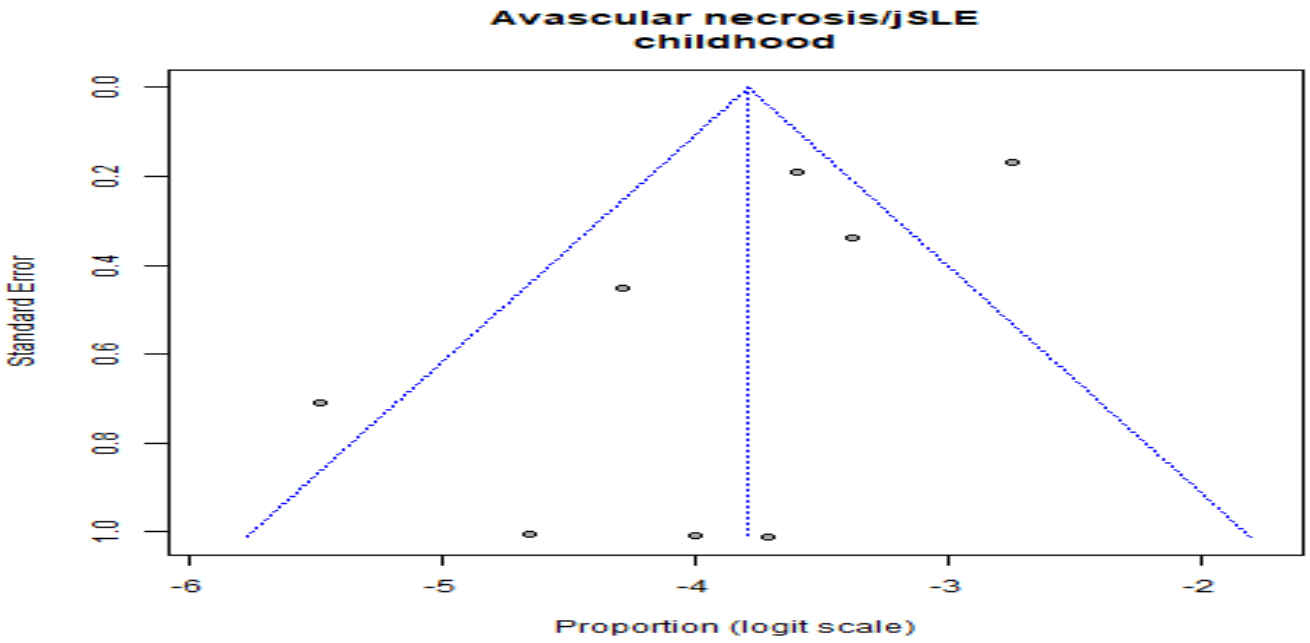
Supplementary Figure S4.36: Forest and Funnel Plots of studies reporting the prevalence of avascular necrosis in children with juvenile-onset Systemic Lupus Erythematosus

Avascular necrosis

Forest



Funnel



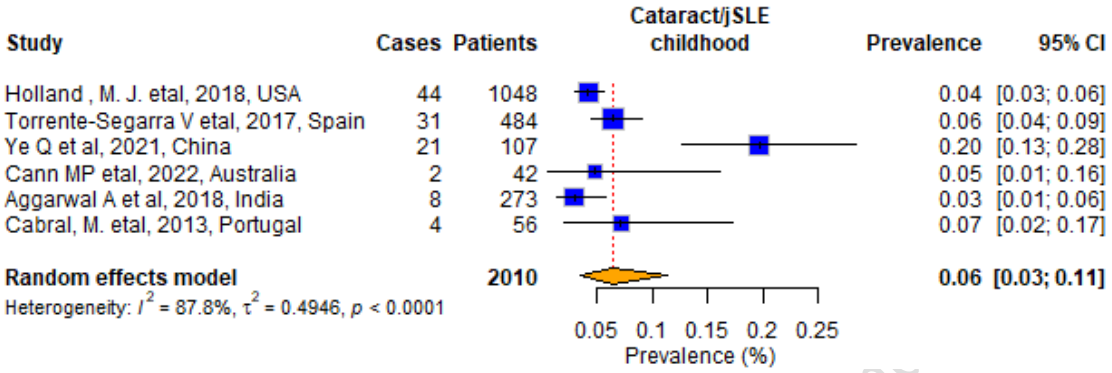
Summary statistics

```
                                    proportion           95%-CI %W(random)
Cann MP etal, 2022, Australia           0.0238 [0.0006; 0.1257]        6.6
Yang, Y.  etal, 2015, Canada            0.0600 [0.0426; 0.0817]       19.5
AlAhmed, O. etal, 2020, USA             0.0136 [0.0044; 0.0314]       14.5
Cabral, M. etal, 2013, Portugal         0.0179 [0.0005; 0.0955]        6.7
Aggarwal A et al, 2018, India           0.0330 [0.0152; 0.0617]       16.7
Holland , M. J. etal, 2018, USA         0.0267 [0.0178; 0.0384]       19.2
Medhat MB et al, 2020, Egypt            0.0094 [0.0002; 0.0514]        6.7
Torrente-Segarra V etal, 2017, Spain    0.0041 [0.0005; 0.0148]       10.1
Number of studies: k = 8; Number of observations: o = 2994; Number of events: e = 84
                     proportion           95%-CI
Random effects model     0.0220 [0.0120; 0.0402]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4856 [0.0757; 2.5590]; tau = 0.6968 [0.2752; 1.5997]
 I^2 = 76.4% [52.8%; 88.2%]; H = 2.06 [1.46; 2.91]
Test of heterogeneity:      Q d.f. p-value
                        29.62    7  0.0001
```

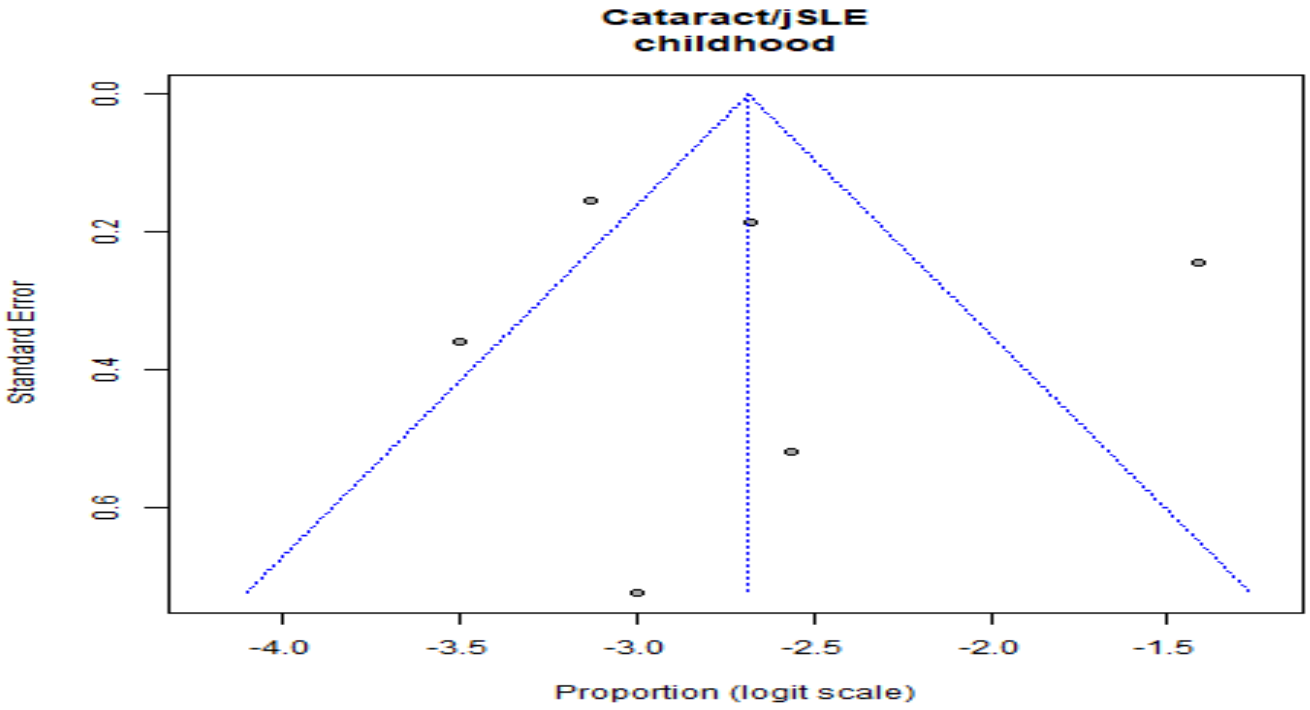
Supplementary Figure S4.37: Forest and Funnel Plots of studies reporting the prevalence of cataract in children with juvenile-onset Systemic Lupus Erythematosus

Cataract

Forest



Funnel



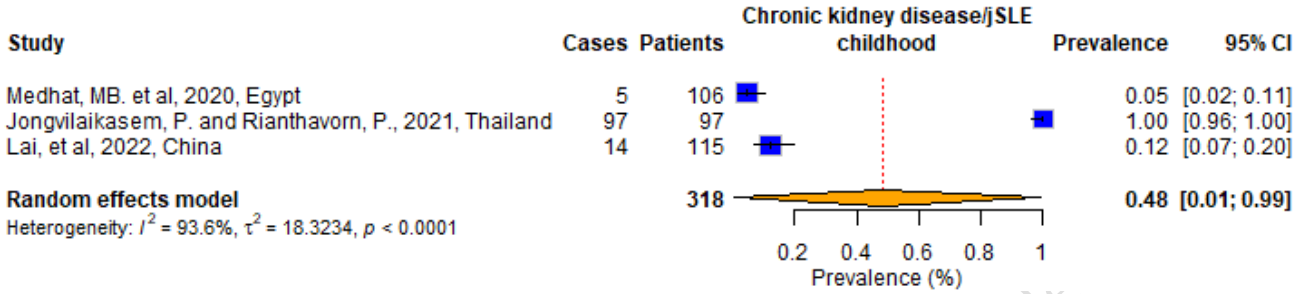
Summary statistics

```
                                    proportion           95%-CI %W(random)
Holland , M. J. etal, 2018, USA         0.0420 [0.0307; 0.0560]       20.3
Torrente-Segarra V etal, 2017, Spain    0.0640 [0.0439; 0.0897]       19.9
Ye Q et al, 2021, China                 0.1963 [0.1258; 0.2842]       19.0
Cann MP etal, 2022, Australia           0.0476 [0.0058; 0.1616]       10.3
Aggarwal A et al, 2018, India           0.0293 [0.0127; 0.0569]       16.9
Cabral, M. etal, 2013, Portugal         0.0714 [0.0198; 0.1729]       13.8
Number of studies: k = 6; Number of observations: o = 2010; Number of events: e = 110
                     proportion           95%-CI
Random effects model     0.0639 [0.0349; 0.1141]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4946 [0.1265; 3.0315]; tau = 0.7032 [0.3557; 1.7411]
 I^2 = 87.8% [75.9%; 93.8%]; H = 2.86 [2.04; 4.02]
Test of heterogeneity:      Q d.f.  p-value
                        40.98    5 < 0.0001
```

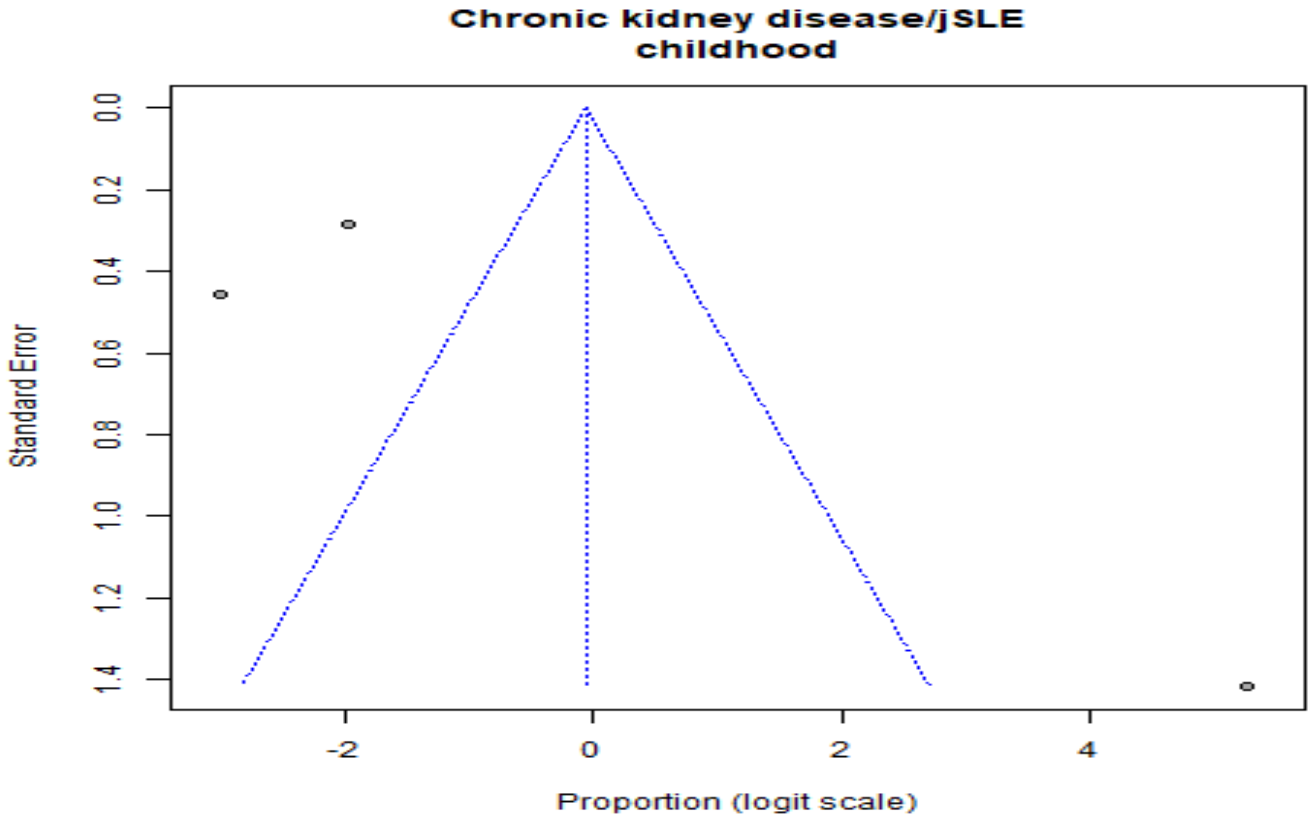
Supplementary Figure S4.38: Forest and Funnel Plots of studies reporting the prevalence of chronic kidney disease in children with juvenile-onset Systemic Lupus Erythematosus

Chronic kidney disease

Forest



Funnel



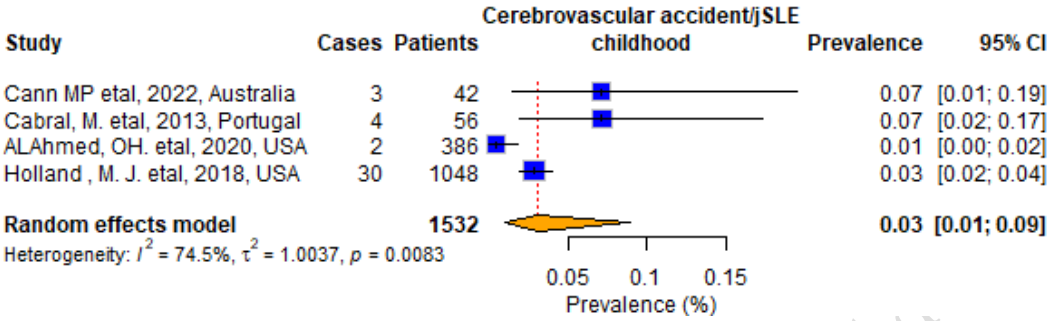
Summary statistics

```
                                                    proportion       95%-CI          %W(random)
Medhat, MB. et al, 2020, Egypt                          0.0472       [0.0155; 0.1067]        34.3
Jongvilaikasem, P. and Rianthavorn, P., 2021, Thailand  1.0000       [0.9627; 1.0000]        31.2
Lai, et al, 2022, China                                 0.1217       [0.0682; 0.1958]        34.5
Number of studies: k = 3; Number of observations: o = 318; Number of events: e = 116
                     proportion           95%-CI
Random effects model     0.4838 [0.0067; 0.9924]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 18.3234 [4.1403; >183.2337]; tau = 4.2806 [2.0348; >13.5364]
 I^2 = 93.6% [84.6%; 97.3%]; H = 3.94 [2.55; 6.10]
Test of heterogeneity:      Q d.f.  p-value
                        31.09    2 < 0.0001
```

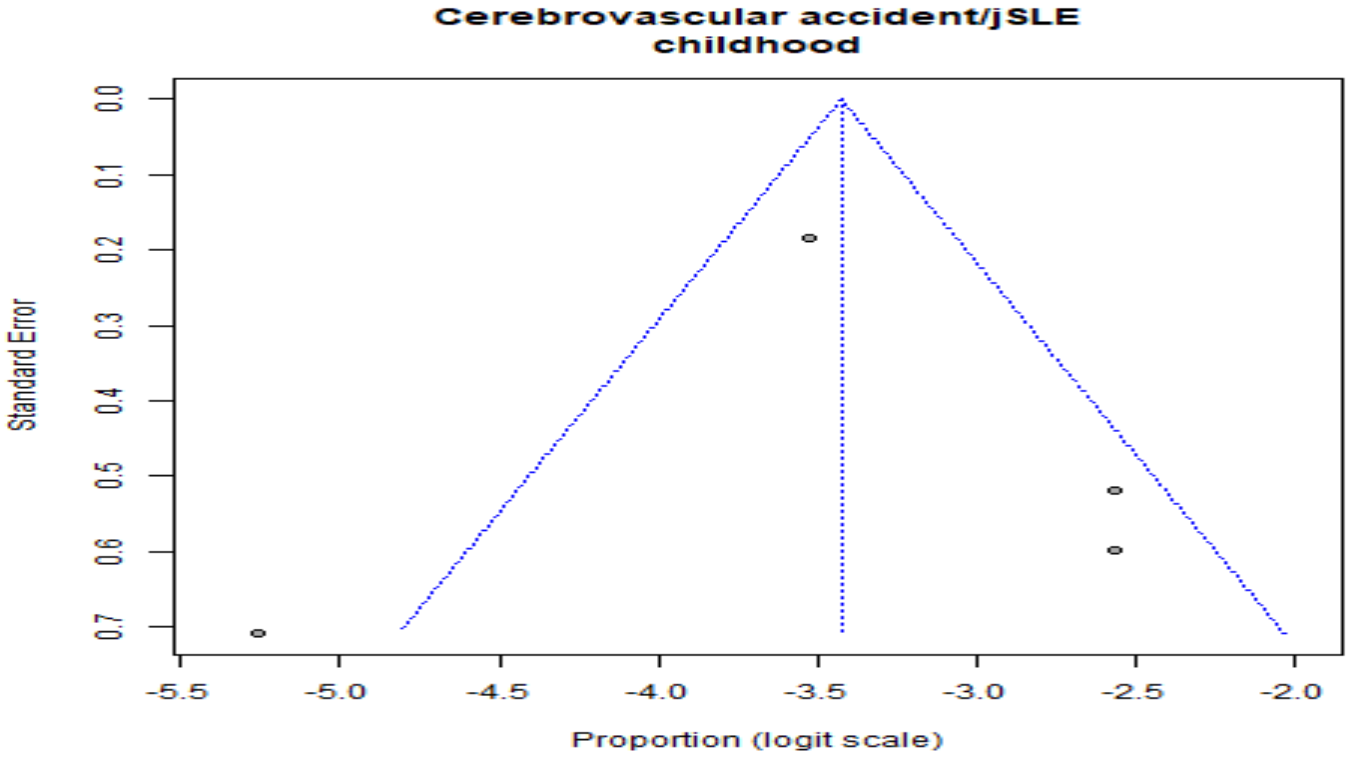
Supplementary Figure S4.39: Forest and Funnel Plots of studies reporting the prevalence of cerebrovascular accident in children with juvenile-onset Systemic Lupus Erythematosus

Cerebrovascular accident

Forest



Funnel



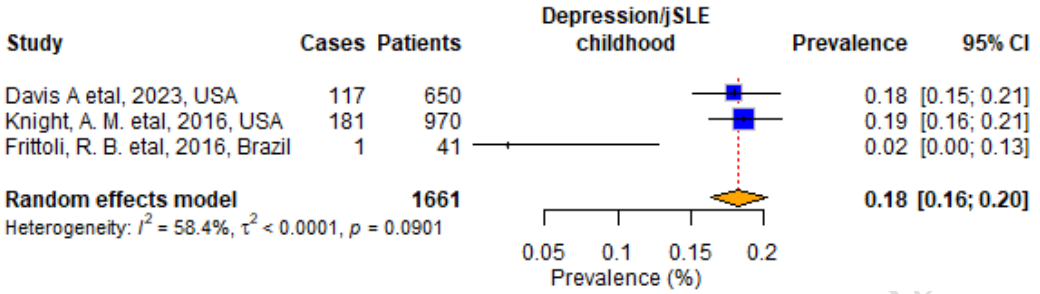
Summary statistics

```
                                proportion      95%-CI          %W(random)
Cann MP etal, 2022, Australia       0.0714      [0.0150; 0.1948]        23.3
Cabral, M. etal, 2013, Portugal     0.0714      [0.0198; 0.1729]        25.0
ALAhmed, OH. etal, 2020, USA        0.0052      [0.0006; 0.0186]        21.1
Holland , M. J. etal, 2018, USA     0.0286      [0.0194; 0.0406]        30.6
Number of studies: k = 4; Number of observations: o = 1532; Number of events: e = 39
                     proportion           95%-CI
Random effects model     0.0315 [0.0106; 0.0893]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.0037 [0.0947; 21.9732]; tau = 1.0018 [0.3078; 4.6876]
 I^2 = 74.5% [28.9%; 90.8%]; H = 1.98 [1.19; 3.30]
Test of heterogeneity:      Q d.f. p-value
                        11.75    3  0.0083
```

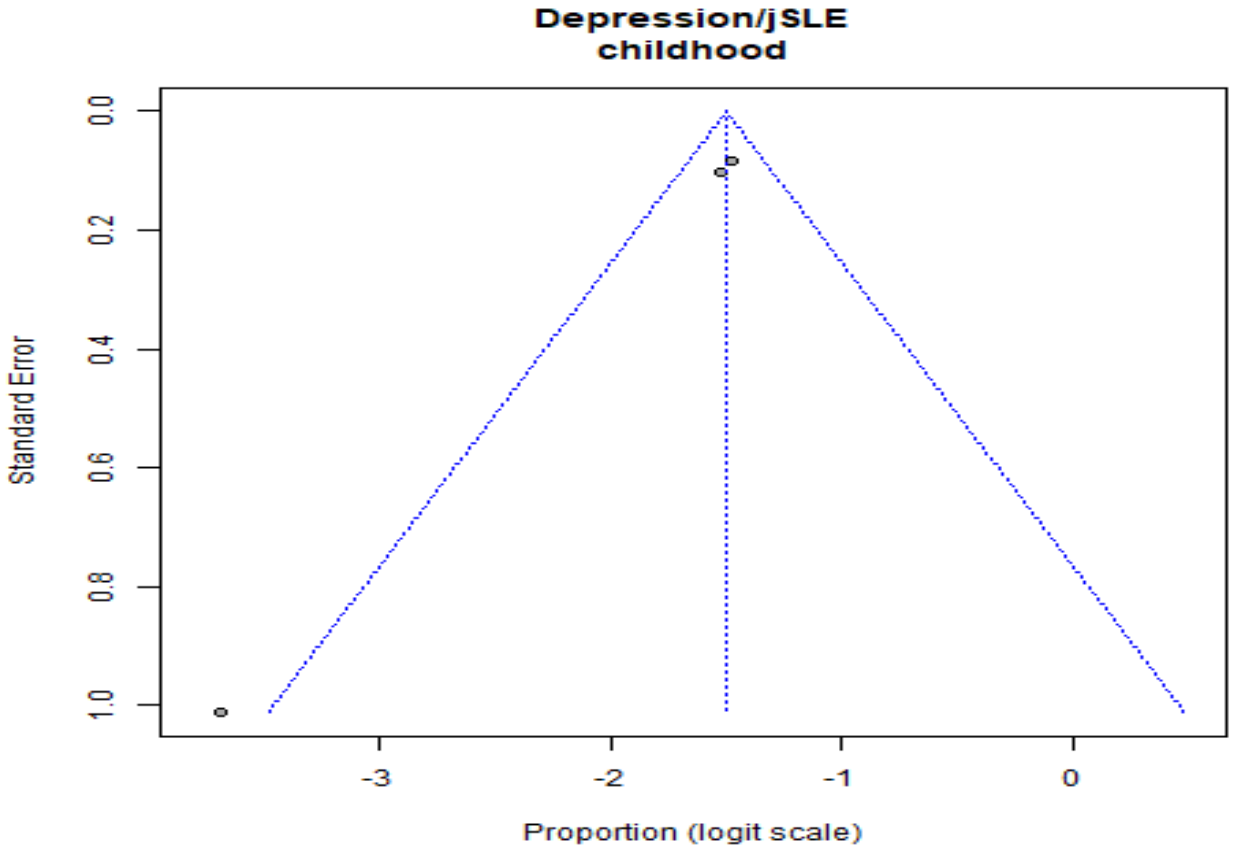
Supplementary Figure S4.40: Forest and Funnel Plots of studies reporting the prevalence of depression in children with juvenile-onset Systemic Lupus Erythematosus

Depression

Forest



Funnel



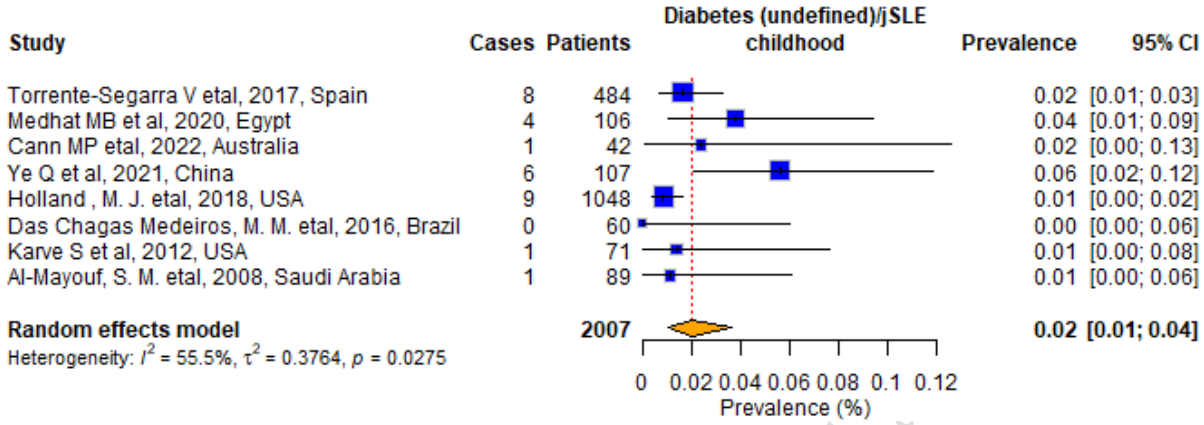
Summary statistics

```
                                proportion   95%-CI          %W(random)
Davis A etal, 2023, USA             0.1800   [0.1512; 0.2117]       39.3
Knight, A. M. etal, 2016, USA       0.1866   [0.1625; 0.2126]       60.3
Frittoli, R. B. etal, 2016, Brazil  0.0244   [0.0006; 0.1286]        0.4
Number of studies: k = 3; Number of observations: o = 1661; Number of events: e = 299
                     proportion           95%-CI
Random effects model     0.1827 [0.1647; 0.2021]
Quantifying heterogeneity (with 95%-CIs):  tau^2 < 0.0001 [0.0000; 62.7424]; tau = 0.0018 [0.0000; 7.9210]
 I^2 = 58.4% [0.0%; 88.2%]; H = 1.55 [1.00; 2.91]
Test of heterogeneity:     Q d.f. p-value
                        4.81    2  0.0901
```

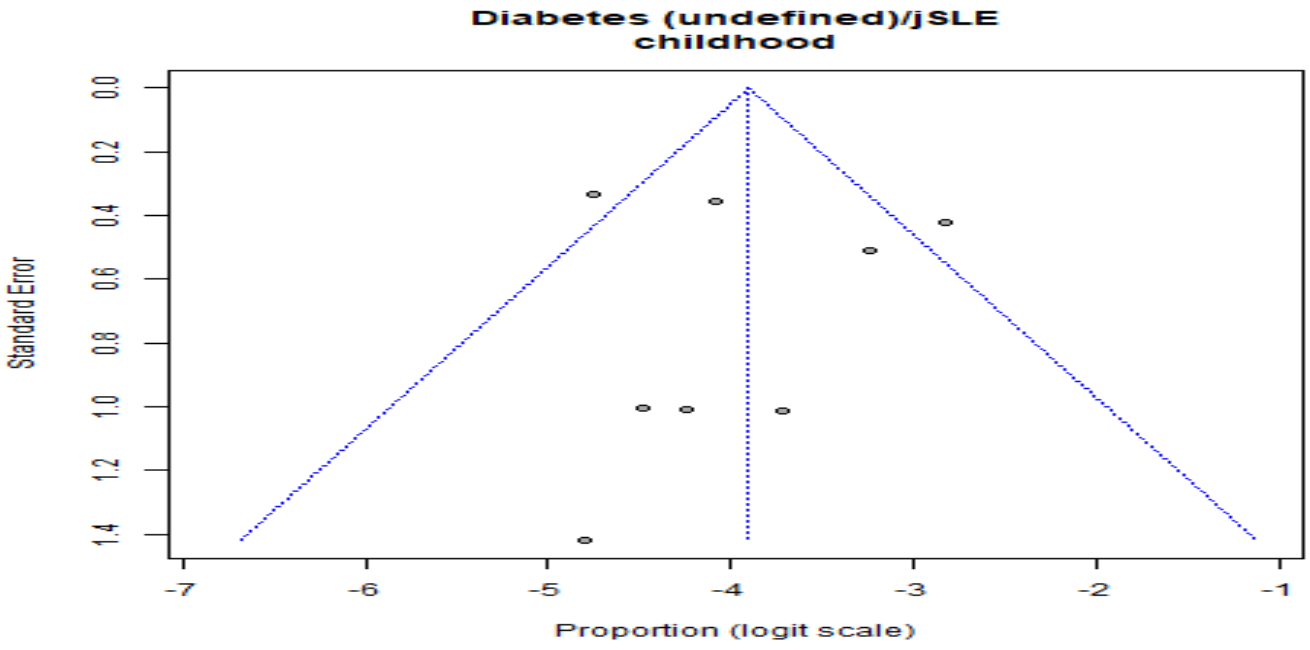
Supplementary Figure S4.41: Forest and Funnel Plots of studies reporting the prevalence of diabetes (undefined) in children with juvenile-onset Systemic Lupus Erythematosus

Diabetes (undefined)

Forest



Funnel



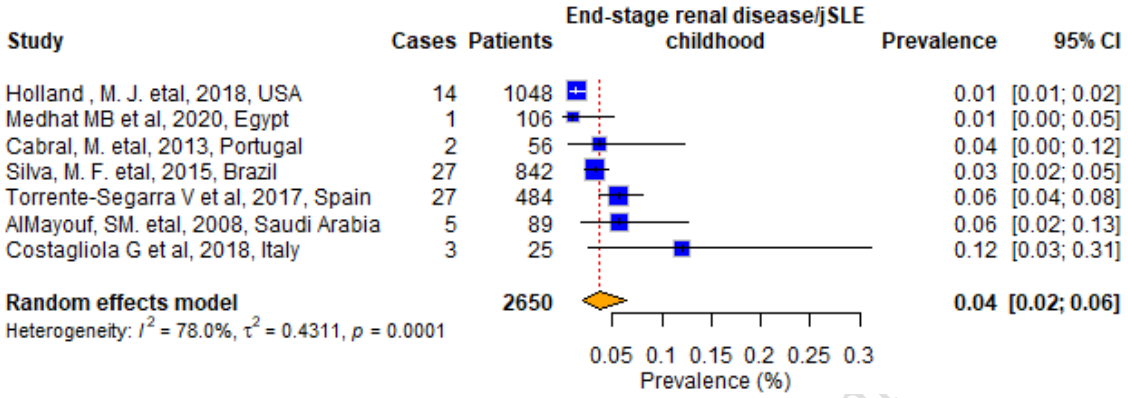
Summary statistics

```
                                          proportion         95%-CI          %W(random)
Torrente-Segarra V etal, 2017, Spain          0.0165        [0.0072; 0.0323]        19.9
Medhat MB et al, 2020, Egypt                  0.0377        [0.0104; 0.0938]        15.7
Cann MP etal, 2022, Australia                 0.0238        [0.0006; 0.1257]         7.1
Ye Q et al, 2021, China                       0.0561        [0.0209; 0.1181]        18.1
Holland , M. J. etal, 2018, USA               0.0086        [0.0039; 0.0162]        20.5
Das Chagas Medeiros, M. M. etal, 2016, Brazil 0.0000        [0.0000; 0.0596]         4.2
Karve S et al, 2012, USA                      0.0141        [0.0004; 0.0760]         7.2
Al-Mayouf, S. M. etal, 2008, Saudi Arabia     0.0112        [0.0003; 0.0610]         7.2
Number of studies: k = 8; Number of observations: o = 2007; Number of events: e = 30
                    proportion           95%-CI
Random effects model     0.0198 [0.0107; 0.0362]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.3764 [0.0000; 1.6458]; tau = 0.6135 [0.0000; 1.2829]
 I^2 = 55.5% [1.9%; 79.9%]; H = 1.50 [1.01; 2.23]
Test of heterogeneity:       Q d.f. p-value
                      15.75    7  0.0275
```

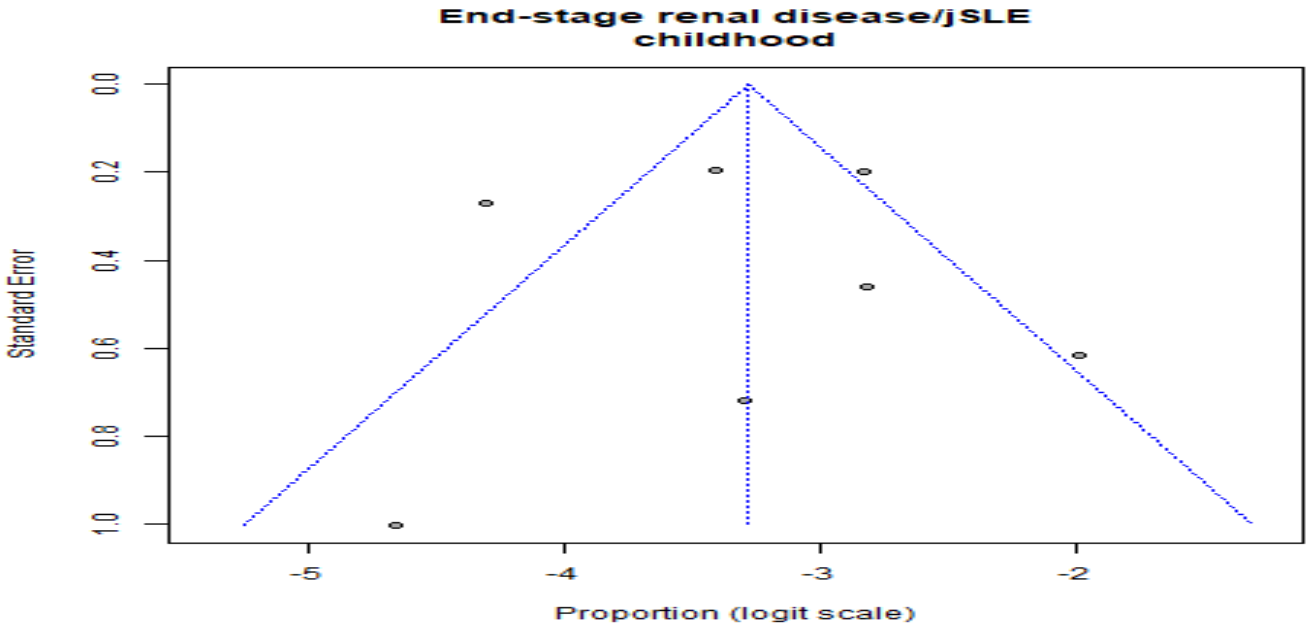
Supplementary Figure S4.42: Forest and Funnel Plots of studies reporting the prevalence of end-stage renal disease in children with juvenile-onset Systemic Lupus Erythematosus

End-stage renal disease

Forest



Funnel



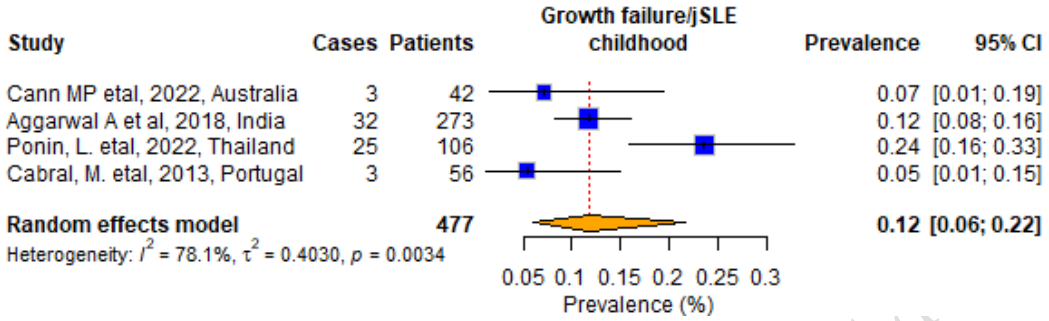
Summary statistics

```
                                    proportion       95%-CI          %W(random)
Holland , M. J. etal, 2018, USA         0.0134       [0.0073; 0.0223]        18.4
Medhat MB et al, 2020, Egypt            0.0094       [0.0002; 0.0514]        6.4
Cabral, M. etal, 2013, Portugal         0.0357       [0.0044; 0.1231]        9.8
Silva, M. F. etal, 2015, Brazil         0.0321       [0.0212; 0.0463]        19.8
Torrente-Segarra V et al, 2017, Spain   0.0558       [0.0371; 0.0801]        19.7
AlMayouf, SM. etal, 2008, Saudi Arabia  0.0562       [0.0185; 0.1263]        14.4
Costagliola G et al, 2018, Italy        0.1200       [0.0255; 0.3122]        11.5
Number of studies: k = 7; Number of observations: o = 2650; Number of events: e = 79
                    proportion           95%-CI
Random effects model     0.0362 [0.0203; 0.0639]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4311 [0.0842; 3.5598]; tau = 0.6566 [0.2901; 1.8867]
 I^2 = 78.0% [54.3%; 89.4%]; H = 2.13 [1.48; 3.07]
Test of heterogeneity:      Q d.f. p-value
                        27.24    6  0.0001
```

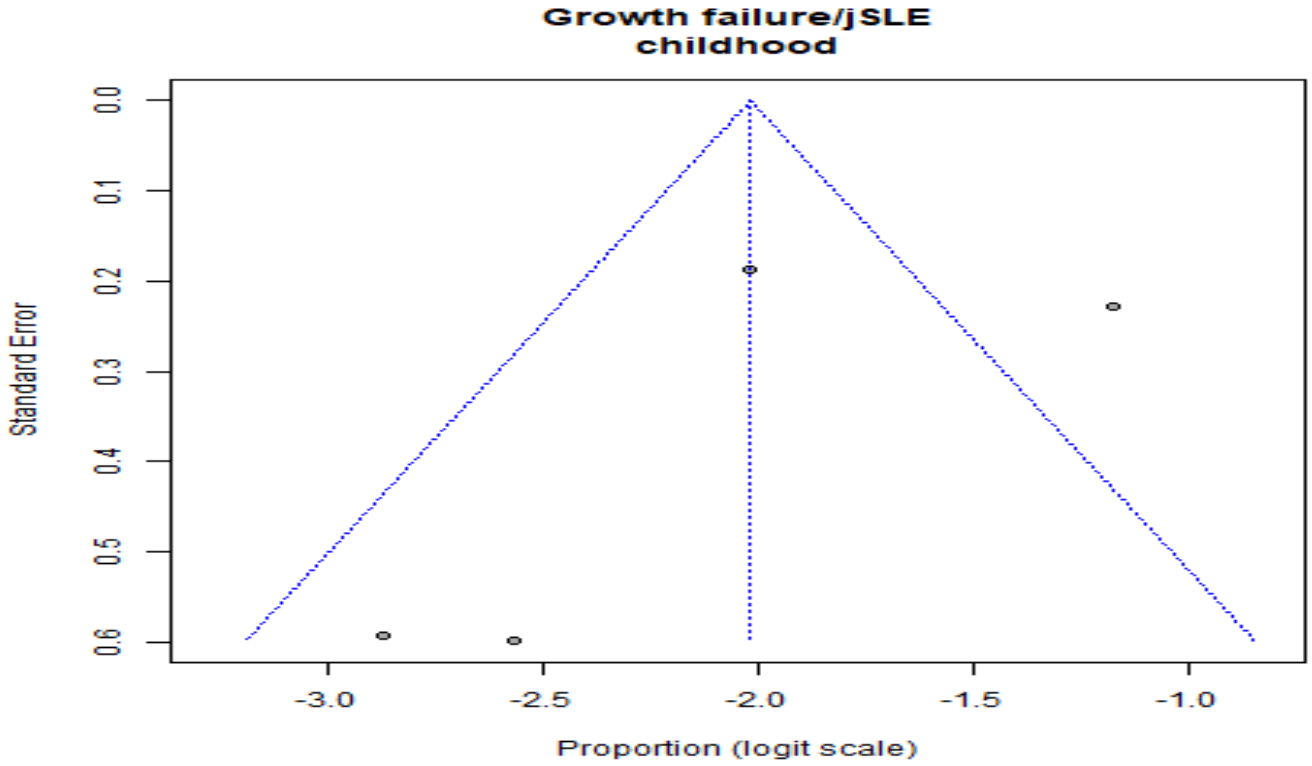
Supplementary Figure S4.43: Forest and Funnel Plots of studies reporting the prevalence of growth failure in children with juvenile-onset Systemic Lupus Erythematosus

Growth failure

Forest



Funnel



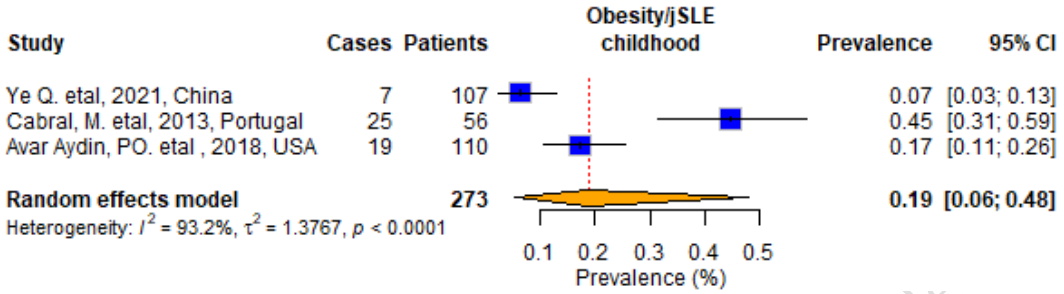
Summary statistics

```
                                proportion      95%-CI          %W(random)
Cann MP etal, 2022, Australia       0.0714      [0.0150; 0.1948]        18.4
Aggarwal A et al, 2018, India       0.1172      [0.0816; 0.1614]        32.1
Ponin, L. etal, 2022, Thailand      0.2358      [0.1588; 0.3282]        30.9
Cabral, M. etal, 2013, Portugal     0.0536      [0.0112; 0.1487]        18.6
Number of studies: k = 4; Number of observations: o = 477; Number of events: e = 63
                     proportion           95%-CI
Random effects model     0.1173 [0.0599; 0.2170]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4030 [0.0383; 7.5155]; tau = 0.6349 [0.1957; 2.7414]
 I^2 = 78.1% [40.8%; 91.9%]; H = 2.14 [1.30; 3.51]
Test of heterogeneity:      Q d.f. p-value
                        13.69    3  0.0034
```

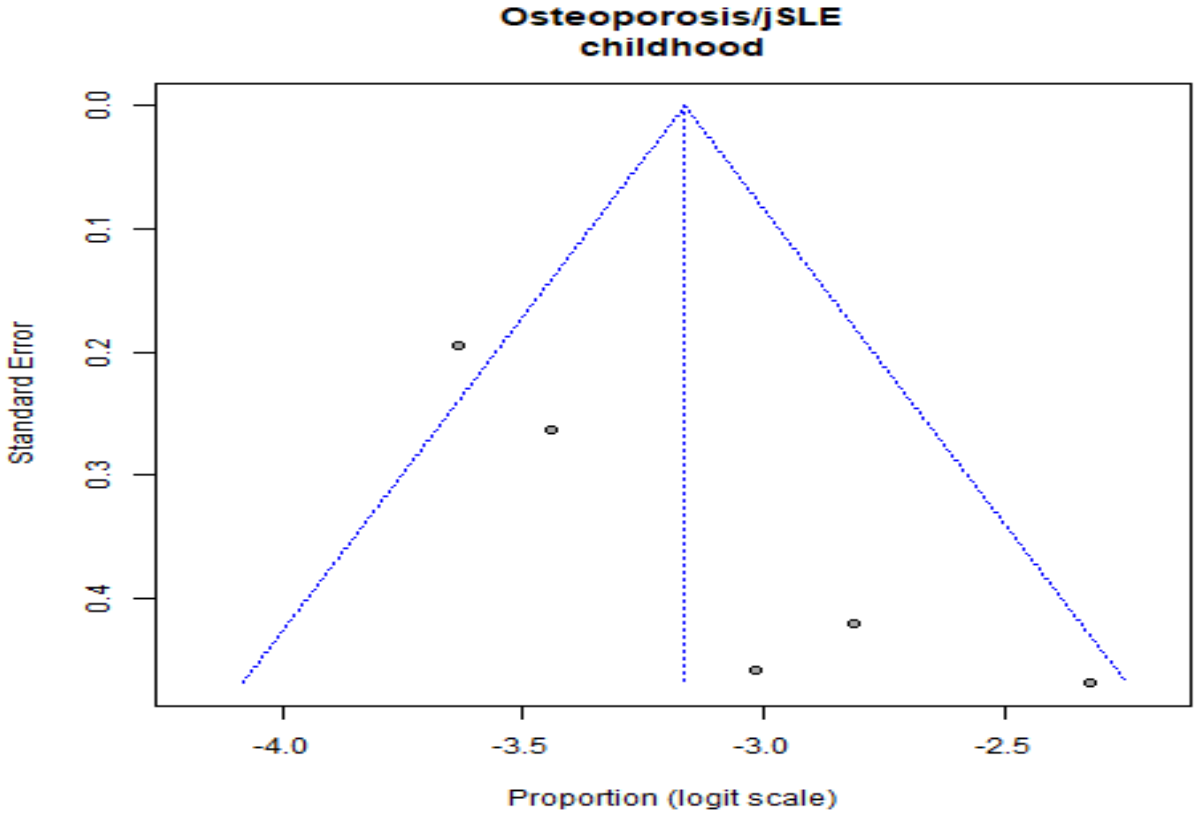
Supplementary Figure S4.44: Forest and Funnel Plots of studies reporting the prevalence of obesity in children with juvenile-onset Systemic Lupus Erythematosus

Obesity

Forest



Funnel



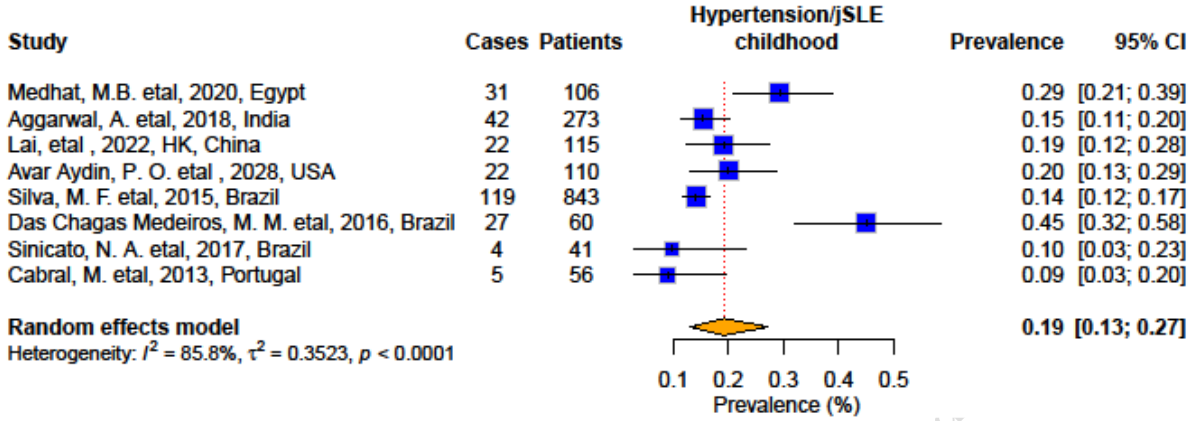
Summary statistics

```
                                proportion     95%-CI          %W(random)
Ye Q. etal, 2021, China             0.0654     [0.0267; 0.1302]        32.1
Cabral, M. etal, 2013, Portugal     0.4464     [0.3134; 0.5853]        33.9
Avar Aydin, PO. etal , 2018, USA    0.1727     [0.1073; 0.2565]        34.1
Number of studies: k = 3; Number of observations: o = 273; Number of events: e = 51
                     proportion           95%-CI
Random effects model     0.1886 [0.0556; 0.4784]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.3767 [0.2984; 59.0990]; tau = 1.1733 [0.5463; 7.6876]
 I^2 = 93.2% [83.5%; 97.2%]; H = 3.83 [2.46; 5.97]
Test of heterogeneity:      Q d.f.  p-value
                        29.39    2 < 0.0001
```

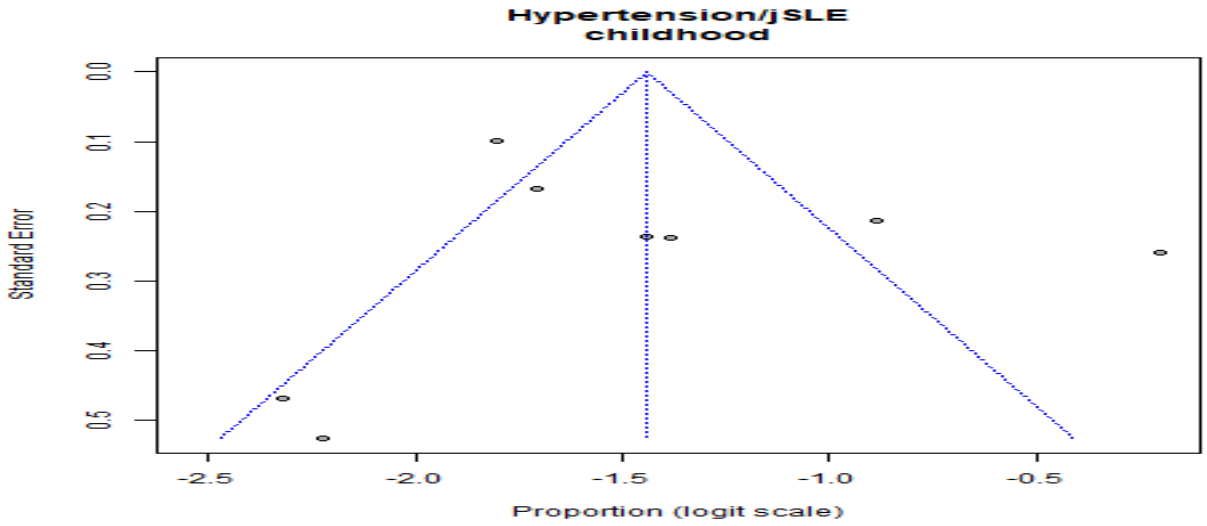
Supplementary Figure S4.45: Forest and Funnel Plots of studies reporting the prevalence of hypertension in children with juvenile-onset Systemic Lupus Erythematosus

Hypertension

Forest



Funnel



Summary statistics

```
                                                proportion         95% CI                 %W(random)
Medhat, M.B. etal, 2020, Egypt                      0.2925       [0.2081; 0.3887]         13.6
Aggarwal, A. etal, 2018, India                      0.1538       [0.1132; 0.2022]         14.2
Lai, etal , 2022, HK, China                         0.1913       [0.1239; 0.2752]         13.2
Avar Aydin, P. O. etal , 2028, USA                  0.2000       [0.1298; 0.2870]         13.2
Silva, M. F. etal, 2015, Brazil                     0.1412       [0.1184; 0.1665]         14.9
Das Chagas Medeiros, M. M. etal, 2016, Brazil       0.4500       [0.3212; 0.5839]         12.9
Sinicato, N. A. etal, 2017, Brazil                  0.0976       [0.0272; 0.2313]         8.6
Cabral, M. etal, 2013, Portugal                     0.0893       [0.0296; 0.1962]         9.4

Number of studies: k = 8
Number of observations: o = 1604
Number of events: e = 272

                        proportion 95% CI
Random effects model        0.1914 [0.1305; 0.2718]

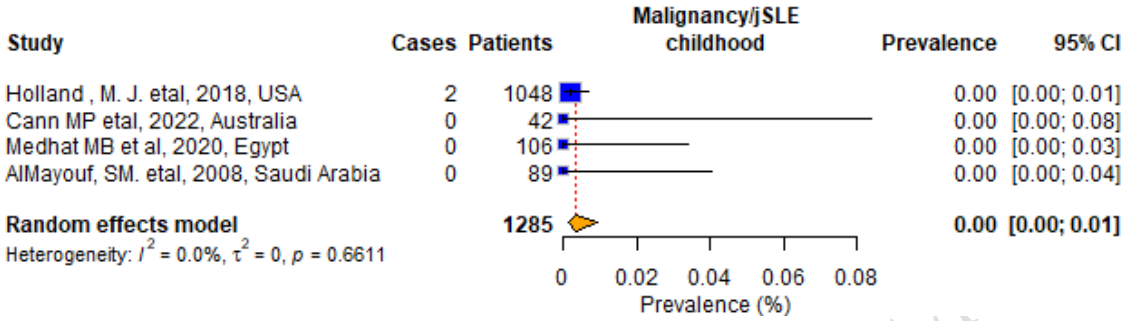
Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.3523 [0.1124; 1.9017]; tau = 0.5936 [0.3352; 1.3790]
 I^2 = 85.8% [74.0%; 92.3%]; H = 2.65 [1.96; 3.59]

Test of heterogeneity:
     Q d.f.  p-value
 49.34    7 < 0.0001
```

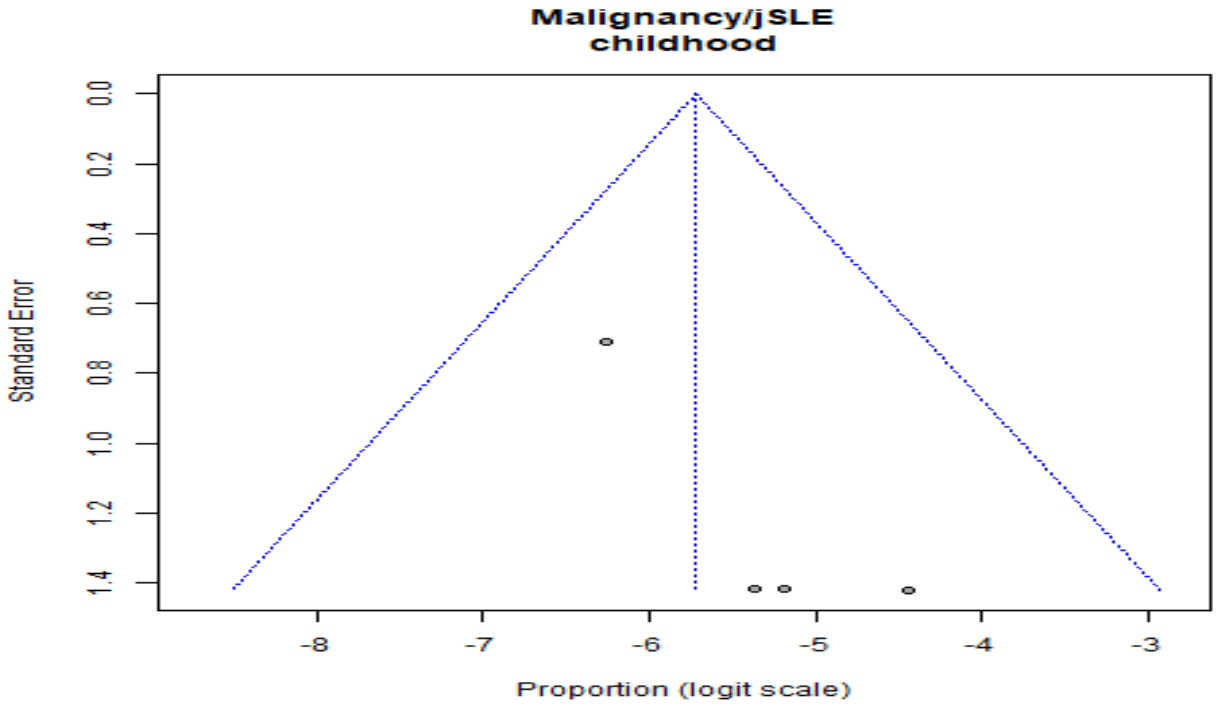
Supplementary Figure S4.46: Forest and Funnel Plots of studies reporting the prevalence of malignancy in children with juvenile-onset Systemic Lupus Erythematosus

Malignancy

Forest



Funnel



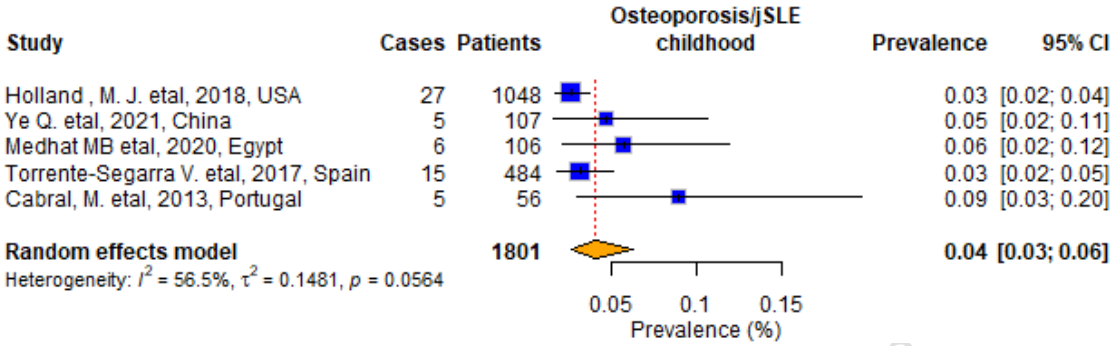
Summary statistics

```
                                        proportion       95%-CI          %W(random)
Holland , M. J. etal, 2018, USA             0.0019       [0.0002; 0.0069]        57.3
Cann MP etal, 2022, Australia               0.0000       [0.0000; 0.0841]        14.2
Medhat MB et al, 2020, Egypt                0.0000       [0.0000; 0.0342]        14.3
AlMayouf, SM. etal, 2008, Saudi Arabia      0.0000       [0.0000; 0.0406]        14.3
Number of studies: k = 4; Number of observations: o = 1285; Number of events: e = 2
                     proportion           95%-CI
Random effects model     0.0033 [0.0011; 0.0093]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 6.5718]; tau = 0 [0.0000; 2.5635]
 I^2 = 0.0% [0.0%; 84.7%]; H = 1.00 [1.00; 2.56]
Test of heterogeneity:     Q d.f. p-value
                        1.59    3  0.6611
```

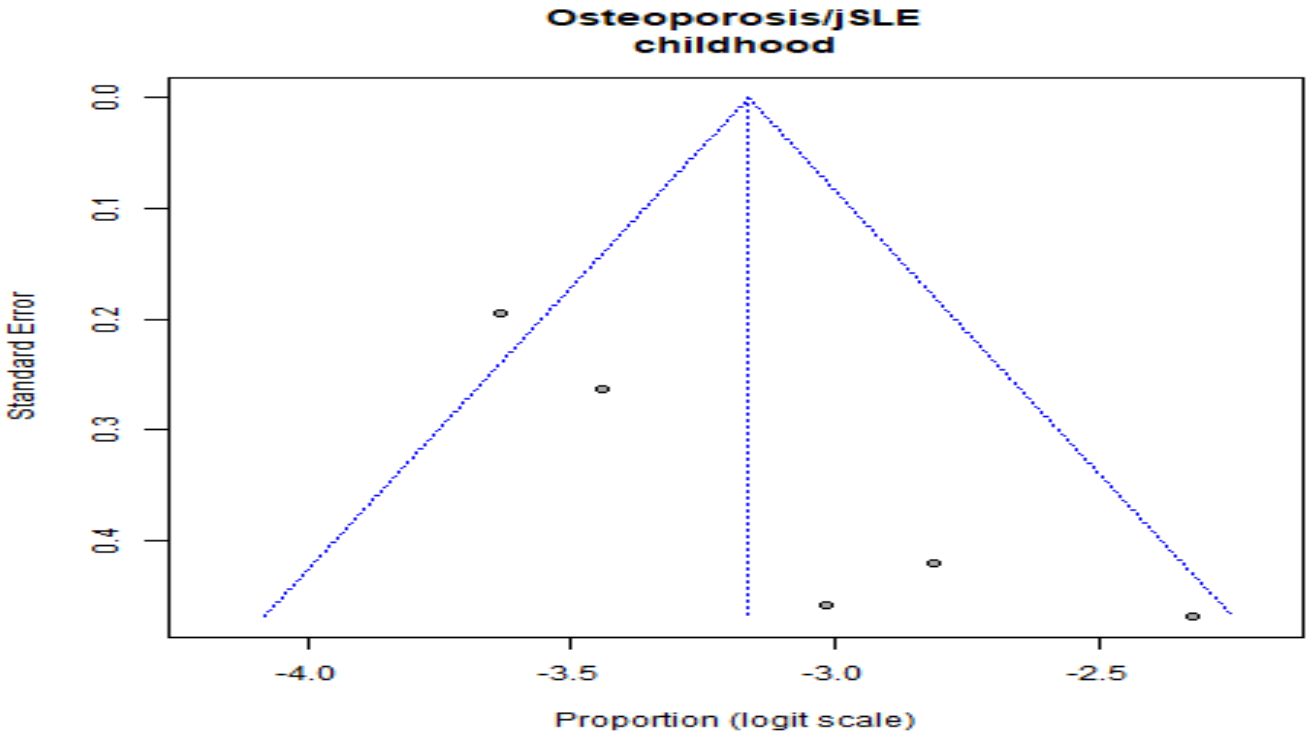
Supplementary Figure S4.47: Forest and Funnel Plots of studies reporting the prevalence of osteoporosis in children with juvenile-onset Systemic Lupus Erythematosus

Osteoporosis

Forest



Funnel



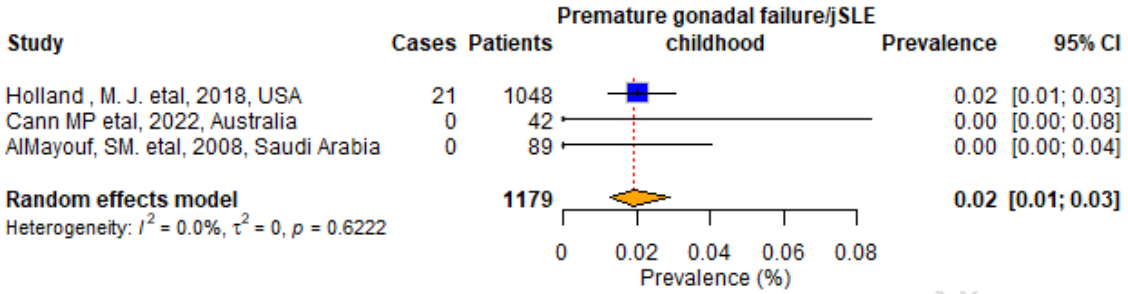
Summary statistics

```
                                    proportion        95%-CI          %W(random)
Holland , M. J. etal, 2018, USA         0.0258       [0.0170; 0.0373]        28.9
Ye Q. etal, 2021, China                 0.0467       [0.0153; 0.1057]        15.0
Medhat MB etal, 2020, Egypt             0.0566       [0.0211; 0.1191]        16.6
Torrente-Segarra V. etal, 2017, Spain   0.0310       [0.0174; 0.0506]        24.8
Cabral, M. etal, 2013, Portugal         0.0893       [0.0296; 0.1962]        14.6
Number of studies: k = 5; Number of observations: o = 1801; Number of events: e = 58
                     proportion           95%-CI
Random effects model     0.0405 [0.0261; 0.0624]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.1481 [0.0000; 2.0934]; tau = 0.3848 [0.0000; 1.4469]
 I^2 = 56.5% [0.0%; 83.9%]; H = 1.52 [1.00; 2.49]
Test of heterogeneity:     Q d.f. p-value
                        9.19    4  0.0564
```

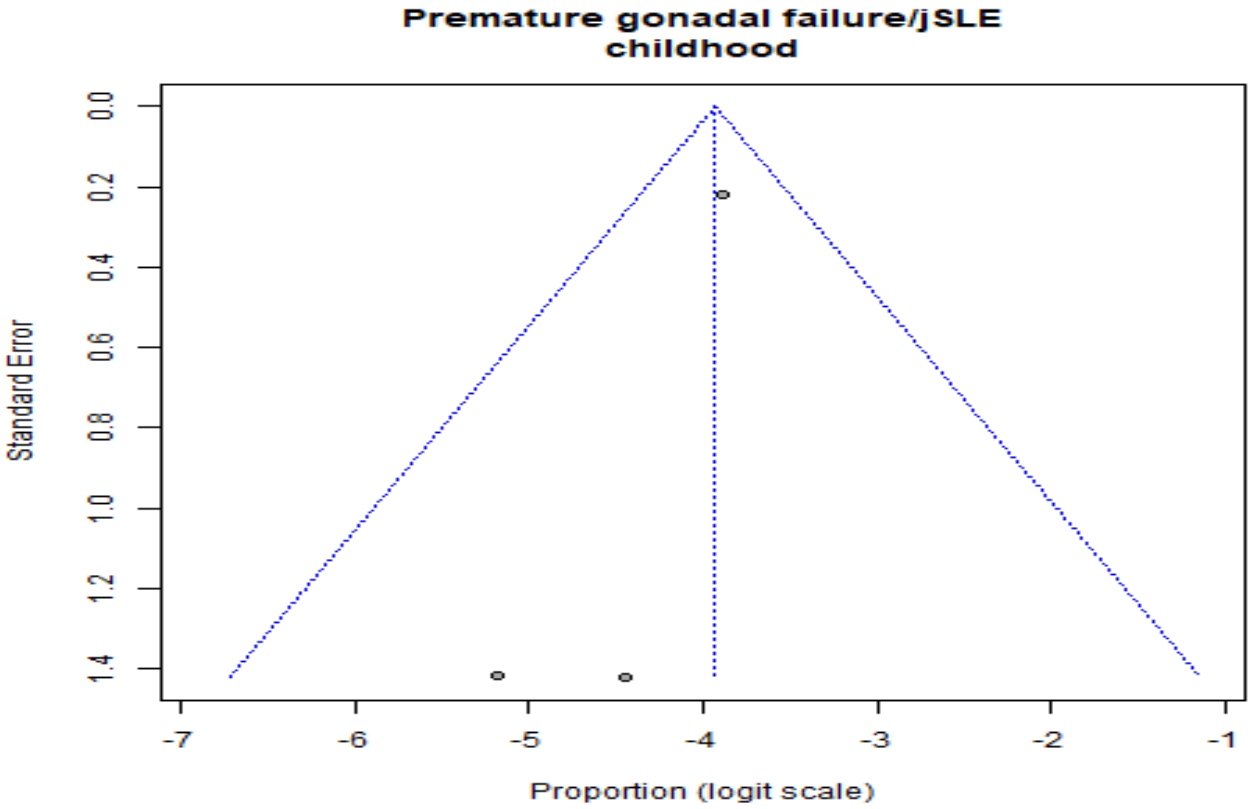
Supplementary Figure S4.48: Forest and Funnel Plots of studies reporting the prevalence of premature gonadal failure in children with juvenile-onset Systemic Lupus Erythematosus

Premature gonadal failure

Forest



Funnel



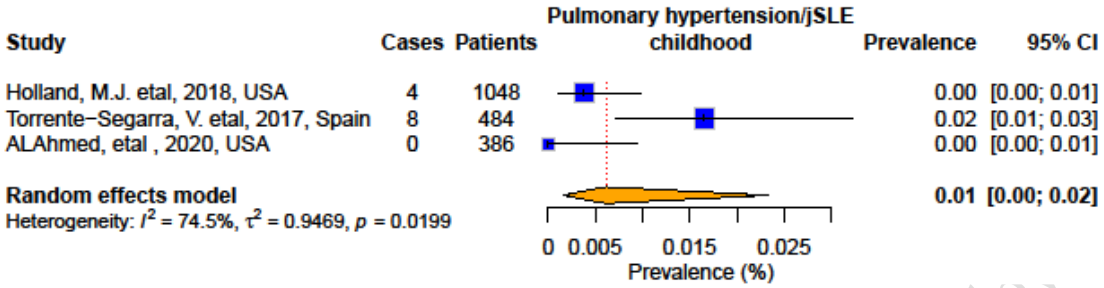
Summary statistics

```
                                        proportion       95%-CI          %W(random)
Holland , M. J. etal, 2018, USA             0.0200       [0.0124; 0.0305]        95.4
Cann MP etal, 2022, Australia               0.0000       [0.0000; 0.0841]        2.3
AlMayouf, SM. etal, 2008, Saudi Arabia      0.0000       [0.0000; 0.0406]        2.3
Number of studies: k = 3; Number of observations: o = 1179; Number of events: e = 21
                      proportion           95%-CI
Random effects model      0.0192 [0.0127; 0.0290]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 15.6365]; tau = 0 [0.0000; 3.9543]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        0.95    2  0.6222
```

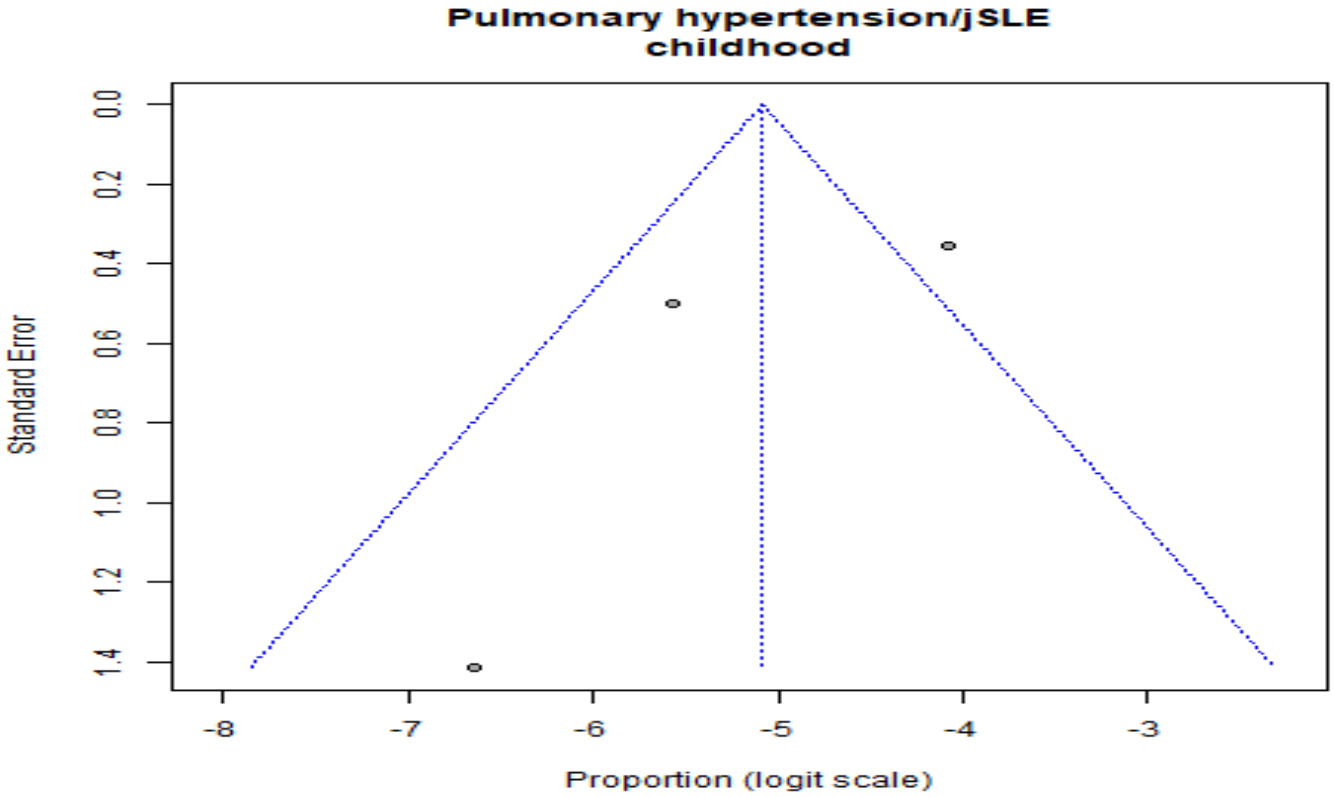
Supplementary Figure S4.49: Forest and Funnel Plots of studies reporting the prevalence of pulmonary hypertension in children with juvenile-onset Systemic Lupus Erythematosus

Pulmonary hypertension

Forest



Funnel



Summary statistics

```
                                          proportion           95%-CI       %W(random)
Holland, MJ. etal, 2018, USA                  0.0038 [0.0010; 0.0097]       39.7
Torrente-Segarra, V. etal, 2017, Spain        0.0165 [0.0072; 0.0323]       44.2
ALAhmed, etal , 2020, USA                     0.0000 [0.0000; 0.0095]       16.1

Number of studies: k = 3
Number of observations: o = 1918
Number of events: e = 12

                         proportion           95%-CI
Random effects model         0.0061 [0.0016; 0.0233]

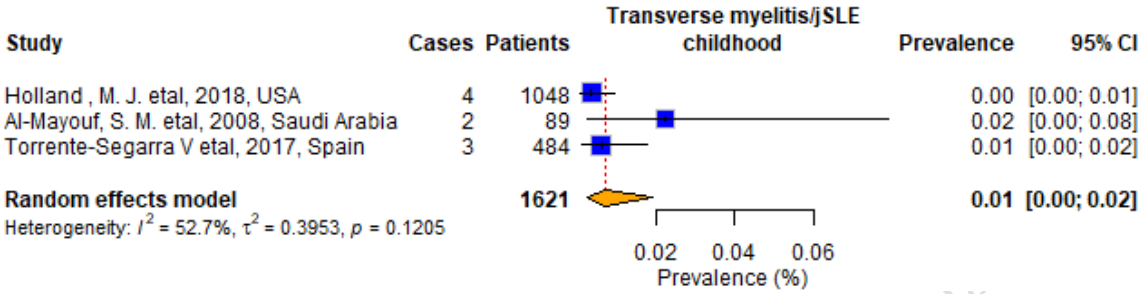
Quantifying heterogeneity (with 95%-CIs):
 tau^2 = 0.9469 [0.0140; 64.4784]; tau = 0.9731 [0.1183; 8.0298]
 I^2 = 74.5% [15.0%; 92.3%]; H = 1.98 [1.08; 3.61]

Test of heterogeneity:
    Q d.f. p-value
 7.83     2  0.0199
```

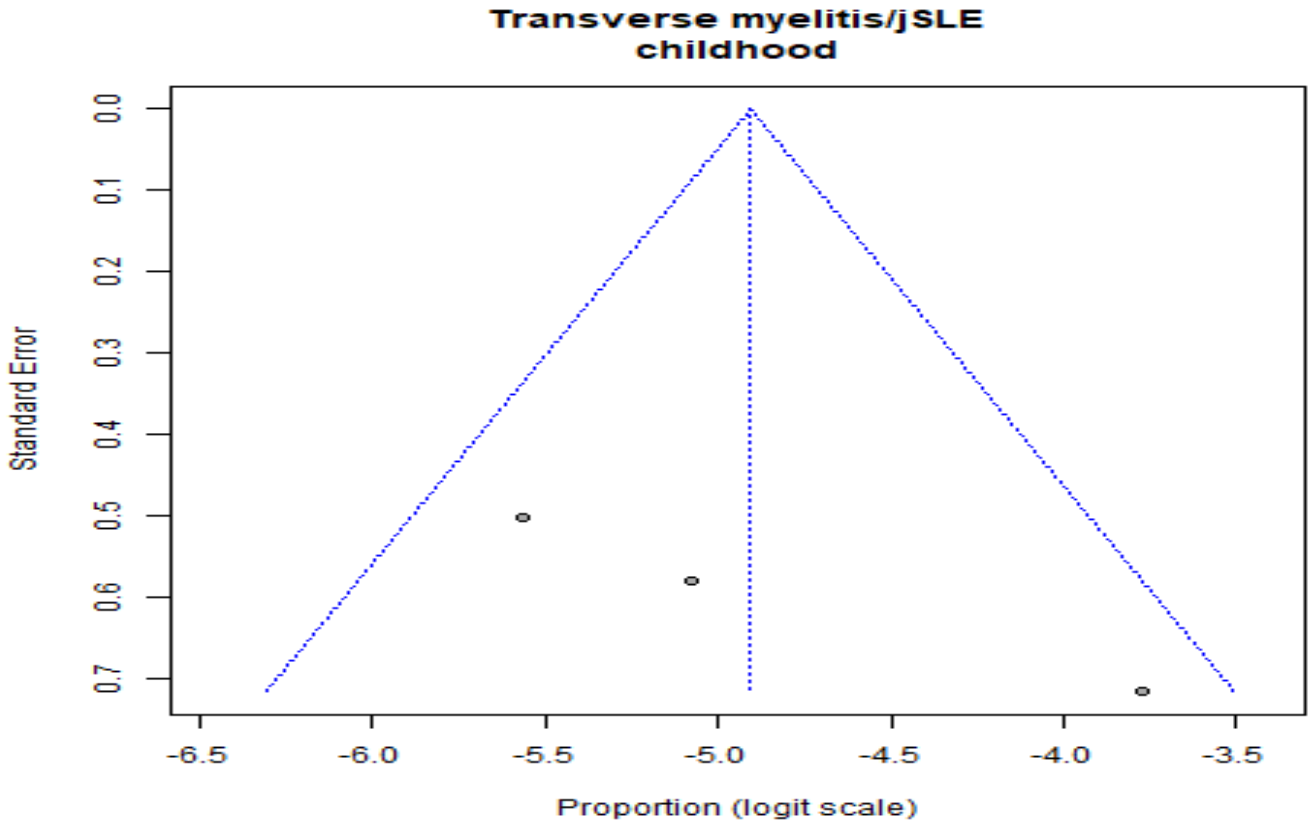
Supplementary Figure S4.50: Forest and Funnel Plots of studies reporting the prevalence of transverse myelitis in children with juvenile-onset Systemic Lupus Erythematosus

Transverse myelitis

Forest



Funnel



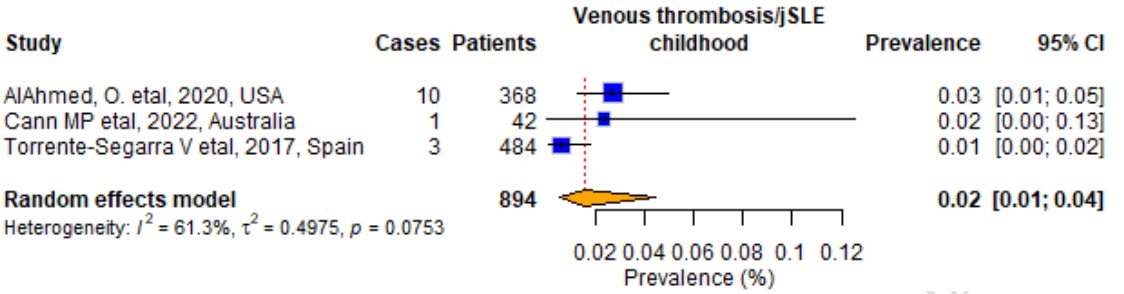
Summary statistics

```
                                        proportion    95%-CI          %W(random)
Holland , M. J. etal, 2018, USA             0.0038    [0.0010; 0.0097]        38.5
Al-Mayouf, S. M. etal, 2008, Saudi Arabia   0.0225    [0.0027; 0.0788]        27.4
Torrente-Segarra V etal, 2017, Spain        0.0062    [0.0013; 0.0180]        34.1
Number of studies: k = 3; Number of observations: o = 1621; Number of events: e = 9
                     proportion           95%-CI
Random effects model     0.0073 [0.0028; 0.0193]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.3953 [0.0000; 33.4831]; tau = 0.6287 [0.0000; 5.7865]
 I^2 = 52.7% [0.0%; 86.4%]; H = 1.45 [1.00; 2.71]
Test of heterogeneity:     Q d.f. p-value
                        4.23    2  0.1205
```

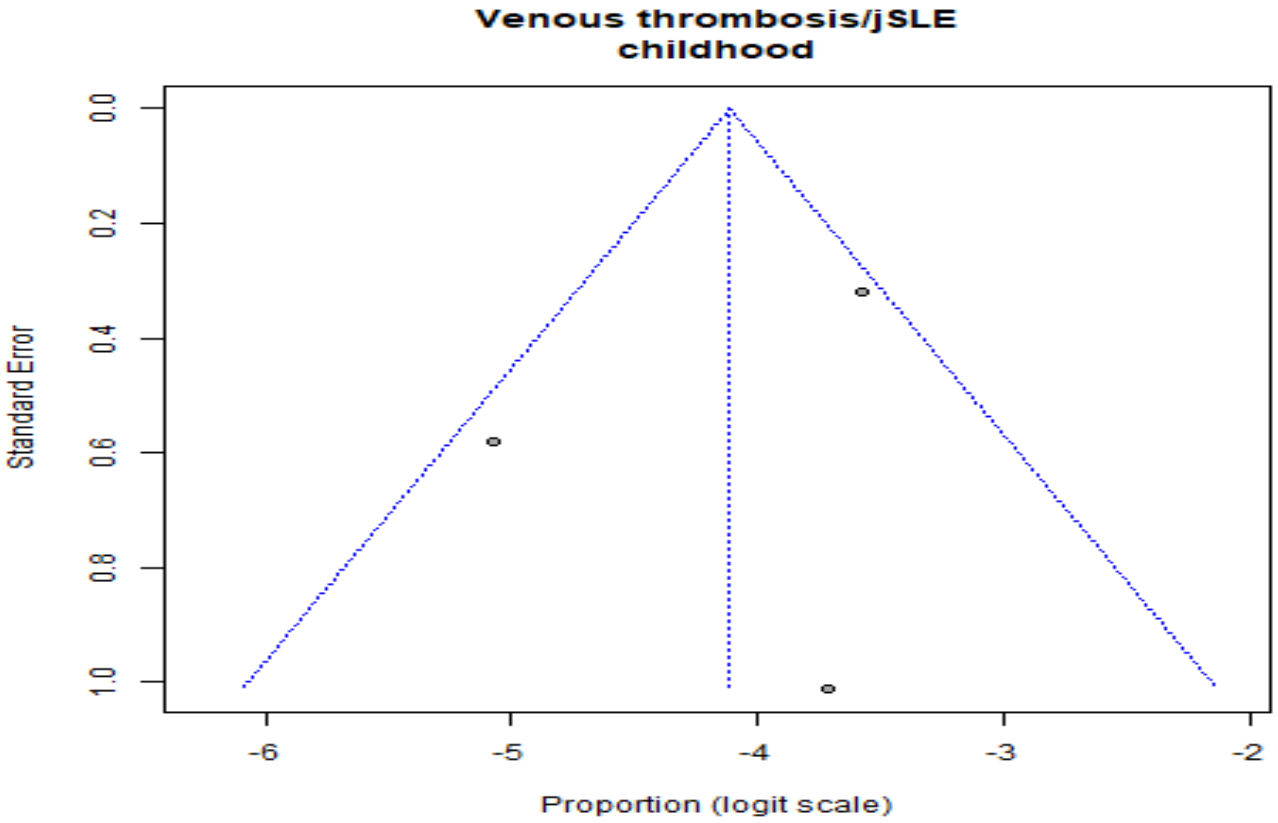
Supplementary Figure S4.51: Forest and Funnel Plots of studies reporting the prevalence of venous thrombosis in children with juvenile-onset Systemic Lupus Erythematosus

Venous thrombosis

Forest



Funnel



Summary statistics

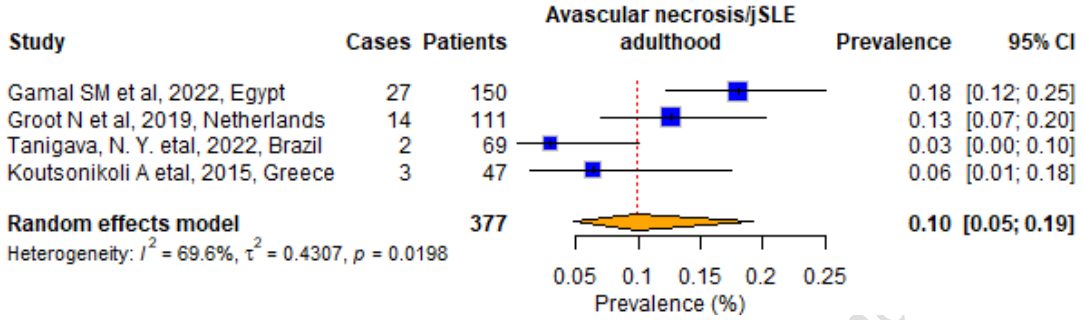
```
                                  proportion         95%-CI  %W(random)
AlAhmed, O. etal, 2020, USA           0.0272 [0.0131; 0.0494]       47.3
Cann MP etal, 2022, Australia         0.0238 [0.0006; 0.1257]       18.6
Torrente-Segarra V etal, 2017, Spain  0.0062 [0.0013; 0.0180]       34.1
Number of studies: k = 3; Number of observations: o = 894; Number of events: e = 14
                     proportion           95%-CI
Random effects model     0.0161 [0.0057; 0.0444]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4975 [0.0000; 26.7942]; tau = 0.7053 [0.0000; 5.1763]
 I^2 = 61.3% [0.0%; 89.0%]; H = 1.61 [1.00; 3.01]
Test of heterogeneity:     Q d.f. p-value
                        5.17    2  0.0753
```

Adulthood/jSLE

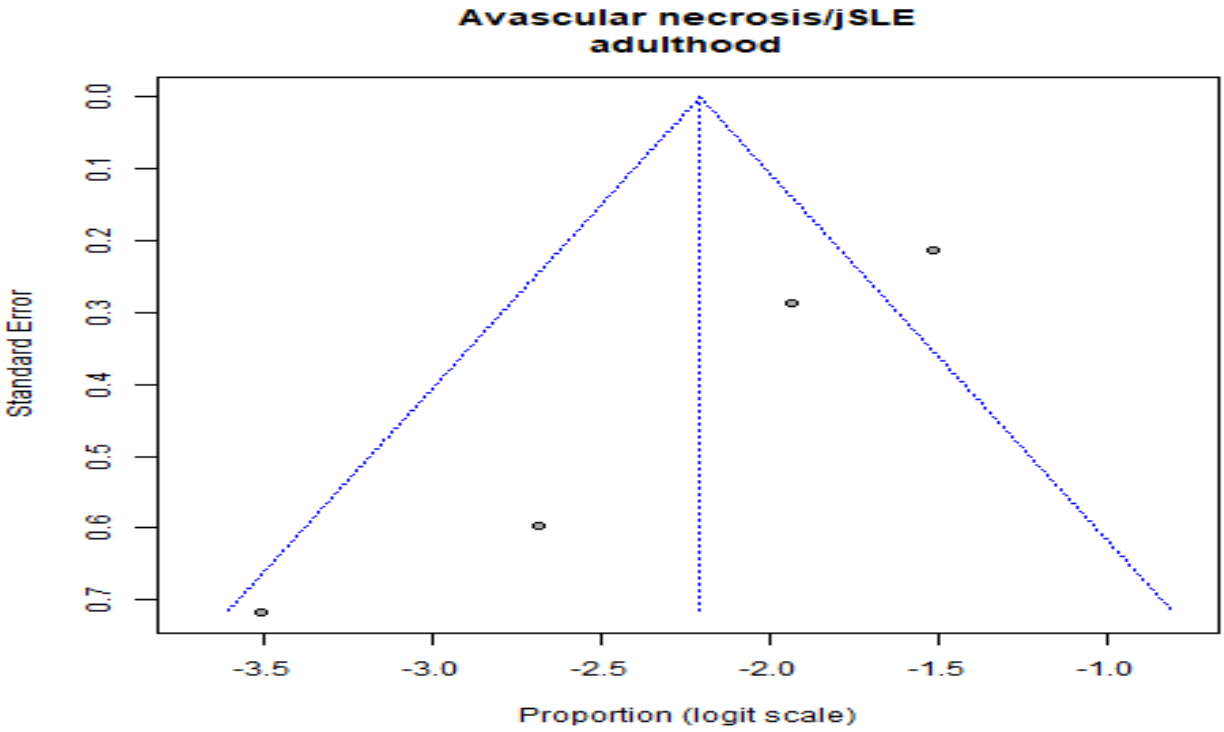
Supplementary Figure S4.52: Forest and Funnel Plots of studies reporting the prevalence of avascular necrosis in adults with juvenile-onset Systemic Lupus Erythematosus

Avascular necrosis

Forest



Funnel



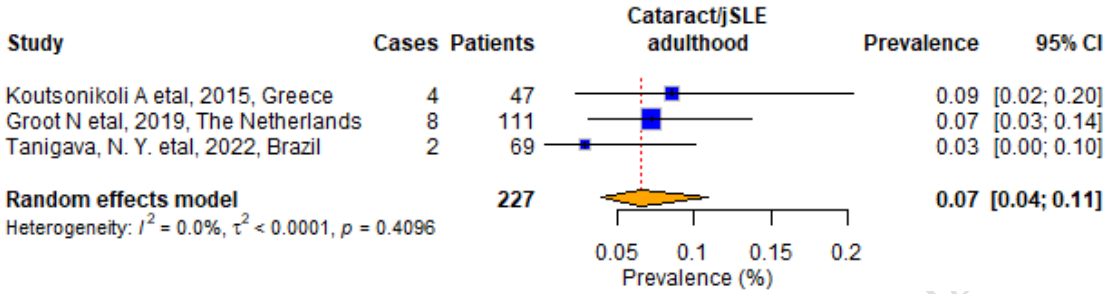
Summary statistics

```
                                proportion   95%-CI          %W(random)
Gamal SM et al, 2022, Egypt         0.1800   [0.1221; 0.2510]       32.9
Groot N et al, 2019, Netherlands    0.1261   [0.0707; 0.2026]       30.6
Tanigava, N. Y. etal, 2022, Brazil  0.0290   [0.0035; 0.1008]       16.6
Koutsonikoli A etal, 2015, Greece   0.0638   [0.0134; 0.1754]       19.9
Number of studies: k = 4; Number of observations: o = 377; Number of events: e = 46
                    proportion           95%-CI
Random effects model     0.0990 [0.0482; 0.1928]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4307 [0.0084; 10.4154]; tau = 0.6563 [0.0918; 3.2273]
 I^2 = 69.6% [12.4%; 89.4%]; H = 1.81 [1.07; 3.08]
Test of heterogeneity:     Q d.f. p-value
                        9.86    3  0.0198
```

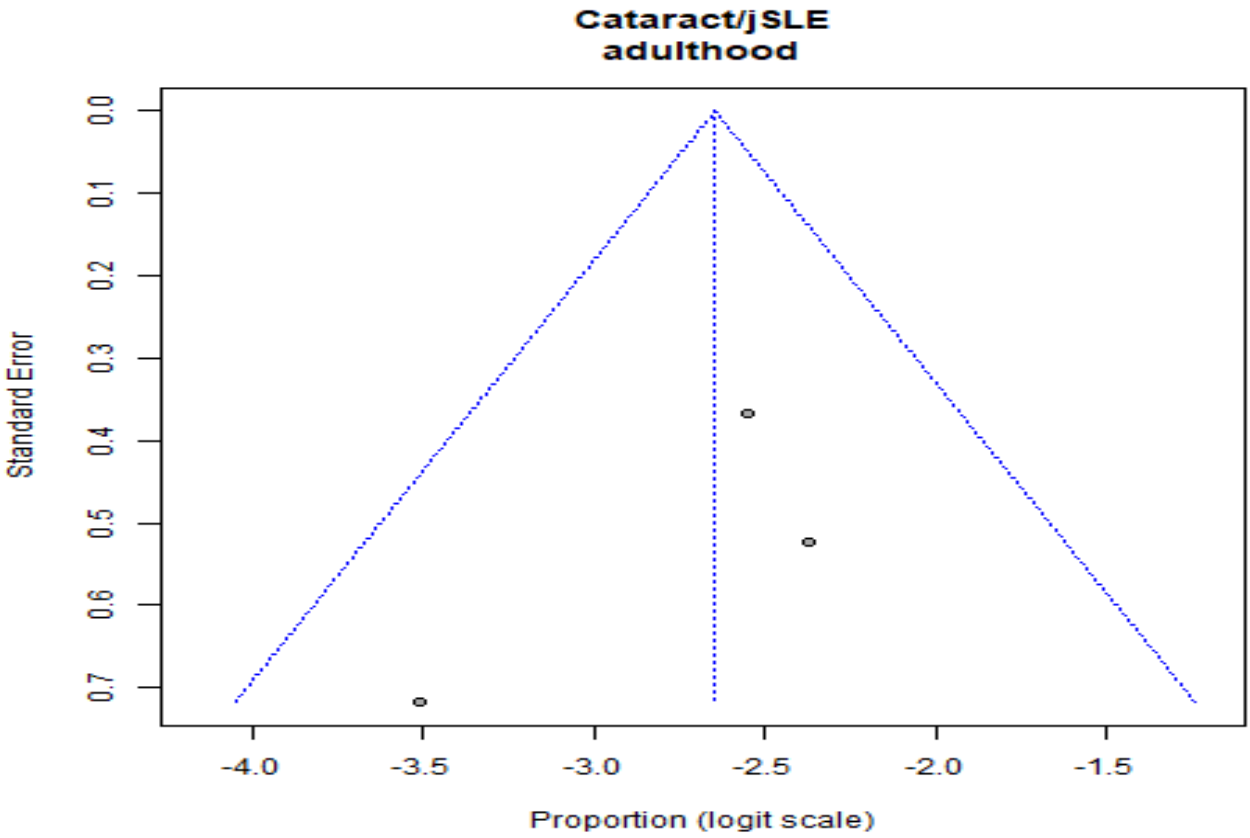
Supplementary Figure S4.53: Forest and Funnel Plots of studies reporting the prevalence of cataract in adults with juvenile-onset Systemic Lupus Erythematosus

Cataract

Forest



Funnel



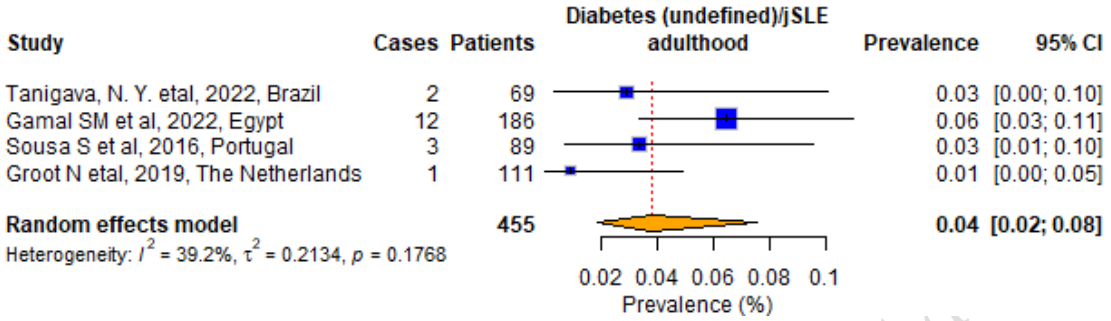
Summary statistics

```
                                    proportion  95%-CI          %W(random)
Koutsonikoli A etal, 2015, Greece       0.0851  [0.0237; 0.2038]        28.1
Groot N etal, 2019, The Netherlands     0.0721  [0.0316; 0.1371]        57.0
Tanigava, N. Y. etal, 2022, Brazil      0.0290  [0.0035; 0.1008]        14.9
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 14
                     proportion           95%-CI
Random effects model     0.0662 [0.0395; 0.1087]
Quantifying heterogeneity (with 95%-CIs):  tau^2 < 0.0001 [0.0000; 14.3203]; tau = 0.0004 [0.0000; 3.7842]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        1.78    2  0.4096
```

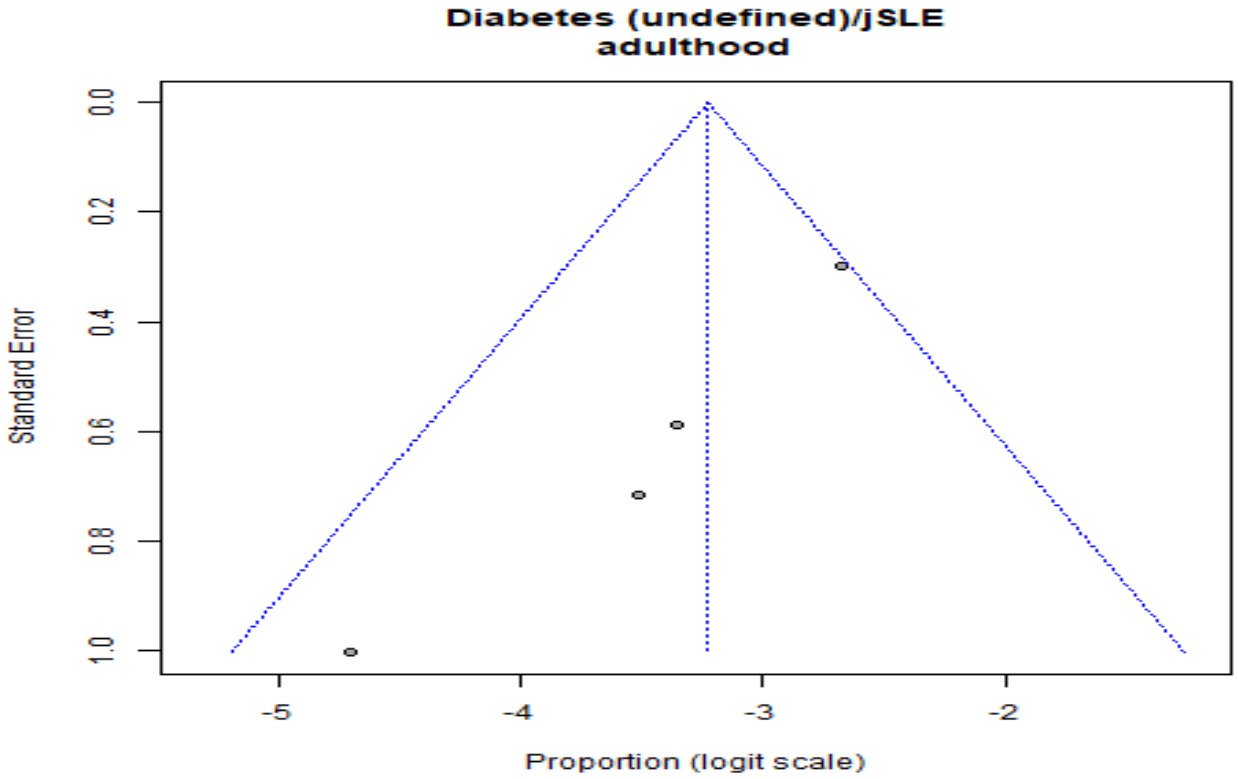
Supplementary Figure S4.54: Forest and Funnel Plots of studies reporting the prevalence of diabetes (undefined) in adults with juvenile-onset Systemic Lupus Erythematosus

Diabetes (undefined)

Forest



Funnel



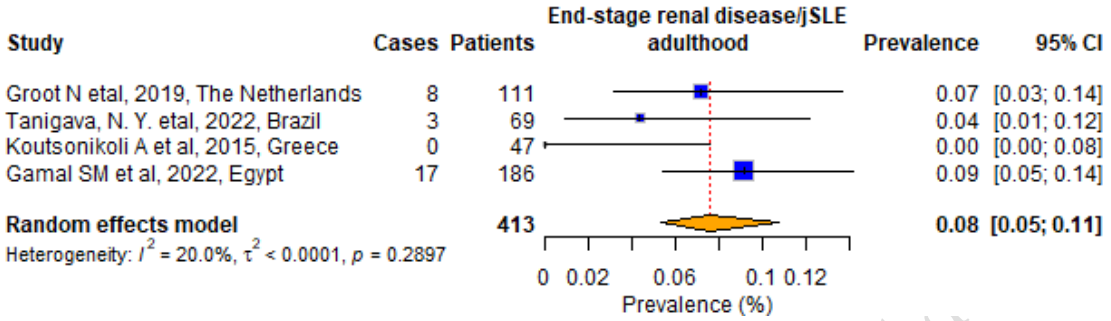
Summary statistics

```
                                  proportion  95%-CI          %W(random)
Tanigava, N. Y. etal, 2022, Brazil     0.0290  [0.0035; 0.1008]       18.8
Gamal SM et al, 2022, Egypt            0.0645  [0.0338; 0.1100]       45.4
Sousa S et al, 2016, Portugal          0.0337  [0.0070; 0.0954]       24.6
Groot N etal, 2019, The Netherlands    0.0090  [0.0002; 0.0492]       11.2
Number of studies: k = 4; Number of observations: o = 455; Number of events: e = 18
                    proportion           95%-CI
Random effects model     0.0382 [0.0188; 0.0758]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.2134 [0.0000; 9.2209]; tau = 0.4620 [0.0000; 3.0366]
 I^2 = 39.2% [0.0%; 79.3%]; H = 1.28 [1.00; 2.20]
Test of heterogeneity:     Q d.f. p-value
                        4.93    3  0.1768
```

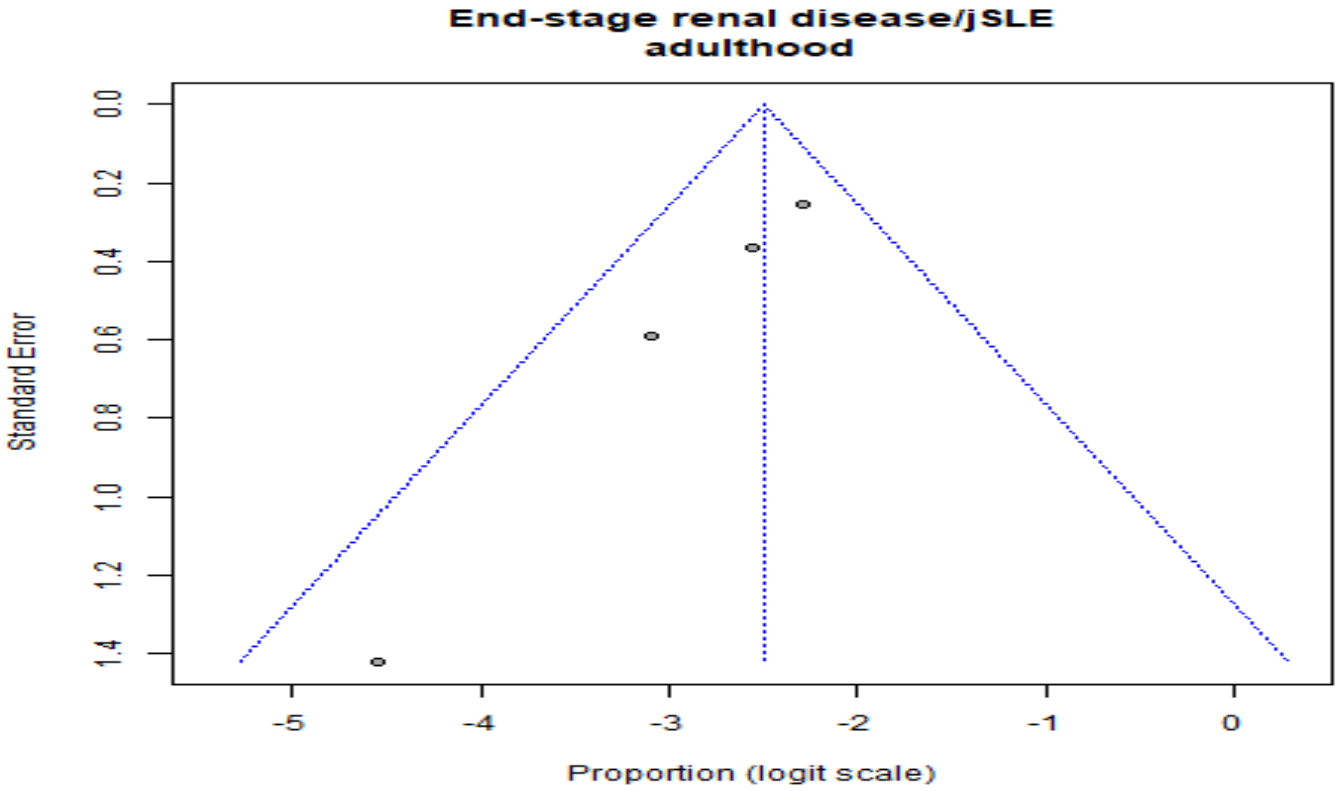
Supplementary Figure S4.55: Forest and Funnel Plots of studies reporting the prevalence of end-stage renal disease in adults with juvenile-onset Systemic Lupus Erythematosus

End-stage renal disease

Forest



Funnel



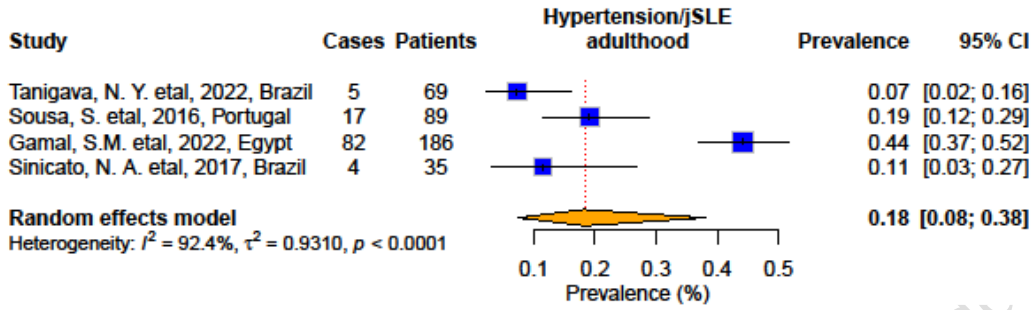
Summary statistics

```
                                  proportion  95%-CI          %W(random)
Groot N etal, 2019, The Netherlands     0.0721  [0.0316; 0.1371]       28.3
Tanigava, N. Y. etal, 2022, Brazil      0.0435  [0.0091; 0.1218]       10.9
Koutsonikoli A et al, 2015, Greece      0.0000  [0.0000; 0.0755]        1.9
Gamal SM et al, 2022, Egypt             0.0914  [0.0541; 0.1423]       58.9
Number of studies: k = 4; Number of observations: o = 413; Number of events: e = 28
                     proportion           95%-CI
Random effects model     0.0759 [0.0531; 0.1075]
Quantifying heterogeneity (with 95%-CIs):  tau^2 < 0.0001 [0.0000; 12.7842]; tau = 0.0006 [0.0000; 3.5755]
 I^2 = 20.0% [0.0%; 87.8%]; H = 1.12 [1.00; 2.86]
Test of heterogeneity:     Q d.f. p-value
                        3.75    3  0.2897
```

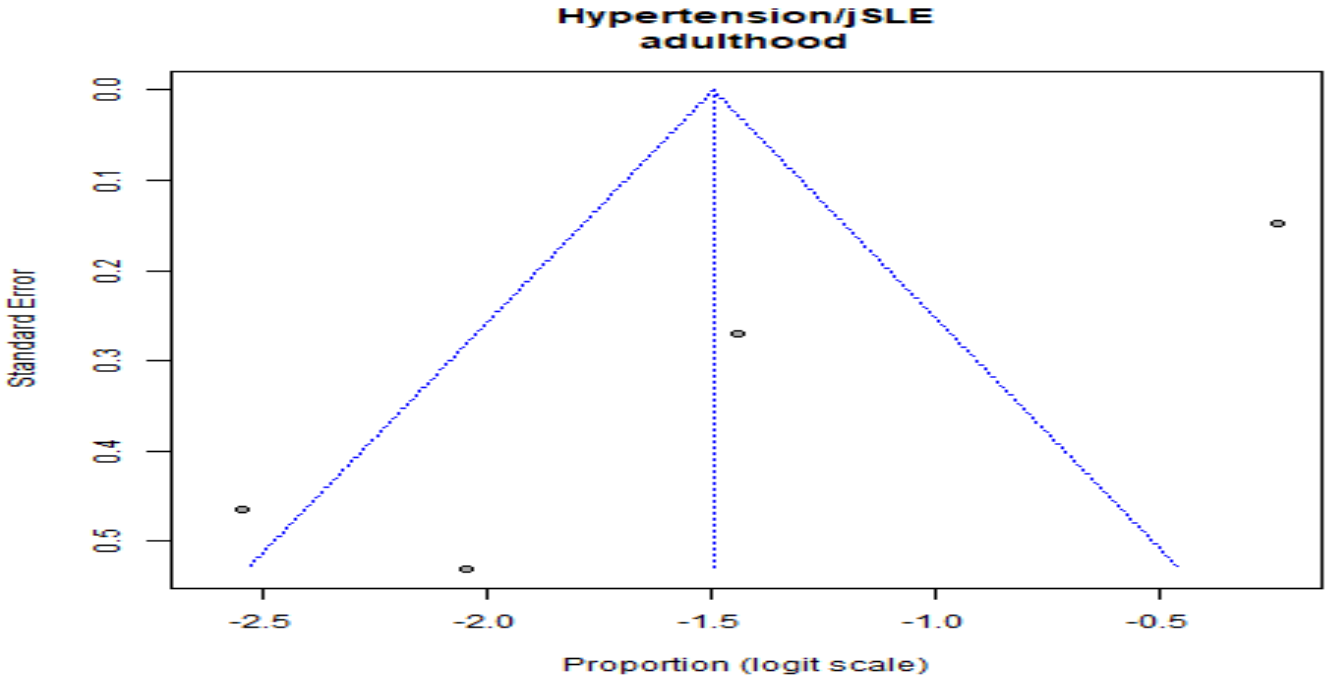
Supplementary Figure S4.56: Forest and Funnel Plots of studies reporting the prevalence of hypertension in adults with juvenile-onset Systemic Lupus Erythematosus

Hypertension

Forest



Funnel



Summary statistics

```
                                    proportion   95% CI                  %W(random)
Tanigava, N. Y. etal, 2022, Brazil      0.0725   [0.0239; 0.1611]        23.3
Sousa, S. etal, 2016, Portugal          0.1910   [0.1154; 0.2881]        26.6
Gamal, S.M. etal, 2022, Egypt           0.4409   [0.3683; 0.5154]        28.0
Sinicato, N. A. etal, 2017, Brazil      0.1143   [0.0320; 0.2674]        22.0

Number of studies: k = 4
Number of observations: o = 379
Number of events: e = 108

                        proportion 95% CI
Random effects model        0.1830 [0.0752; 0.3816]

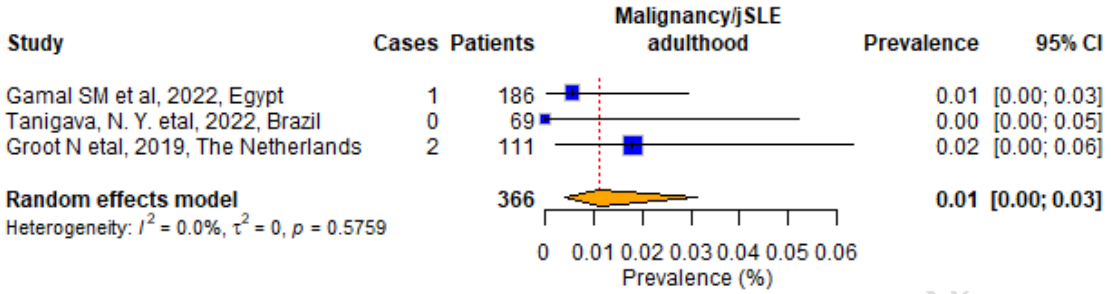
Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.9310 [0.2222; 13.6986]; tau = 0.9649 [0.4714; 3.7012]
 I^2 = 92.4% [83.8%; 96.5%]; H = 3.63 [2.49; 5.31]

Test of heterogeneity:
     Q d.f.  p-value
 39.62    3 < 0.0001
```

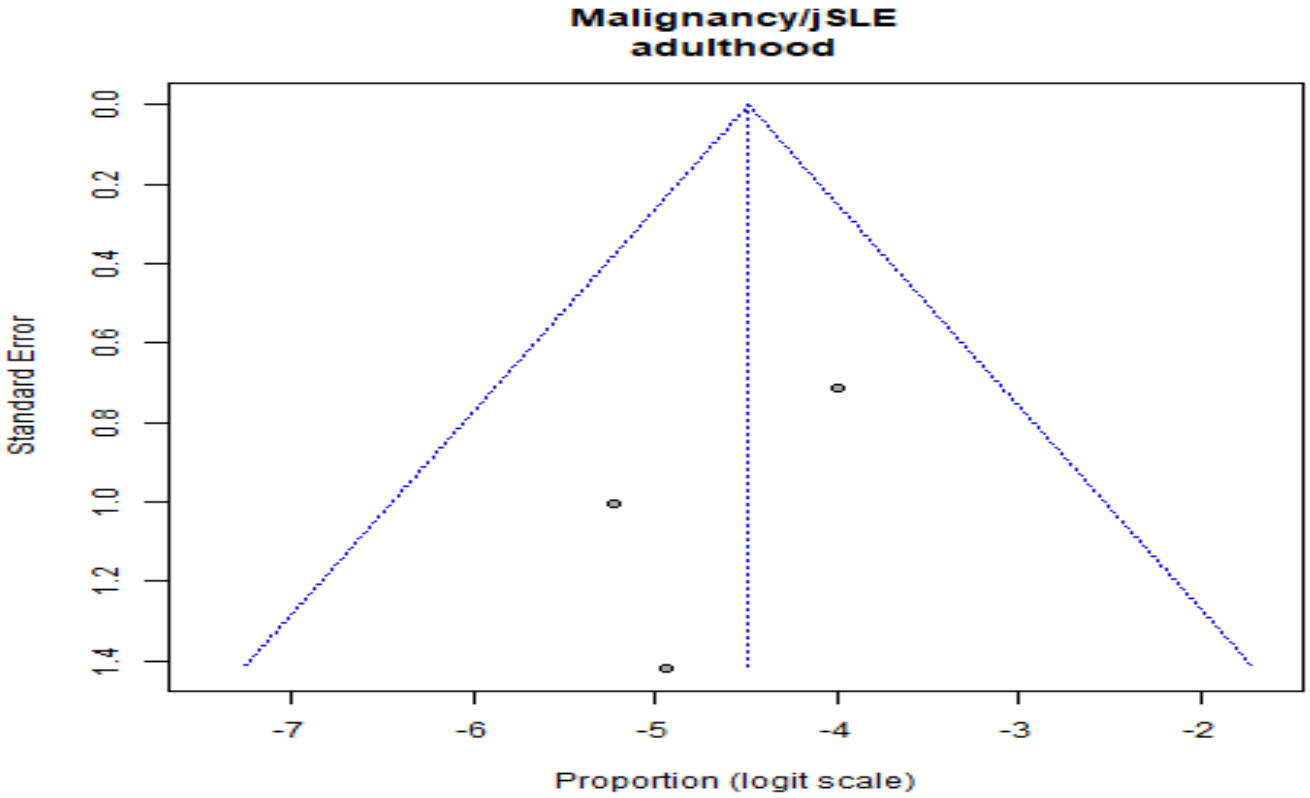
Supplementary Figure S4.57: Forest and Funnel Plots of studies reporting the prevalence of malignancy in adults with juvenile-onset Systemic Lupus Erythematosus

Malignancy

Forest



Funnel



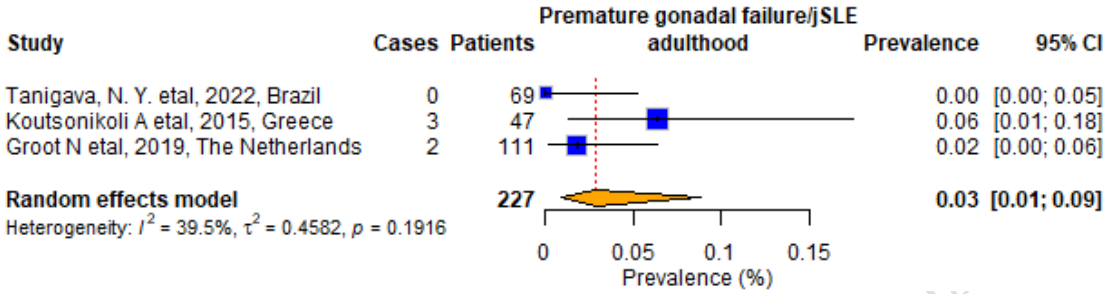
Summary statistics

```
                                  proportion  95%-CI          %W(random)
Gamal SM et al, 2022, Egypt           0.0054  [0.0001; 0.0296]       28.8
Tanigava, N. Y. etal, 2022, Brazil    0.0000  [0.0000; 0.0521]       14.4
Groot N etal, 2019, The Netherlands   0.0180  [0.0022; 0.0636]       56.8
Number of studies: k = 3; Number of observations: o = 366; Number of events: e = 3
                     proportion           95%-CI
Random effects model     0.0112 [0.0039; 0.0314]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 15.3931]; tau = 0 [0.0000; 3.9234]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        1.10    2  0.5759
```

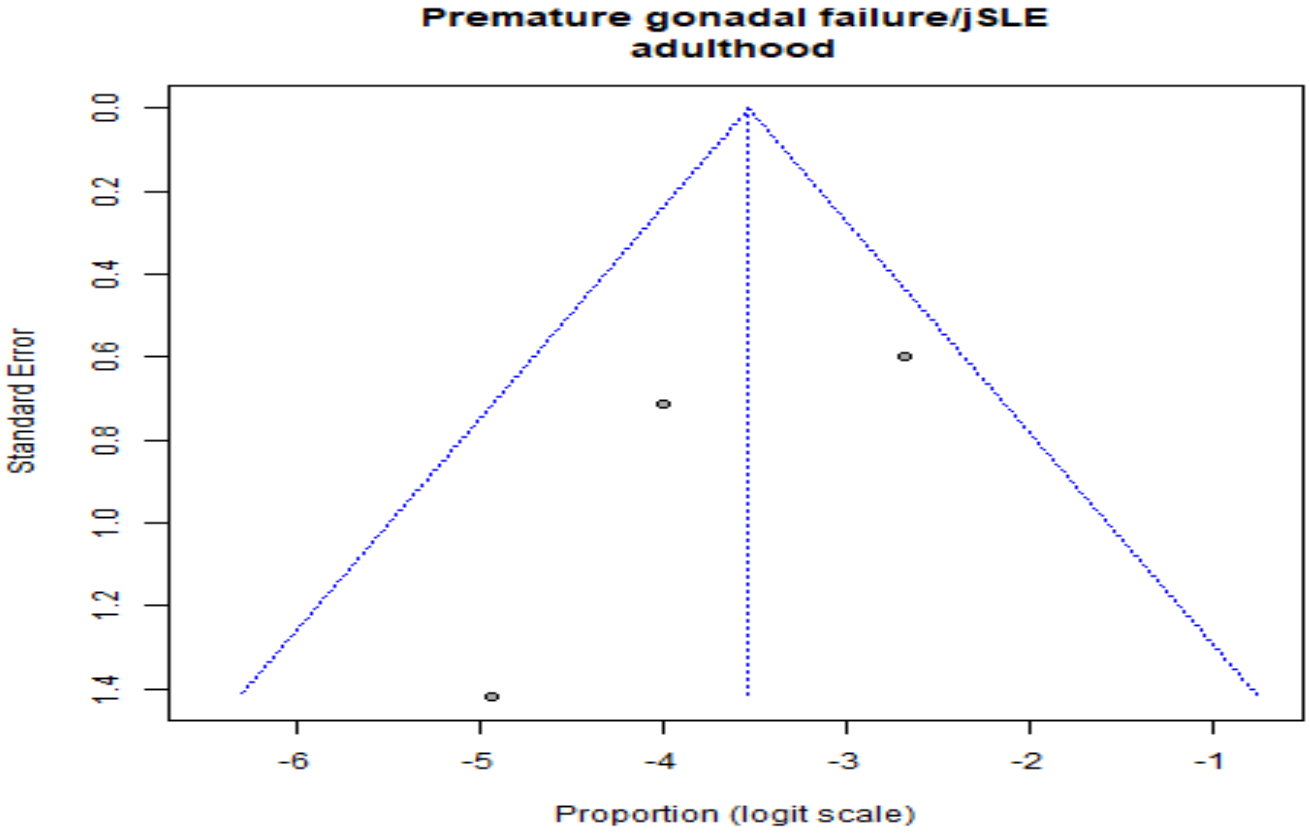
Supplementary Figure S4.58: Forest and Funnel Plots of studies reporting the prevalence of premature gonadal failure in adults with juvenile-onset Systemic Lupus Erythematosus

Premature gonadal failure

Forest



Funnel



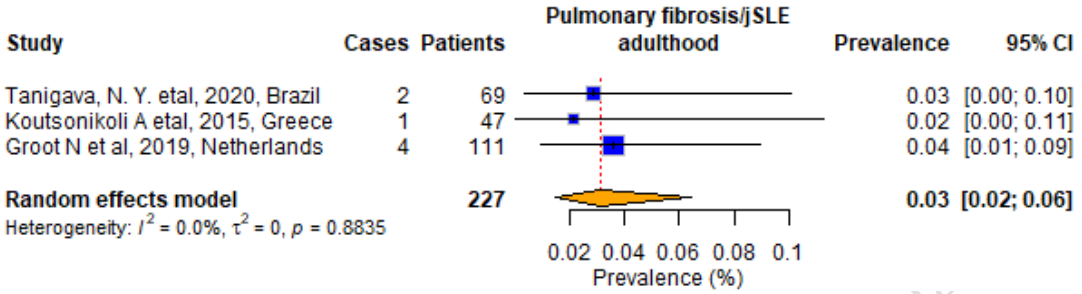
Summary statistics

```
                                  proportion  95%-CI          %W(random)
Tanigava, N. Y. etal, 2022, Brazil     0.0000  [0.0000; 0.0521]       15.2
Koutsonikoli A etal, 2015, Greece      0.0638  [0.0134; 0.1754]       46.1
Groot N etal, 2019, The Netherlands    0.0180  [0.0022; 0.0636]       38.8
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 5
                    proportion           95%-CI
Random effects model     0.0283 [0.0087; 0.0882]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4582 [0.0000; 49.3226]; tau = 0.6769 [0.0000; 7.0230]
 I^2 = 39.5% [0.0%; 81.3%]; H = 1.29 [1.00; 2.31]
Test of heterogeneity:     Q d.f. p-value
                        3.30    2  0.1916
```

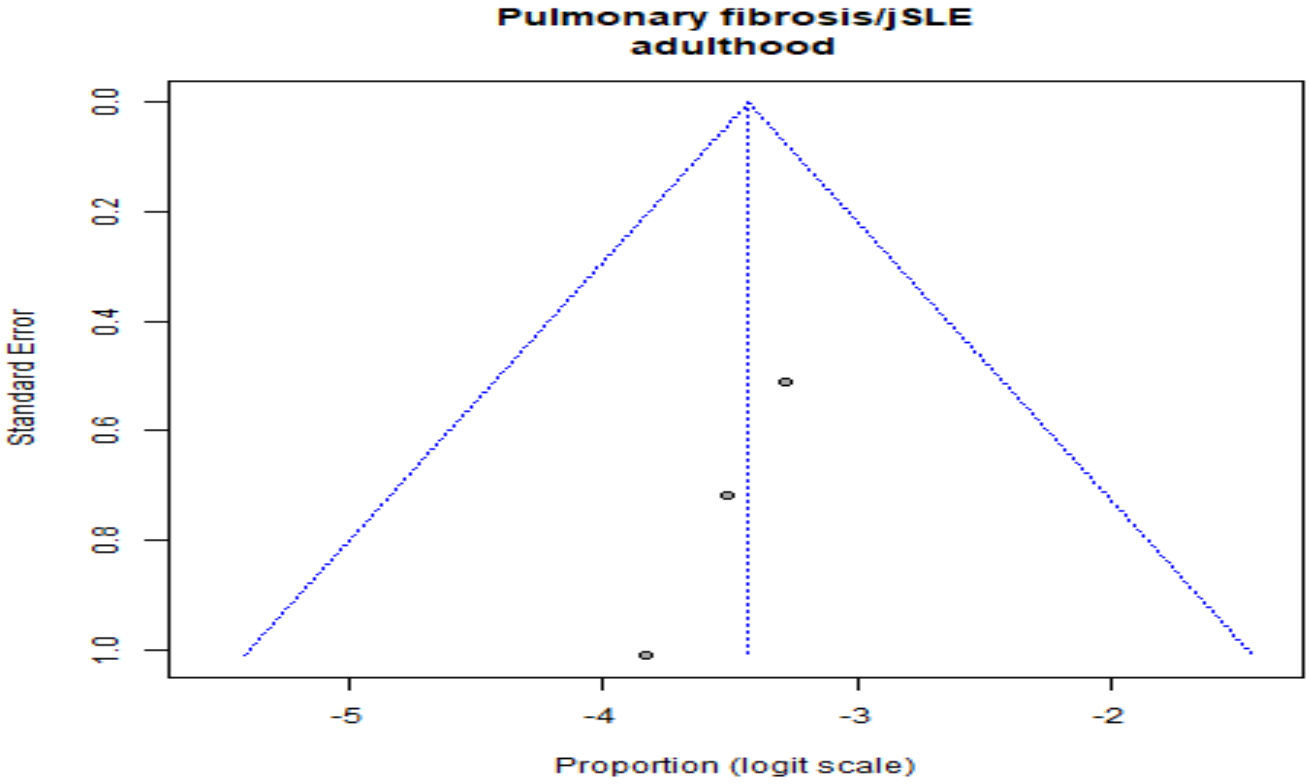
Supplementary Figure S4.59: Forest and Funnel Plots of studies reporting the prevalence of pulmonary fibrosis in adults with juvenile-onset Systemic Lupus Erythematosus

Pulmonary fibrosis

Forest



Funnel



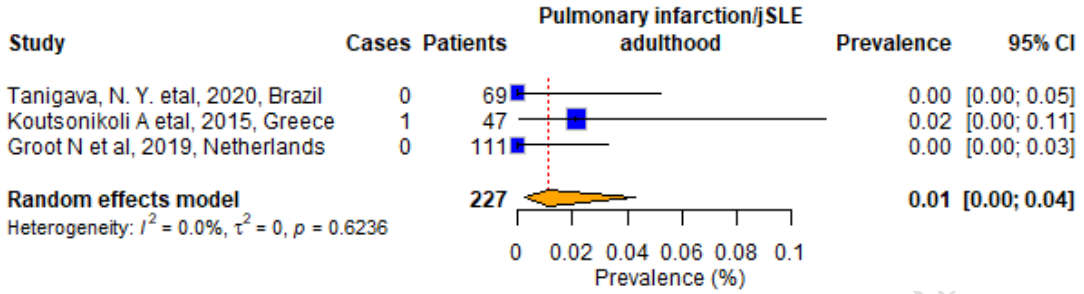
Summary statistics

```
                                    proportion   95%-CI          %W(random)
Tanigava, N. Y. etal, 2020, Brazil     0.0290   [0.0035; 0.1008]       28.7
Koutsonikoli A etal, 2015, Greece      0.0213   [0.0005; 0.1129]       14.4
Groot N et al, 2019, Netherlands       0.0360   [0.0099; 0.0897]       56.9
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 7
                     proportion           95%-CI
Random effects model     0.0314 [0.0150; 0.0644]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 2.2629]; tau = 0 [0.0000; 1.5043]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:    Q d.f. p-value
                       0.25    2  0.8835
```

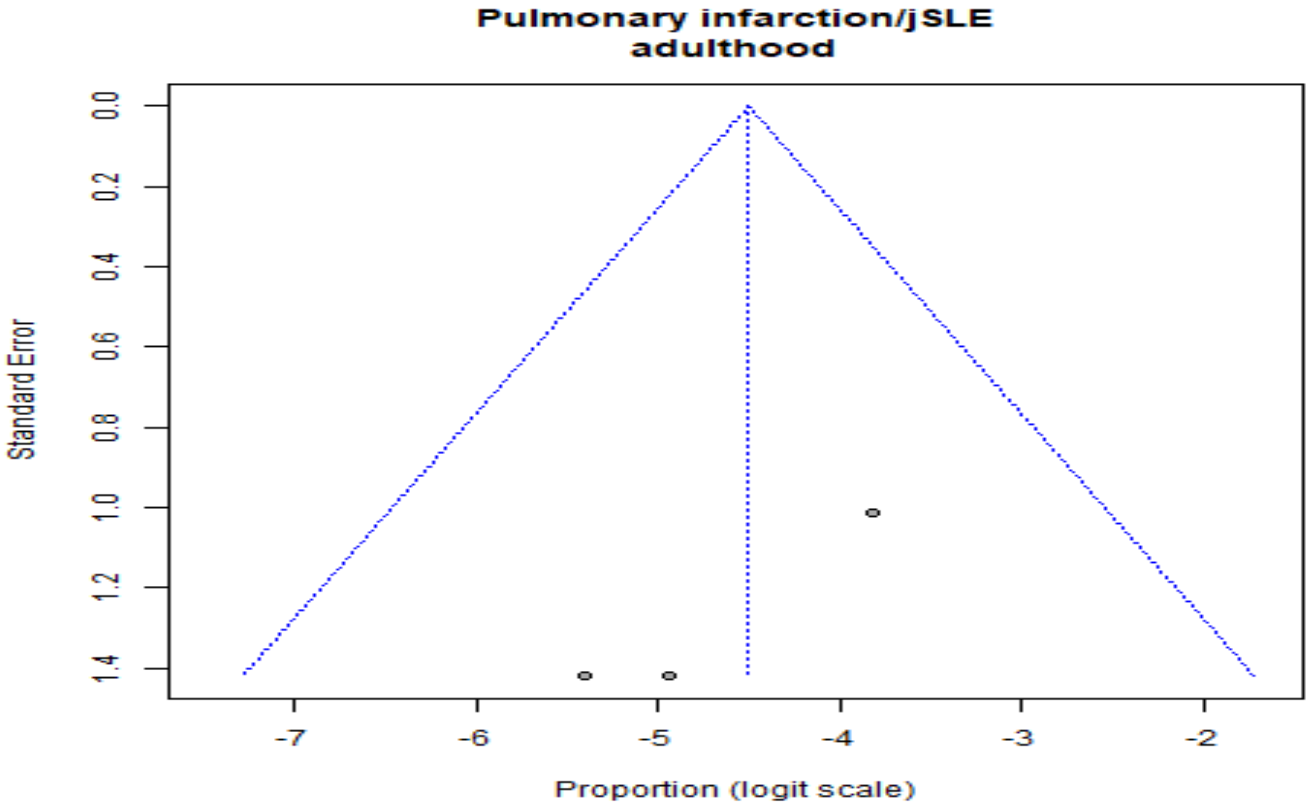
Supplementary Figure S4.60: Forest and Funnel Plots of studies reporting the prevalence of pulmonary infarction in adults with juvenile-onset Systemic Lupus Erythematosus

Pulmonary infarction

Forest



Funnel



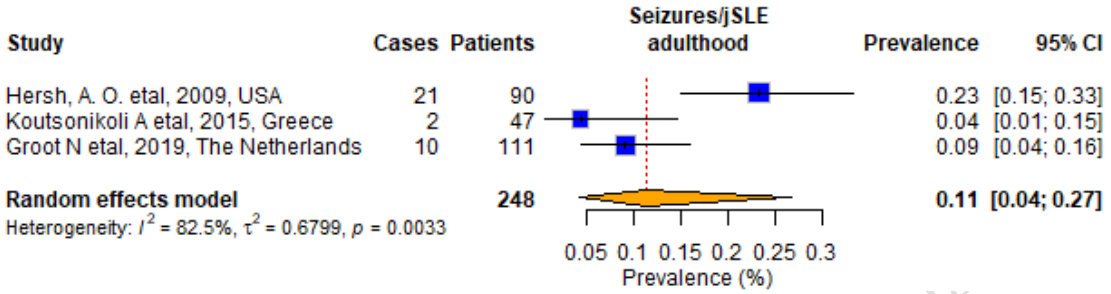
Summary statistics

```
                                proportion   95%-CI          %W(random)
Tanigava, N. Y. etal, 2020, Brazil     0.0000   [0.0000; 0.0521]       25.2
Koutsonikoli A etal, 2015, Greece      0.0213   [0.0005; 0.1129]       49.6
Groot N et al, 2019, Netherlands       0.0000   [0.0000; 0.0327]       25.2
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 1
                    proportion           95%-CI
Random effects model     0.0109 [0.0027; 0.0427]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 24.5185]; tau = 0 [0.0000; 4.9516]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        0.94    2  0.6236
```

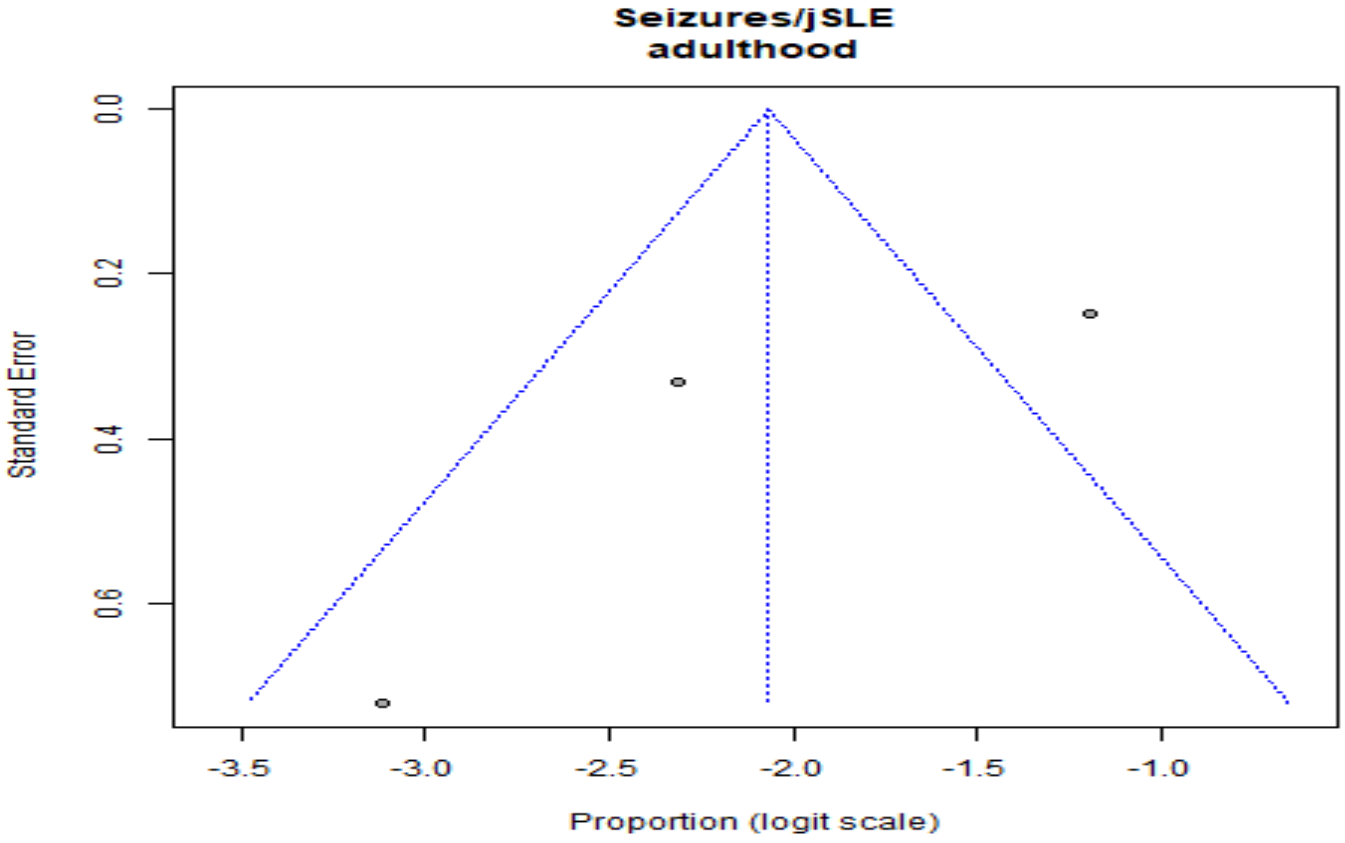
Supplementary Figure S4.61: Forest and Funnel Plots of studies reporting the prevalence of seizures in adults with juvenile-onset Systemic Lupus Erythematosus

Seizures

Forest



Funnel



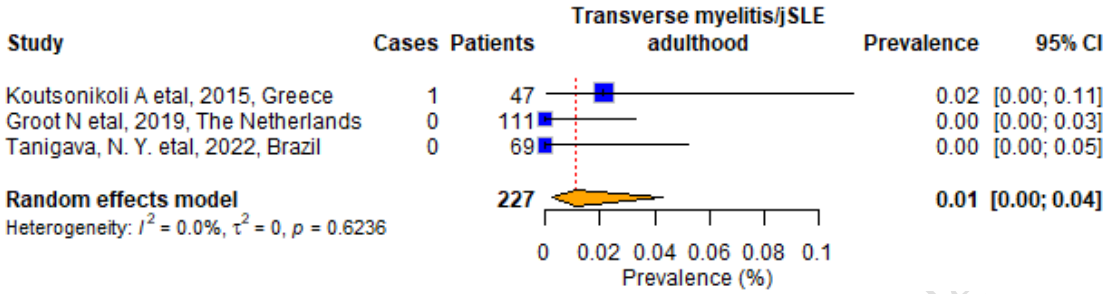
Summary statistics

```
                                  proportion  95%-CI          %W(random)
Hersh, A. O. etal, 2009, USA          0.2333  [0.1506; 0.3343]       39.1
Koutsonikoli A etal, 2015, Greece     0.0426  [0.0052; 0.1454]       24.1
Groot N etal, 2019, The Netherlands   0.0901  [0.0441; 0.1594]       36.7
Number of studies: k = 3; Number of observations: o = 248; Number of events: e = 33
                     proportion           95%-CI
Random effects model     0.1124 [0.0422; 0.2668]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.6799 [0.0646; 36.6266]; tau = 0.8246 [0.2542; 6.0520]
 I^2 = 82.5% [46.4%; 94.3%]; H = 2.39 [1.37; 4.19]
Test of heterogeneity:      Q d.f. p-value
                        11.43    2  0.0033
```

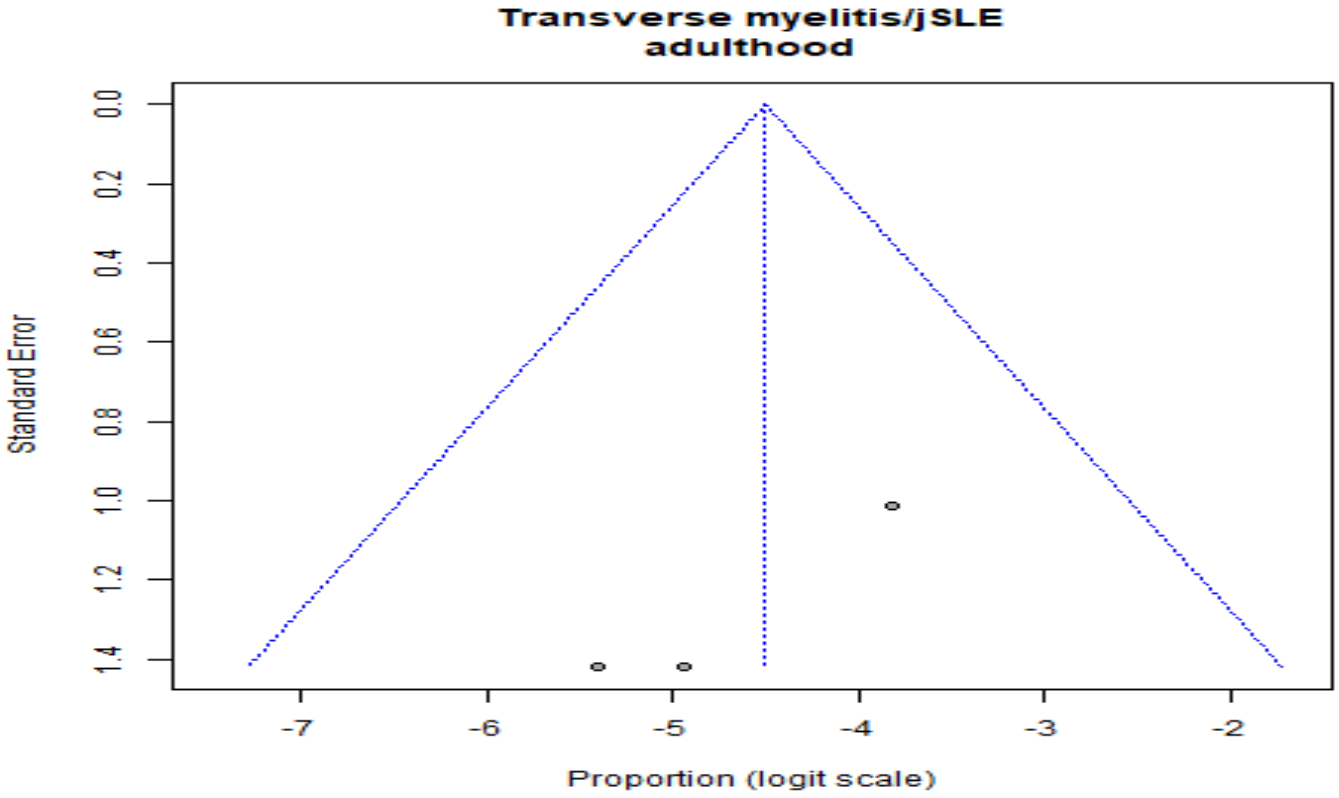
Supplementary Figure S4.62: Forest and Funnel Plots of studies reporting the prevalence of transverse myelitis in adults with juvenile-onset Systemic Lupus Erythematosus

Transverse myelitis

Forest



Funnel



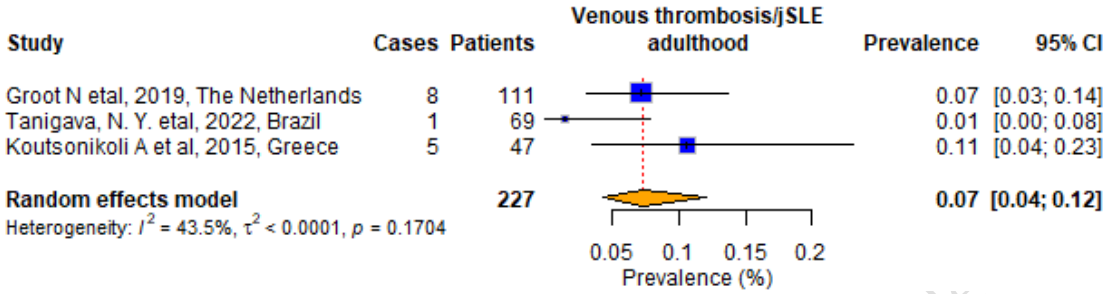
Summary statistics

```
                                  proportion  95%-CI          %W(random)
Koutsonikoli A etal, 2015, Greece     0.0213  [0.0005; 0.1129]       49.6
Groot N etal, 2019, The Netherlands   0.0000  [0.0000; 0.0327]       25.2
Tanigava, N. Y. etal, 2022, Brazil    0.0000  [0.0000; 0.0521]       25.2
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 1
                     proportion          95%-CI
Random effects model     0.0109 [0.0027; 0.0427]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 24.5185]; tau = 0 [0.0000; 4.9516]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        0.94    2  0.6236
```

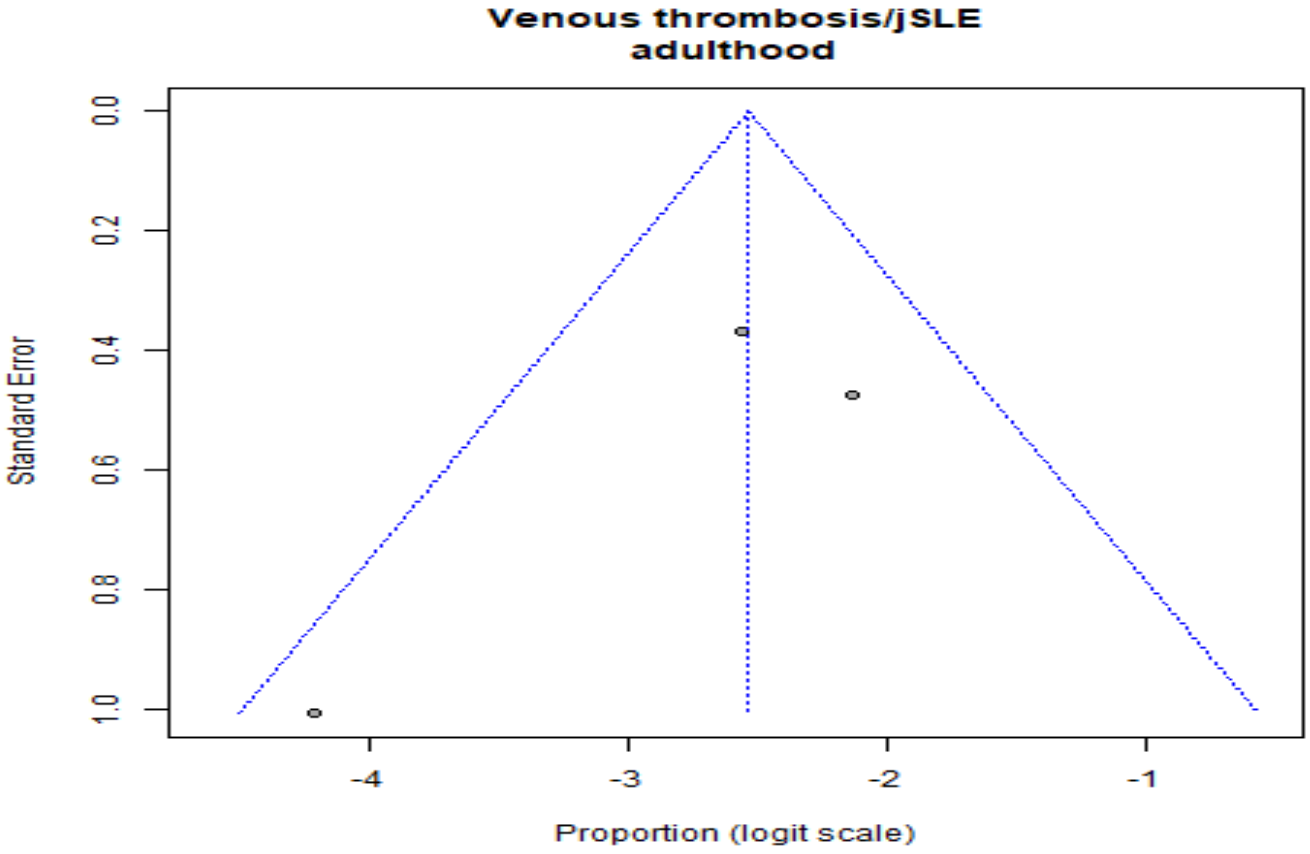
Supplementary Figure S4.63: Forest and Funnel Plots of studies reporting the prevalence of venous thrombosis in adults with juvenile-onset Systemic Lupus Erythematosus

Venous thrombosis

Forest



Funnel



Summary statistics

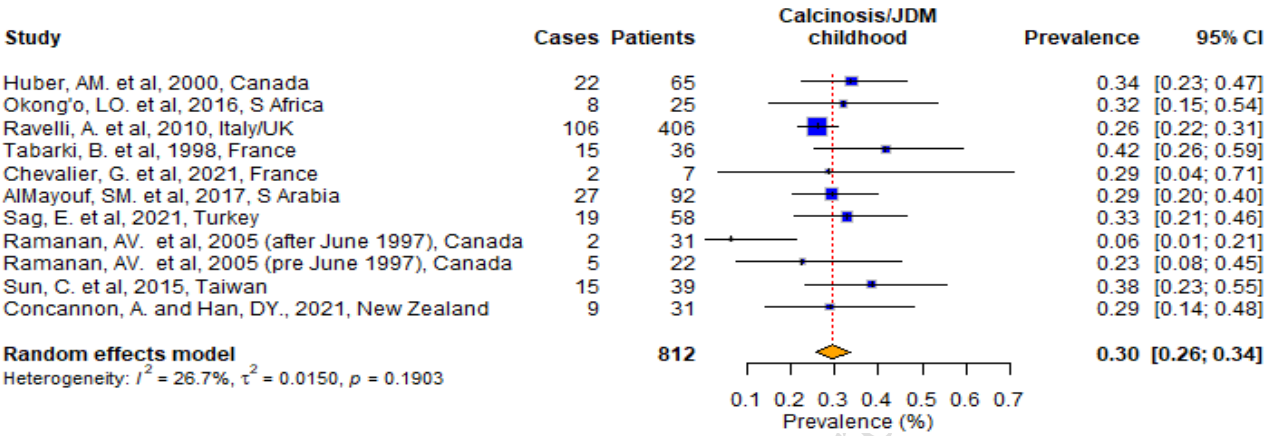
```
                                    proportion  95%-CI          %W(random)
Groot N etal, 2019, The Netherlands     0.0721  [0.0316; 0.1371]        57.6
Tanigava, N. Y. etal, 2022, Brazil      0.0145  [0.0004; 0.0781]         7.7
Koutsonikoli A et al, 2015, Greece      0.1064  [0.0355; 0.2310]        34.7
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 14
                     proportion           95%-CI
Random effects model     0.0735 [0.0439; 0.1204]
Quantifying heterogeneity (with 95%-CIs):  tau^2 < 0.0001 [0.0000; 47.4982]; tau = 0.0006 [0.0000; 6.8919]
 I^2 = 43.5% [0.0%; 83.1%]; H = 1.33 [1.00; 2.43]
Test of heterogeneity:     Q d.f. p-value
                        3.54    2  0.1704
```

Childhood/JDM

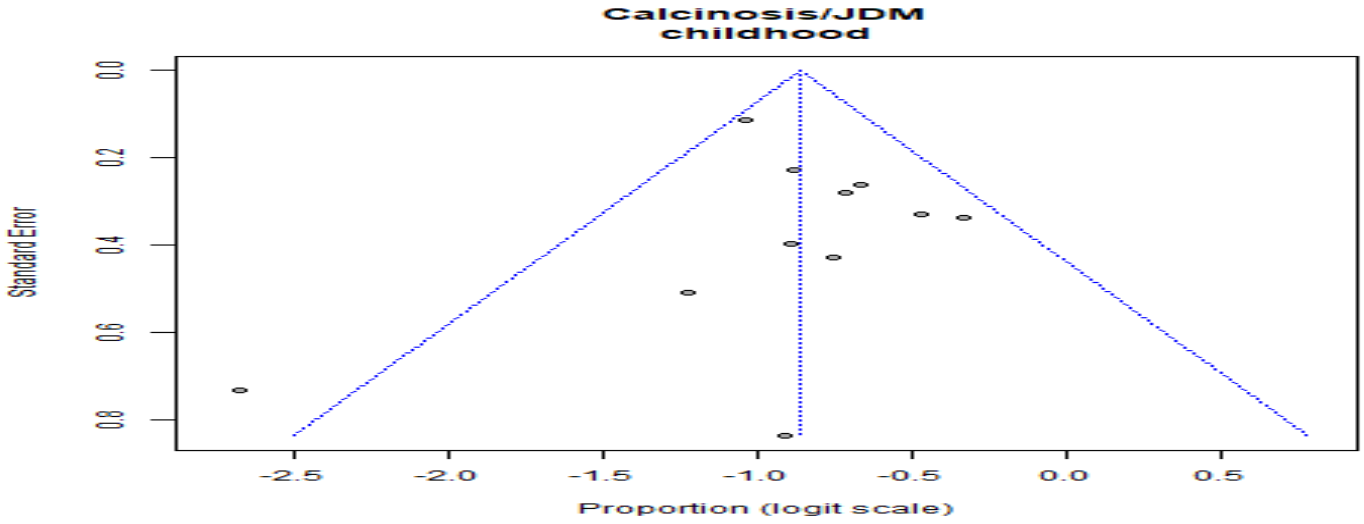
Supplementary Figure S4.64: Forest and Funnel Plots of studies reporting the prevalence of calcinosis in children with Juvenile Dermatomyositis

Calcinosis

Forest



Funnel



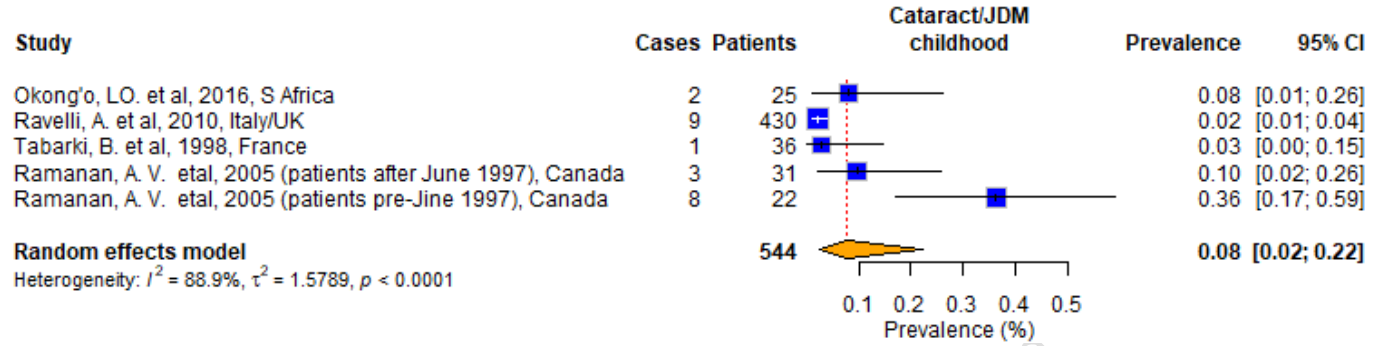
Summary statistics

```
                                                    proportion  95%-CI          %W(random)
Huber, AM. et al, 2000, Canada                          0.3385  [0.2257; 0.4665]       11.2
Okong'o, LO. et al, 2016, S Africa                      0.3200  [0.1495; 0.5350]        4.7
Ravelli, A. et al, 2010, Italy/UK                       0.2611  [0.2190; 0.3067]       33.6
Tabarki, B. et al, 1998, France                         0.4167  [0.2551; 0.5924]        7.2
Chevalier, G. et al, 2021, France                       0.2857  [0.0367; 0.7096]        1.3
AlMayouf, SM. et al, 2017, S Arabia                     0.2935  [0.2031; 0.3976]       13.9
Sag, E. et al, 2021, Turkey                             0.3276  [0.2101; 0.4634]       10.0
Ramanan, AV.  et al, 2005 (after June 1997), Canada     0.0645  [0.0079; 0.2142]        1.7
Ramanan, AV.  et al, 2005 (pre June 1997), Canada       0.2273  [0.0782; 0.4537]        3.4
Sun, C. et al, 2015, Taiwan                             0.3846  [0.2336; 0.5538]        7.6
Concannon, A. and Han, DY., 2021, New Zealand           0.2903  [0.1422; 0.4804]        5.4
Number of studies: k = 11; Number of observations: o = 812; Number of events: e = 230
                     proportion           95%-CI
Random effects model     0.2970 [0.2591; 0.3381]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.0150 [0.0000; 0.7653]; tau = 0.1226 [0.0000; 0.8748]
 I^2 = 26.7% [0.0%; 63.7%]; H = 1.17 [1.00; 1.66]
Test of heterogeneity:       Q d.f. p-value
                         13.63   10  0.1903
```

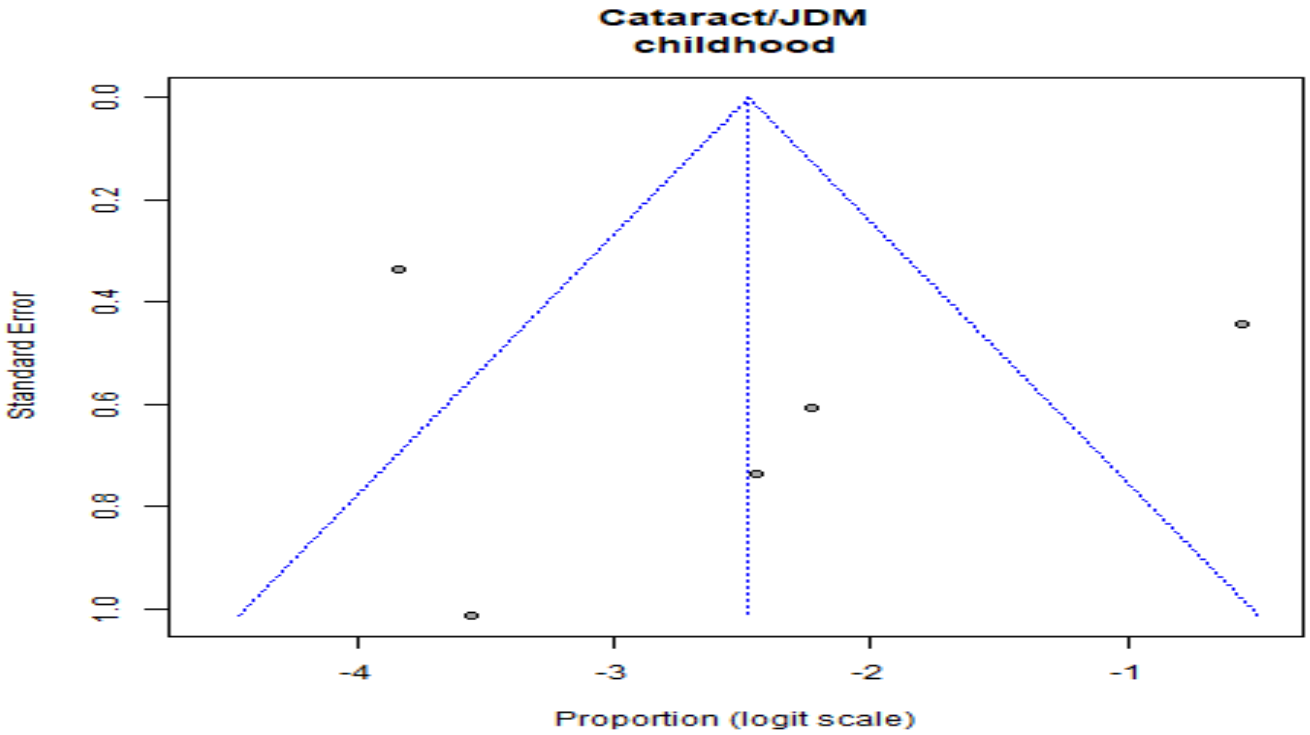
Supplementary Figure S4.65: Forest and Funnel Plots of studies reporting the prevalence of cataract in children with Juvenile Dermatomyositis

Cataract

Forest



Funnel



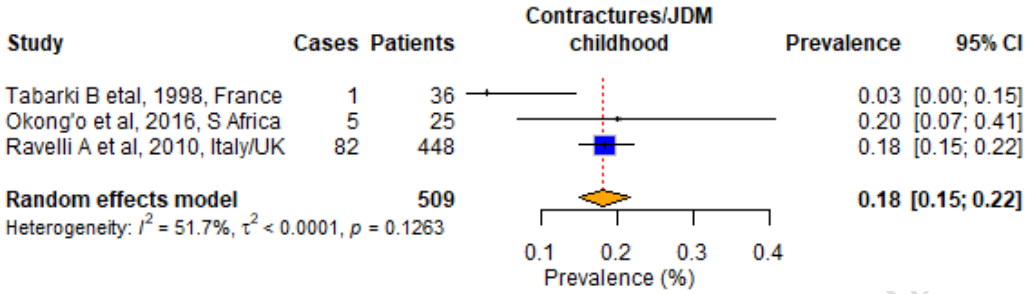
Summary statistics

```
                                                          proportion        95%-CI          %W(random)
Okong'o, LO. et al, 2016, S Africa                            0.0800       [0.0098; 0.2603]        18.7
Ravelli, A. et al, 2010, Italy/UK                             0.0209       [0.0096; 0.0394]        23.4
Tabarki, B. et al, 1998, France                               0.0278       [0.0007; 0.1453]        15.2
Ramanan, A. V.  etal, 2005 (patients after June 1997), Canada 0.0968       [0.0204; 0.2575]        20.4
Ramanan, A. V.  etal, 2005 (patients pre-Jine 1997), Canada   0.3636       [0.1720; 0.5934]        22.3
Number of studies: k = 5; Number of observations: o = 544; Number of events: e = 23
                     proportion           95%-CI
Random effects model     0.0775 [0.0239; 0.2239]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.5789 [0.3765; 13.6521]; tau = 1.2565 [0.6136; 3.6949]
 I^2 = 88.9% [76.9%; 94.7%]; H = 3.01 [2.08; 4.35]
Test of heterogeneity:      Q d.f.  p-value
                        36.16    4 < 0.0001
```

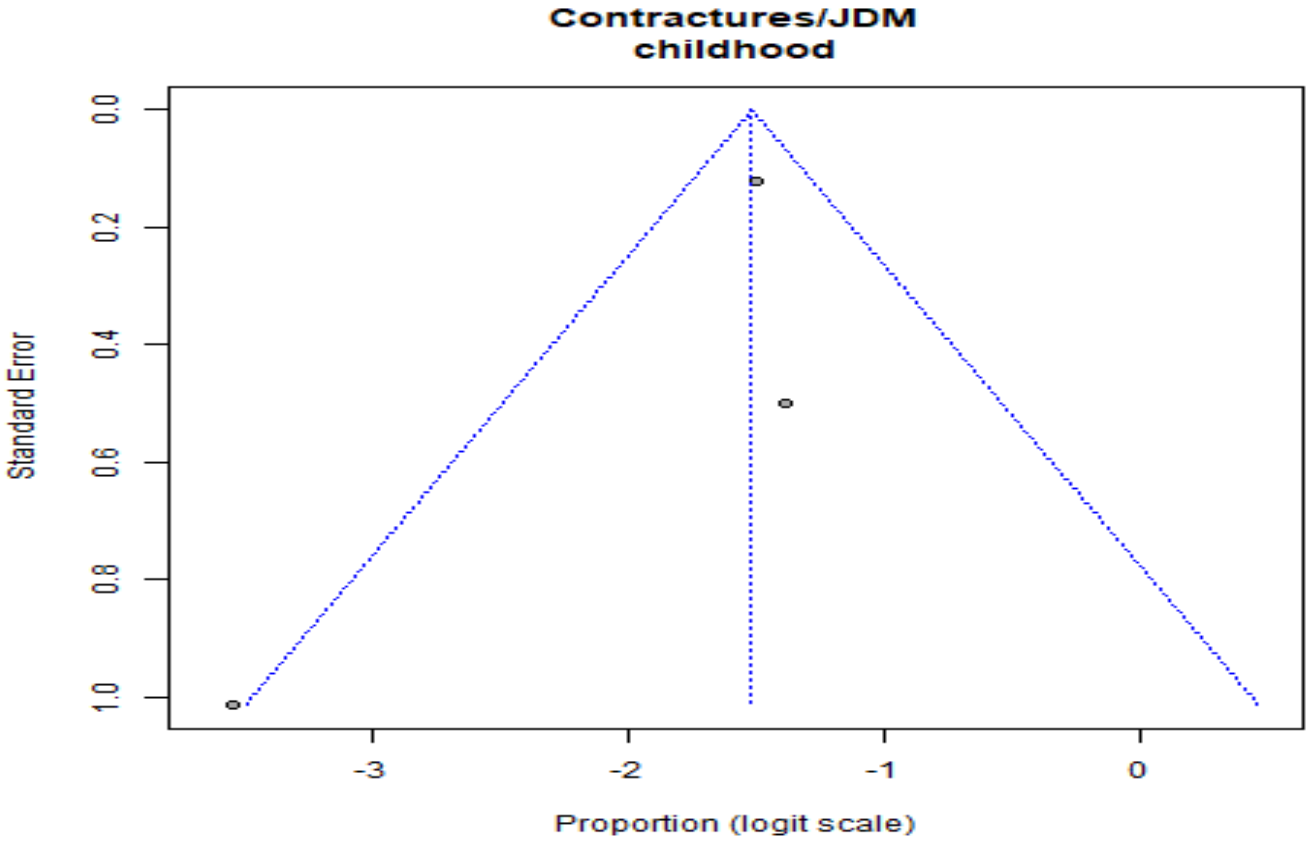
Supplementary Figure S4.66: Forest and Funnel Plots of studies reporting the prevalence of contractures in children with Juvenile Dermatomyositis

Contractures

Forest



Funnel



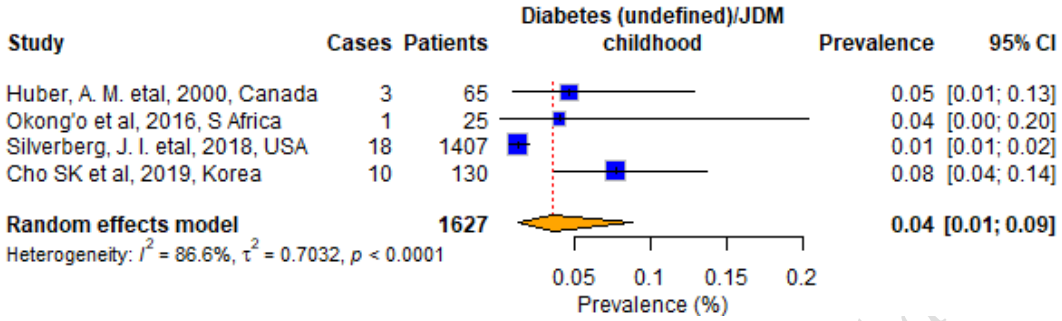
Summary statistics

```
                                proportion      95%-CI          %W(random)
Tabarki B etal, 1998, France        0.0278      [0.0007; 0.1453]        1.4
Okong'o et al, 2016, S Africa       0.2000      [0.0683; 0.4070]        5.6
Ravelli A et al, 2010, Italy/UK     0.1830      [0.1483; 0.2220]       93.1
Number of studies: k = 3; Number of observations: o = 509; Number of events: e = 88
                     proportion           95%-CI
Random effects model     0.1798 [0.1482; 0.2164]
Quantifying heterogeneity (with 95%-CIs):  tau^2 < 0.0001 [0.0000; 58.2361]; tau = 0.0020 [0.0000; 7.6313]
 I^2 = 51.7% [0.0%; 86.1%]; H = 1.44 [1.00; 2.68]
Test of heterogeneity:     Q d.f. p-value
                        4.14    2  0.1263
```

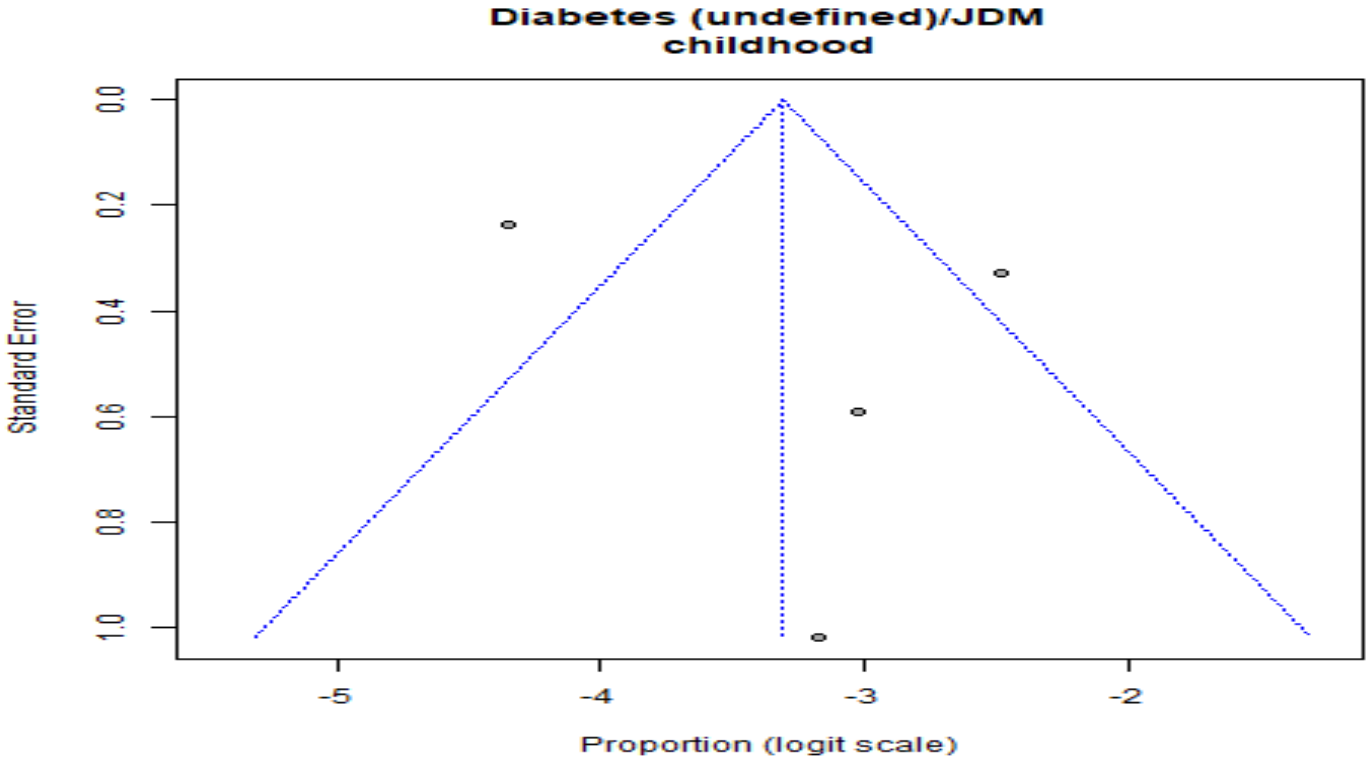
Supplementary Figure S4.67: Forest and Funnel Plots of studies reporting the prevalence of diabetes (undefined) in children with Juvenile Dermatomyositis

Diabetes (undefined)

Forest



Funnel



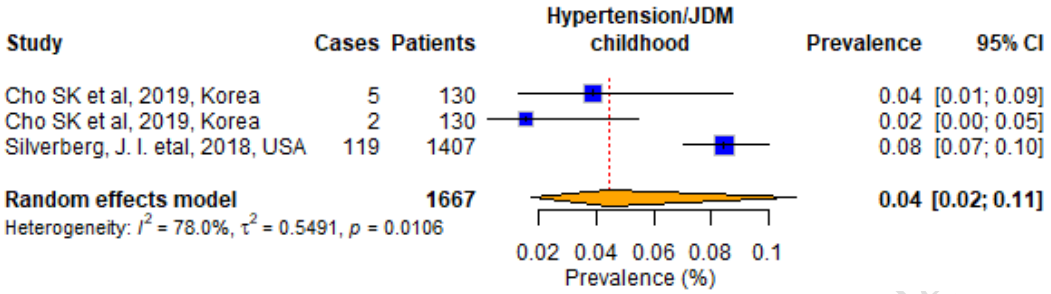
Summary statistics

```
                                  proportion    95%-CI          %W(random)
Huber, A. M. etal, 2000, Canada       0.0462    [0.0096; 0.1290]        23.3
Okong'o et al, 2016, S Africa         0.0400    [0.0010; 0.2035]        14.1
Silverberg, J. I. etal, 2018, USA     0.0128    [0.0076; 0.0201]        32.3
Cho SK et al, 2019, Korea             0.0769    [0.0375; 0.1369]        30.3
Number of studies: k = 4; Number of observations: o = 1627; Number of events: e = 32
                     proportion           95%-CI
Random effects model     0.0352 [0.0136; 0.0879]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.7032 [0.1139; 8.4433]; tau = 0.8386 [0.3376; 2.9057]
 I^2 = 86.6% [67.7%; 94.5%]; H = 2.74 [1.76; 4.25]
Test of heterogeneity:     Q d.f.  p-value
                       22.44    3 < 0.0001
```

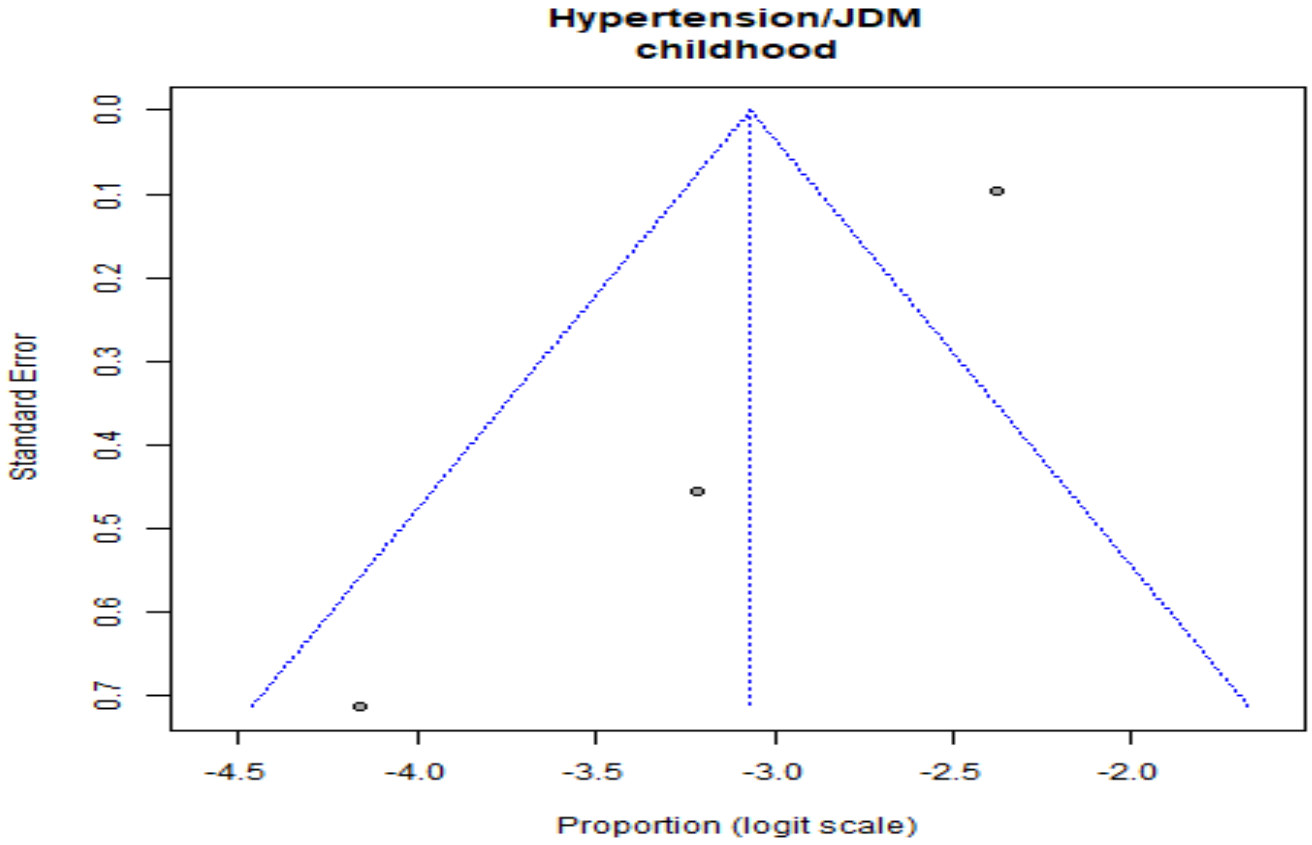
Supplementary Figure S4.68: Forest and Funnel Plots of studies reporting the prevalence of hypertension in children with Juvenile Dermatomyositis

Hypertension

Forest



Funnel



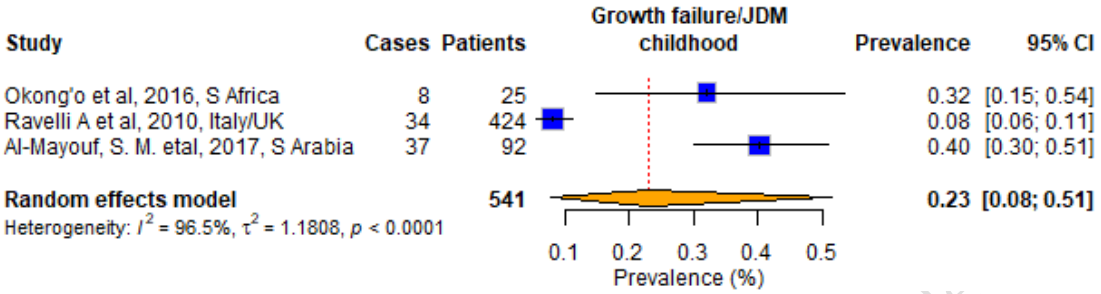
Summary statistics

```
                                proportion    95%-CI          %W(random)
Cho SK et al, 2019, Korea           0.0385    [0.0126; 0.0875]       32.5
Cho SK et al, 2019, Korea           0.0154    [0.0019; 0.0545]       23.3
Silverberg, J. I. etal, 2018, USA   0.0846    [0.0706; 0.1004]       44.1
Number of studies: k = 3; Number of observations: o = 1667; Number of events: e = 126
                     proportion           95%-CI
Random effects model     0.0444 [0.0173; 0.1095]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.5491 [0.0293; 30.9540]; tau = 0.7410 [0.1711; 5.5636]
 I^2 = 78.0% [29.1%; 93.2%]; H = 2.13 [1.19; 3.83]
Test of heterogeneity:     Q d.f. p-value
                        9.10    2  0.0106
```

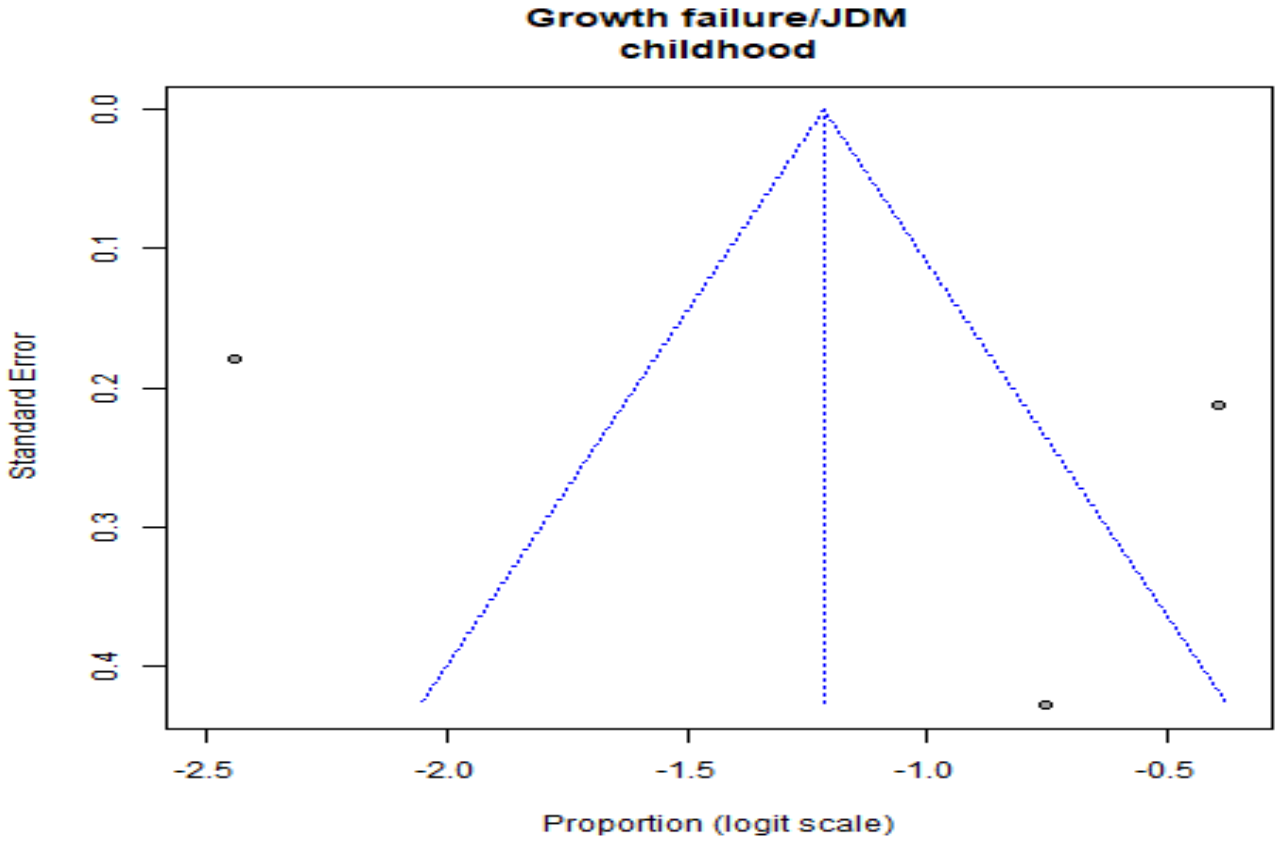
Supplementary Figure S4.69: Forest and Funnel Plots of studies reporting the prevalence of growth failure in children with Juvenile Dermatomyositis

Growth failure

Forest



Funnel



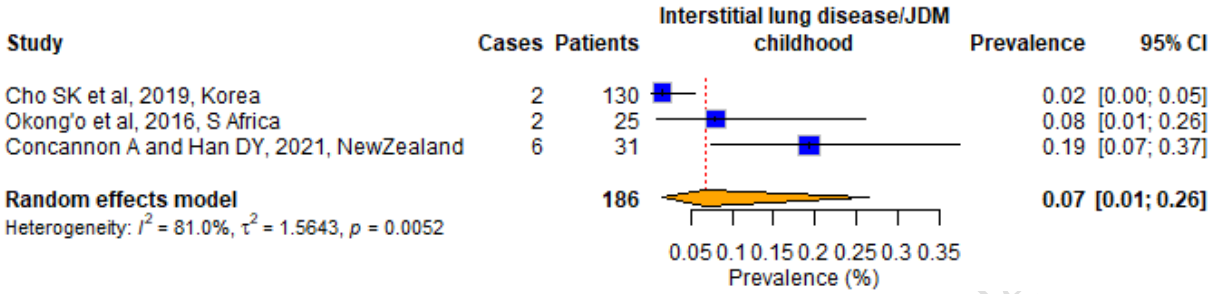
Summary statistics

```
                                    proportion        95%-CI          %W(random)
Okong'o et al, 2016, S Africa           0.3200        [0.1495; 0.5350]        30.9
Ravelli A et al, 2010, Italy/UK         0.0802        [0.0562; 0.1103]        34.7
Al-Mayouf, S. M. etal, 2017, S Arabia   0.4022        [0.3012; 0.5096]        34.4
Number of studies: k = 3; Number of observations: o = 541; Number of events: e = 79
                     proportion           95%-CI
Random effects model     0.2285 [0.0766; 0.5139]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.1808 [0.2769; 46.9921]; tau = 1.0866 [0.5262; 6.8551]
 I^2 = 96.5% [92.8%; 98.3%]; H = 5.37 [3.72; 7.74]
Test of heterogeneity:      Q d.f.  p-value
                        57.58    2 < 0.0001
```

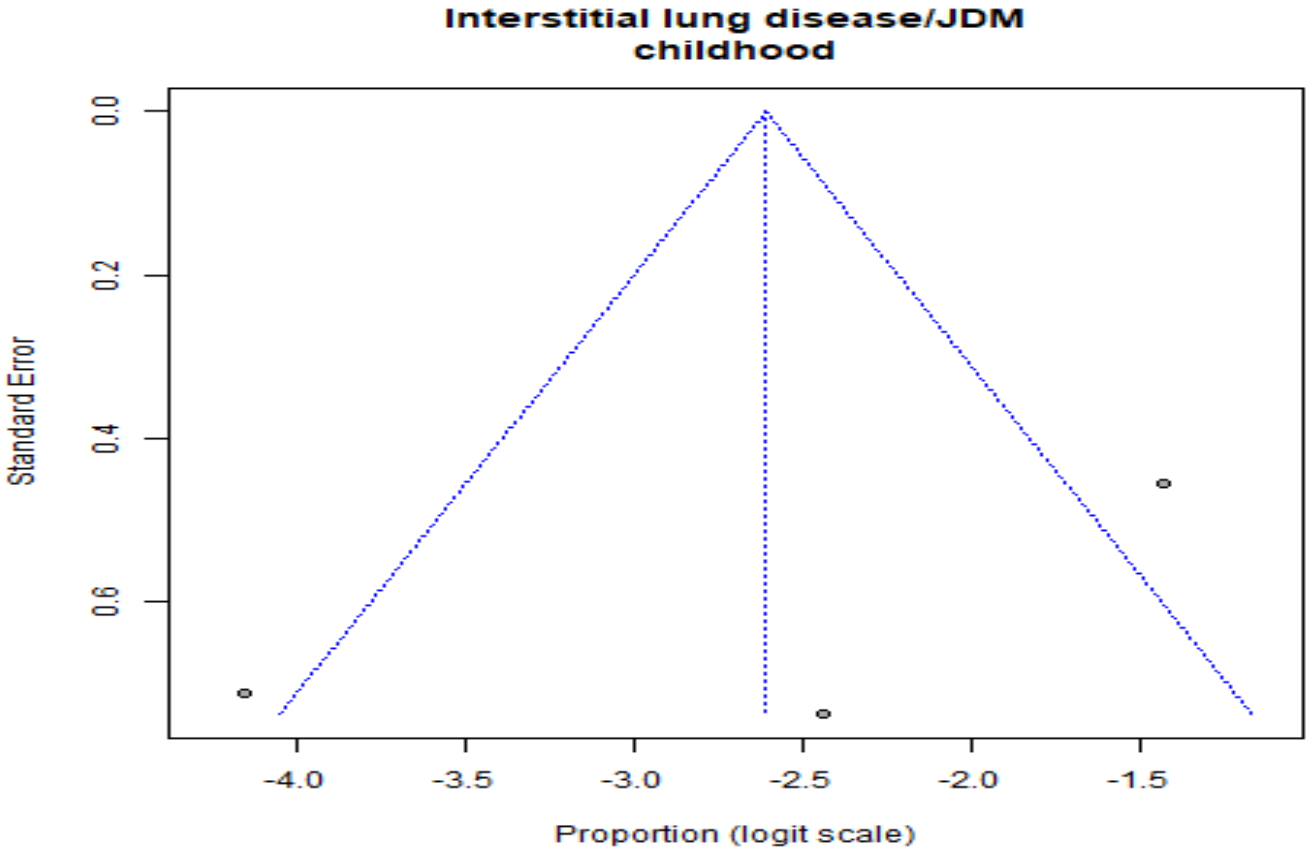
Supplementary Figure S4.70: Forest and Funnel Plots of studies reporting the prevalence of Interstitial lung disease in children with Juvenile Dermatomyositis

Interstitial lung disease

Forest



Funnel



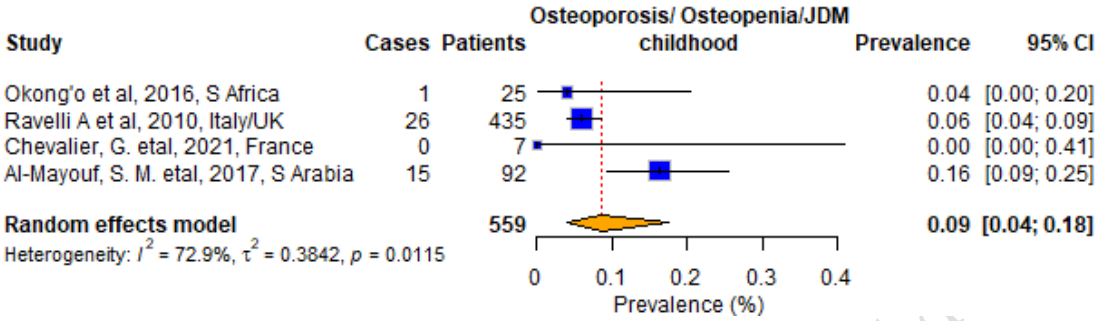
Summary statistics

```
                                    proportion    95%-CI          %W(random)
Cho SK et al, 2019, Korea               0.0154    [0.0019; 0.0545]        31.7
Okong'o et al, 2016, S Africa           0.0800    [0.0098; 0.2603]        31.2
Concannon A and Han DY, 2021, NewZealand 0.1935   [0.0745; 0.3747]        37.1
Number of studies: k = 3; Number of observations: o = 186; Number of events: e = 10
                    proportion           95%-CI
Random effects model     0.0685 [0.0148; 0.2648]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.5643 [0.1487; 74.9226]; tau = 1.2507 [0.3856; 8.6558]
 I^2 = 81.0% [40.6%; 93.9%]; H = 2.29 [1.30; 4.05]
Test of heterogeneity:      Q d.f. p-value
                        10.52    2  0.0052
```

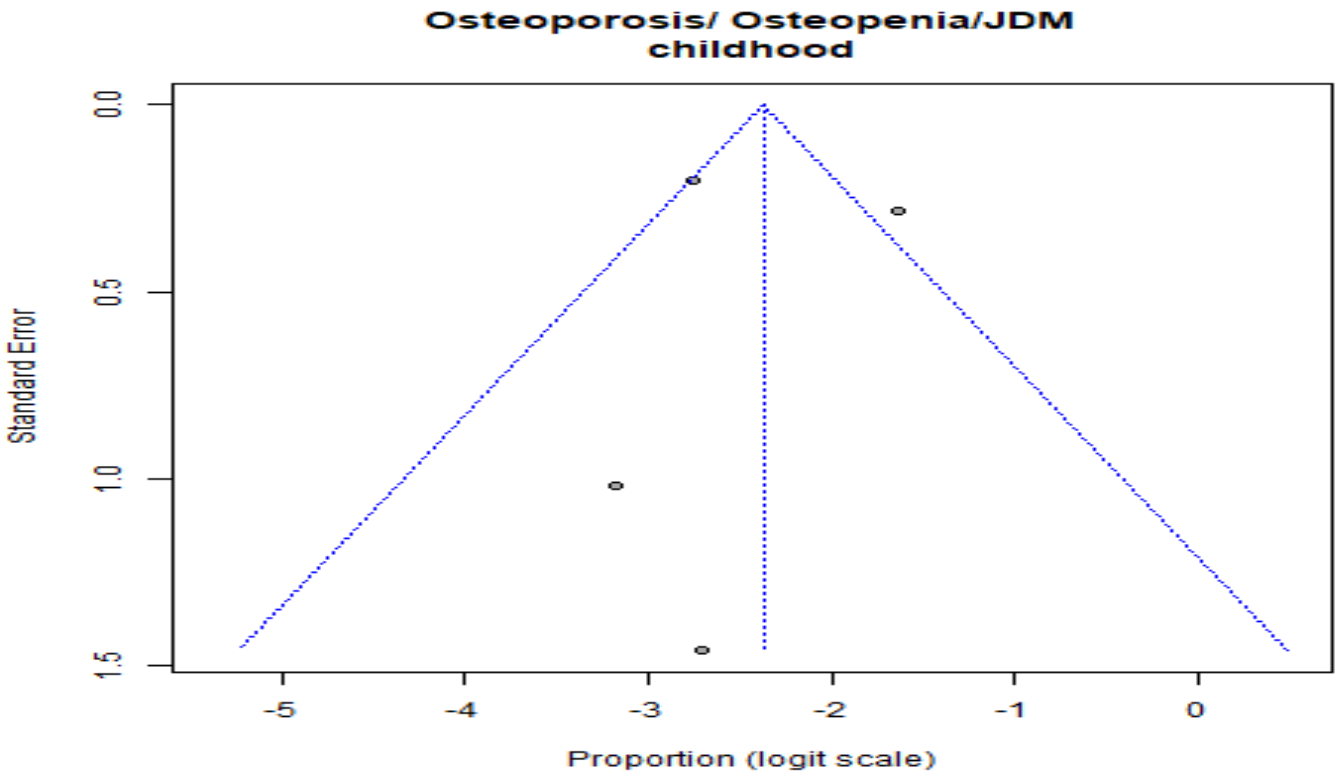
Supplementary Figure S4.71: Forest and Funnel Plots of studies reporting the prevalence of Osteoporosis/Osteopenia in children with Juvenile Dermatomyositis

Osteoporosis/ Osteopenia

Forest



Funnel



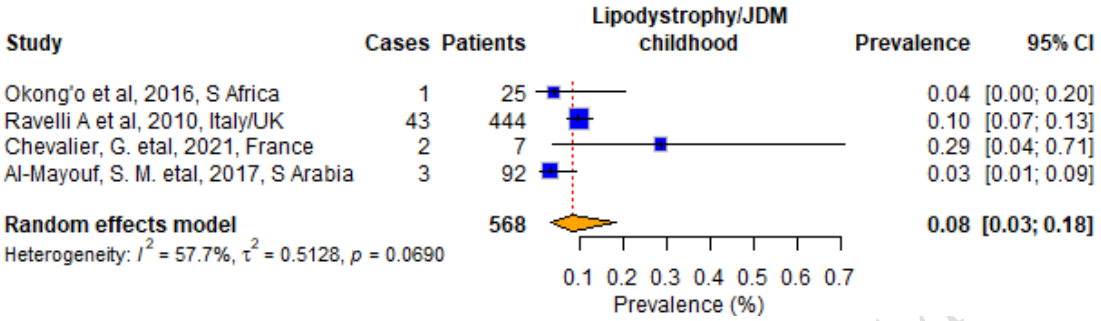
Summary statistics

```
                                    proportion        95%-CI          %W(random)
Okong'o et al, 2016, S Africa           0.0400       [0.0010; 0.2035]        12.5
Ravelli A et al, 2010, Italy/UK         0.0598       [0.0394; 0.0864]        42.0
Chevalier, G. etal, 2021, France        0.0000       [0.0000; 0.4096]         7.1
Al-Mayouf, S. M. etal, 2017, S Arabia   0.1630       [0.0942; 0.2546]        38.5
Number of studies: k = 4; Number of observations: o = 559; Number of events: e = 42
                     proportion           95%-CI
Random effects model     0.0851 [0.0391; 0.1756]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.3842 [0.0122; 5.6370]; tau = 0.6199 [0.1103; 2.3742]
 I^2 = 72.9% [23.4%; 90.4%]; H = 1.92 [1.14; 3.22]
Test of heterogeneity:      Q d.f. p-value
                        11.05    3  0.0115
```

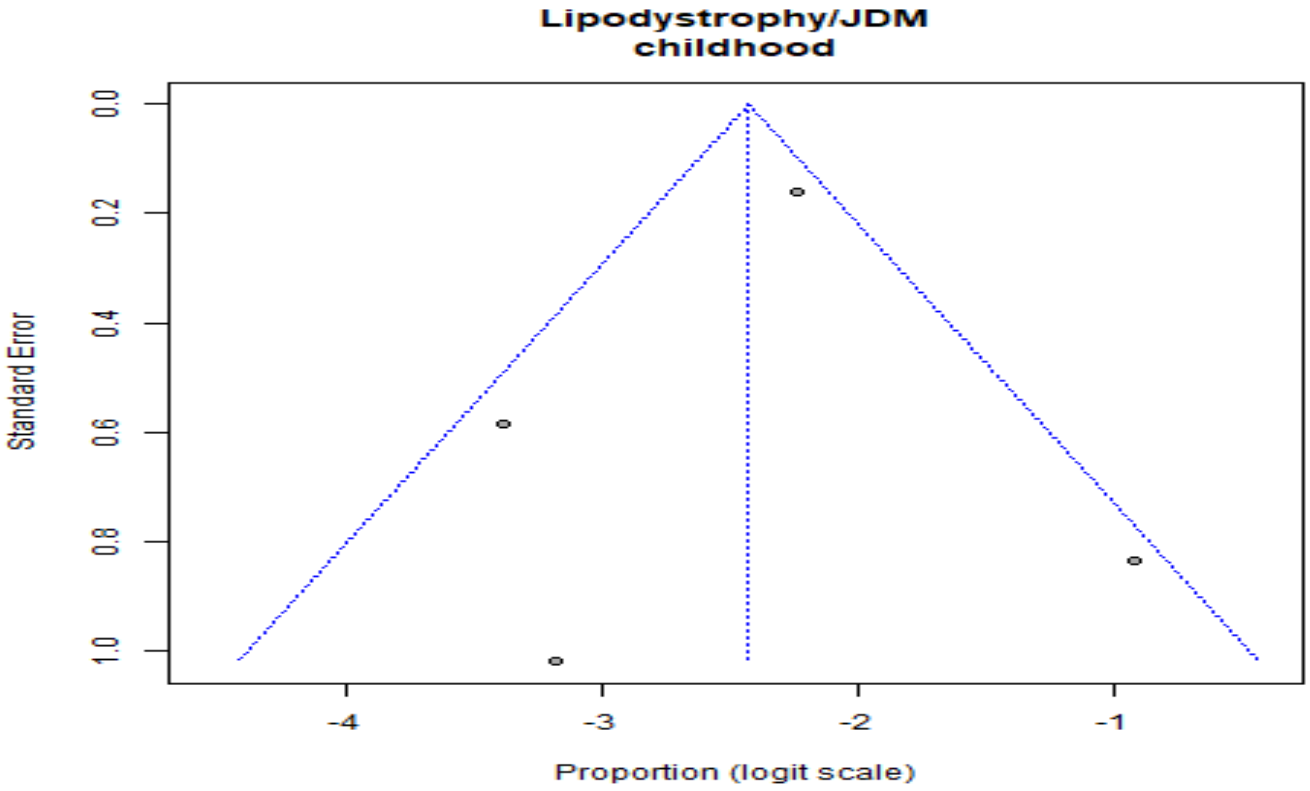
Supplementary Figure S4.72: Forest and Funnel Plots of studies reporting the prevalence of lipodystrophy in children with Juvenile Dermatomyositis

Lipodystrophy

Forest



Funnel



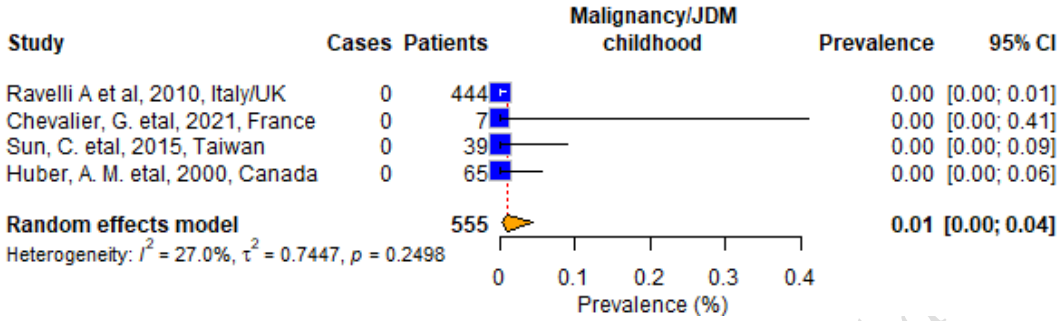
Summary statistics

```
                                    proportion        95%-CI          %W(random)
Okong'o et al, 2016, S Africa           0.0400       [0.0010; 0.2035]        14.3
Ravelli A et al, 2010, Italy/UK         0.0968       [0.0710; 0.1282]        41.3
Chevalier, G. etal, 2021, France        0.2857       [0.0367; 0.7096]        18.4
Al-Mayouf, S. M. etal, 2017, S Arabia   0.0326       [0.0068; 0.0923]        26.0
Number of studies: k = 4; Number of observations: o = 568; Number of events: e = 49
                     proportion           95%-CI
Random effects model     0.0811 [0.0338; 0.1821]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.5128 [0.0000; 17.0078]; tau = 0.7161 [0.0000; 4.1241]
 I^2 = 57.7% [0.0%; 85.9%]; H = 1.54 [1.00; 2.67]
Test of heterogeneity:     Q d.f. p-value
                        7.09    3  0.0690
```

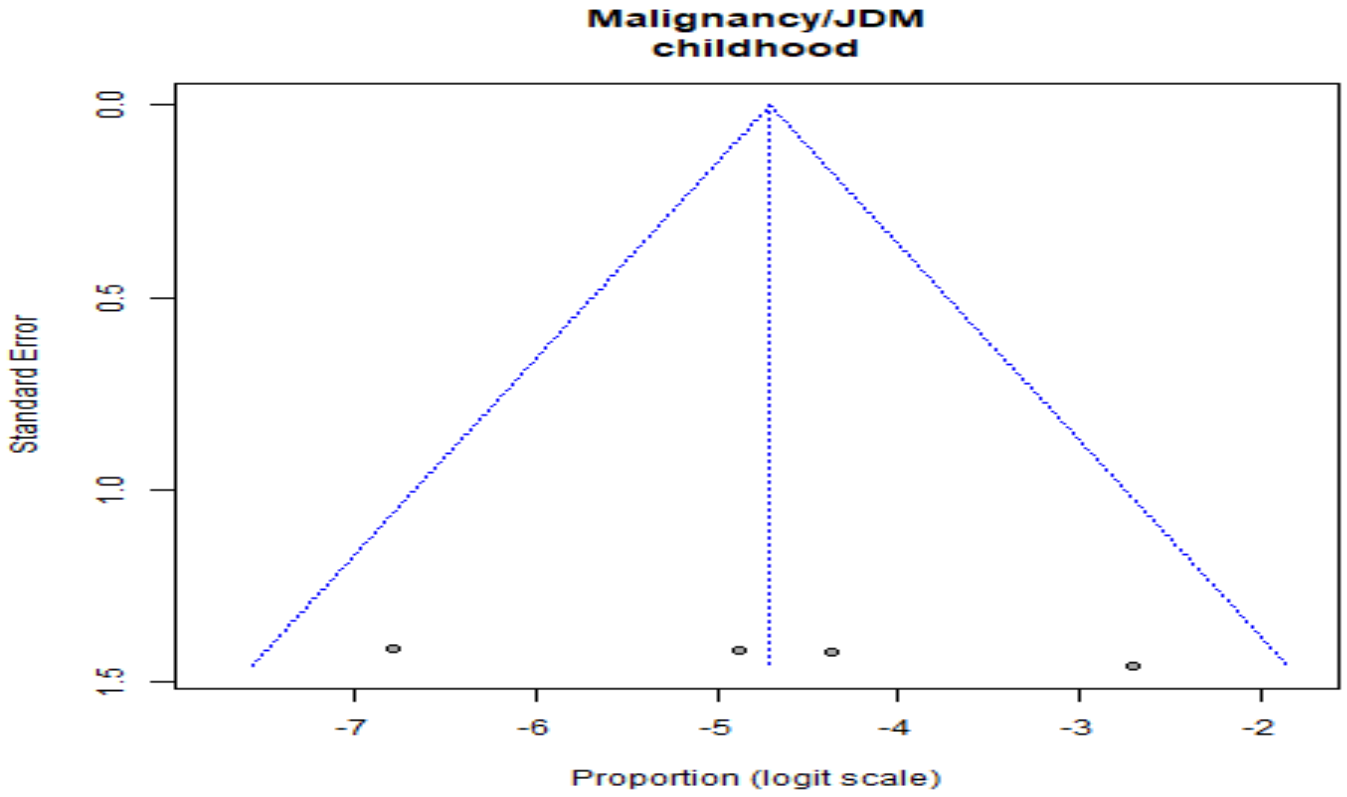
Supplementary Figure S4.73: Forest and Funnel Plots of studies reporting the prevalence of malignancy in children with Juvenile Dermatomyositis

Malignancy

Forest



Funnel



Summary statistics

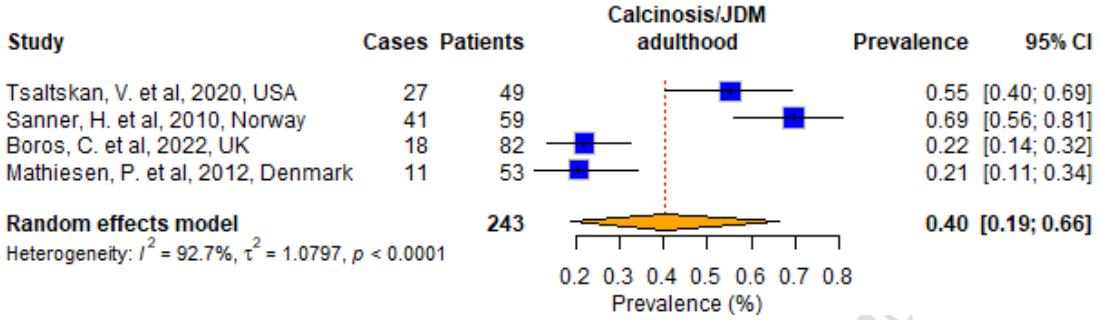
```
                                proportion     95%-CI          %W(random)
Ravelli A et al, 2010, Italy/UK     0.0000     [0.0000; 0.0083]
Chevalier, G. etal, 2021, France    0.0000     [0.0000; 0.4096]
Sun, C. etal, 2015, Taiwan          0.0000     [0.0000; 0.0903]
Huber, A. M. etal, 2000, Canada     0.0000     [0.0000; 0.0552]
Number of studies: k = 4; Number of observations: o = 555; Number of events: e = 0
                     proportion           95%-CI
Random effects model     0.0089 [0.0018; 0.0442]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.7447 [0.0000; 37.2127]; tau = 0.8629 [0.0000; 6.1002]
 I^2 = 27.0% [0.0%; 72.6%]; H = 1.17 [1.00; 1.91]
Test of heterogeneity:     Q d.f. p-value
                        4.11    3  0.2498
```

Adulthood/JDM

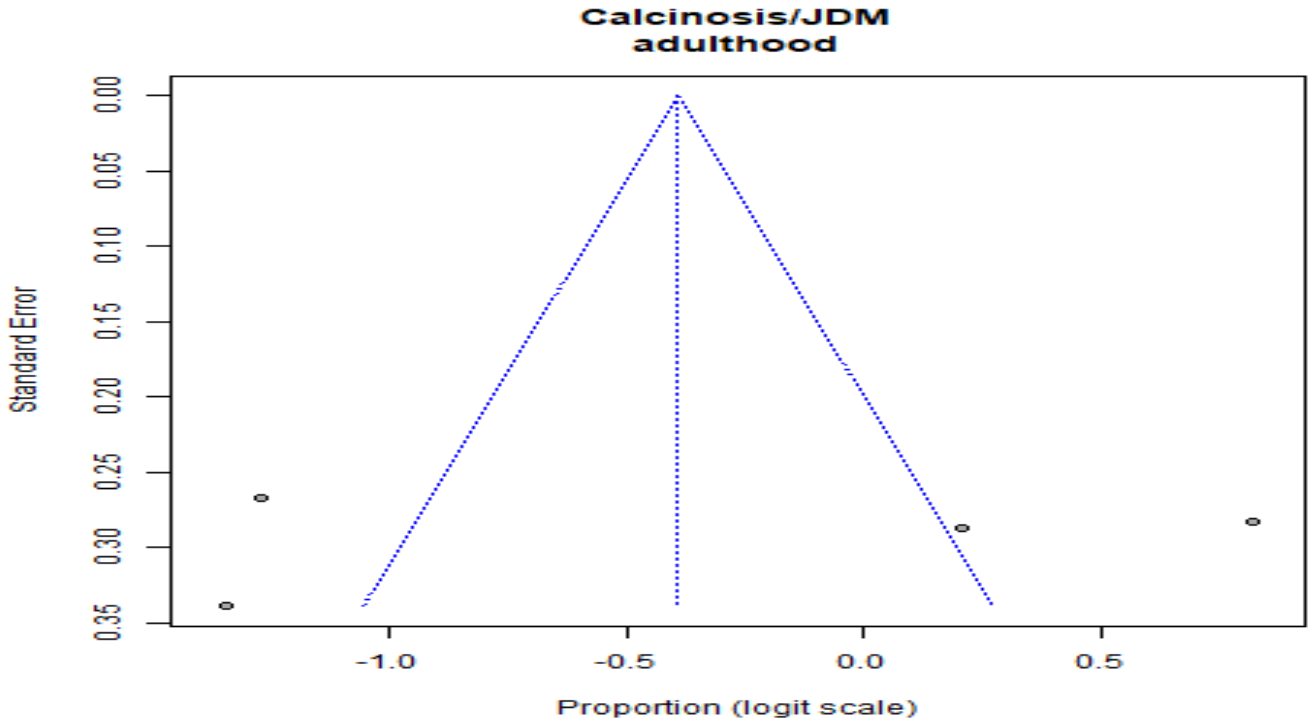
Supplementary Figure S4.74: Forest and Funnel Plots of studies reporting the prevalence of calcinosis in adults with Juvenile Dermatomyositis

Calcinosis

Forest



Funnel



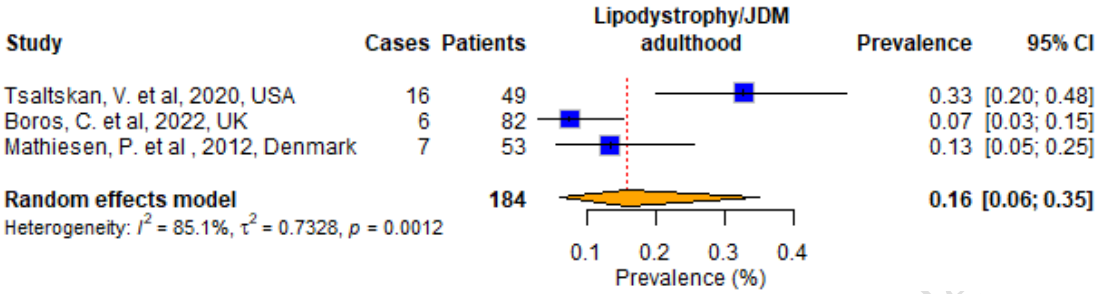
Summary statistics

```
                                  proportion   95%-CI          %W(random)
Tsaltskan, V. et al, 2020, USA        0.5510   [0.4023; 0.6933]       25.1
Sanner, H. et al, 2010, Norway        0.6949   [0.5613; 0.8081]       25.1
Boros, C. et al, 2022, UK             0.2195   [0.1356; 0.3246]       25.3
Mathiesen, P. et al, 2012, Denmark    0.2075   [0.1084; 0.3411]       24.4
Number of studies: k = 4; Number of observations: o = 243; Number of events: e = 97
                     proportion           95%-CI
Random effects model     0.4037 [0.1902; 0.6611]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.0797 [0.2878; 16.1291]; tau = 1.0391 [0.5365; 4.0161]
 I^2 = 92.7% [84.6%; 96.6%]; H = 3.71 [2.55; 5.40]
Test of heterogeneity:      Q d.f.  p-value
                        41.34    3 < 0.0001
```

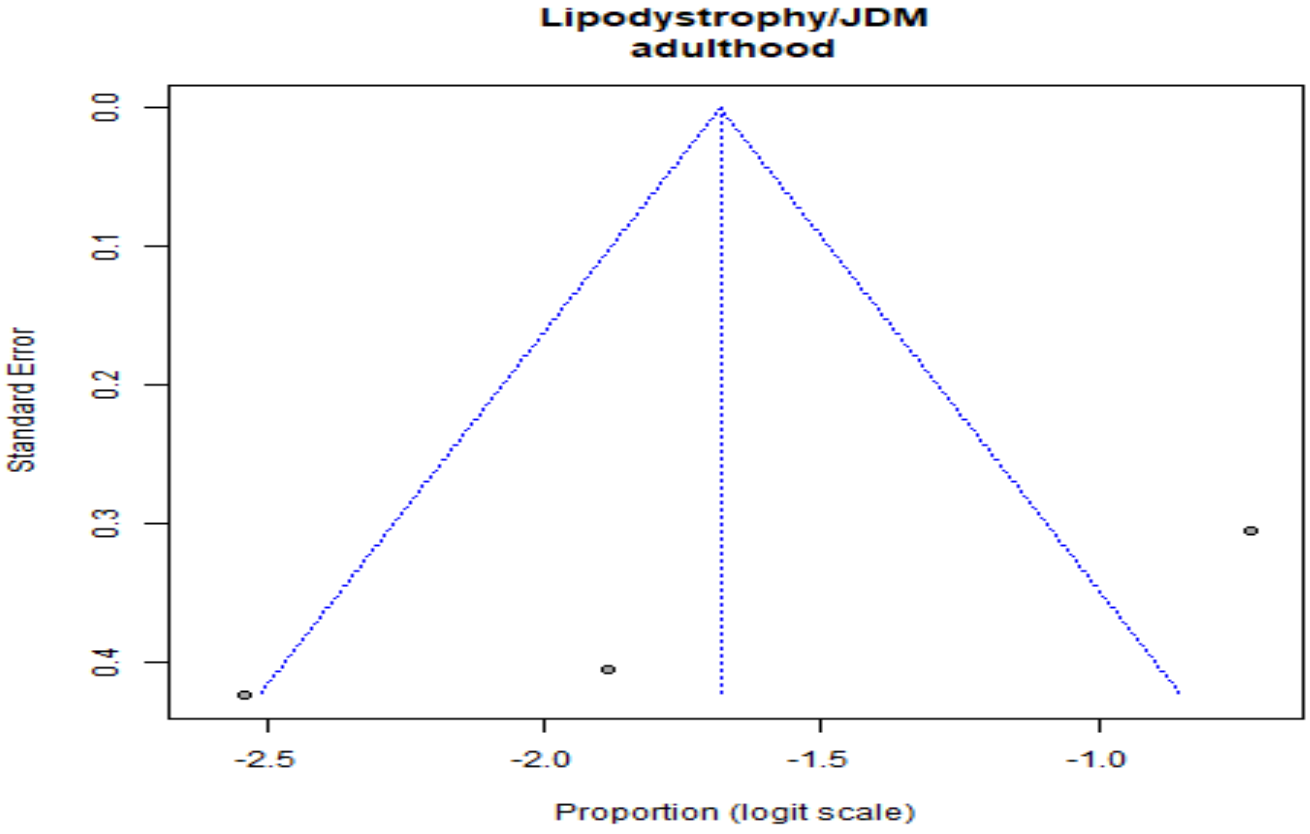
Supplementary Figure S4.75: Forest and Funnel Plots of studies reporting the prevalence of lipodystrophy in adults with Juvenile Dermatomyositis

Lipodystrophy

Forest



Funnel



Summary statistics

```
                                  proportion  95%-CI          %W(random)
Tsaltskan, V. et al, 2020, USA        0.3265  [0.1995; 0.4754]       35.4
Boros, C. et al, 2022, UK             0.0732  [0.0273; 0.1525]       32.0
Mathiesen, P. et al , 2012, Denmark   0.1321  [0.0548; 0.2534]       32.6
Number of studies: k = 3; Number of observations: o = 184; Number of events: e = 29
                     proportion           95%-CI
Random effects model     0.1567 [0.0605; 0.3491]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.7328 [0.1017; 33.2331]; tau = 0.8560 [0.3189; 5.7648]
 I^2 = 85.1% [55.9%; 94.9%]; H = 2.59 [1.51; 4.44]
Test of heterogeneity:      Q d.f. p-value
                        13.38    2  0.0012
```